

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 2

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\Boiler feed Water Preheater-AK.pvdb

Date of Analysis : Aug 31,2026 9:57am

PV Elite 28, January 2026

Vessel Design Summary:

ASME Code, Section VIII Division 2, 2025

Diameter Spec : 1220.000 mm ID
 Vessel Design Length, Tangent to Tangent 542.58 cm
 Specified Datum Line Distance 0.00 cm
 Shell Side Design Temperature 280 °C
 Channel Side Design Temperature 310 °C
 Shell Side Design Pressure 85.658 kgf/cm²
 Channel Side Design Pressure 194.770 kgf/cm²
 Shop Shell Side Test Pressure 107.669 kgf/cm²
 Shop Channel Side Test Pressure 248.027 kgf/cm²
 Wind Design Code ASCE/SEI 7-22
 Earthquake Design Code ASCE/SEI 7-22

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell		...	...		+IT
Head		...	...		+IT
Head		...	...		+IT
Nozzle	SA-336 F11	3	...	K11572	
Nozzle	SA-266 2	...	...	K03506	
Nozzle Flg	SA-105	...	...	K03504	
Tubes	SA-213 T11	...	...	K11597	
Tubesheet	SA-336 F11	3	...	K11572	
Hrz Bolting	SA-193 B7	...	<= 64	G41400	
Saddles	SA-387 11	2	...	K11789	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (kgf/cm² & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Ch Dish Head	194.775	0.00	200.000	2.0000	N/A	No
Ch Shell	194.775	0.00	198.000	2.0000	N/A	No
Main Shell	85.780	0.00	86.000	2.0000	N/A	No
Shell Dish End	85.780	0.00	94.000	2.0000	No	No

Liquid Level: 122.00 cm Dens.: 999.542 kg/m³ Sp. Gr.: 1.000

Element Types and Properties:

Element Type	"To" Elev cm	Element Length cm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Sphere	0.0	0.0	52.0	44.0	42.8	...	1.00	1.00
Cylinder	177.5	177.5	88.0	88.0	86.3	...	1.00	1.00
Cylinder	537.5	330.0	38.0	38.0	37.8	...	1.00	1.00
Ellipse	542.6	5.1	45.0	37.0	33.7	...	1.00	1.00

Local Stress Analysis Results:

Analysis	Max Stress	High Stress	Pass
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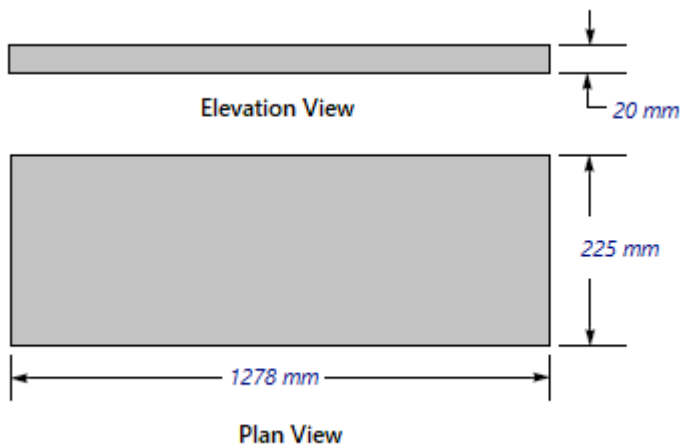
Description	Type	Ratio	Location	Fail
A1-Tube Inlet	WRC-297	0.987	Shell	Passed
B1-Tube Outlet	WRC-297	0.987	Shell	Passed
A2-Shell inlet	WRC-297	0.994	Shell	Passed
B2-Shell outle	WRC-297	0.995	Shell	Passed

Foundation/Support Summary

Saddle Parameters:

Saddle Width	200.000	mm
Saddle Bearing Angle	120.000	deg.
Centerline Dimension	1180.000	mm
Wear Pad Width	580.000	mm
Wear Pad Thickness	12.000	mm
Wear Pad Bearing Angle	132.000	deg.
Distance from Saddle to Tangent	800.000	mm
Baseplate Length	1278.000	mm
Baseplate Thickness	20.000	mm
Baseplate Width	225.000	mm
Number of Ribs (including outside ribs)	4	
Rib Thickness	12.000	mm
Web Thickness	12.000	mm
Height of Center Web	570.000	mm
Number of Bolts in Baseplate	2	

Baseplate Sketch



Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	41498.94	kgf
Maximum Transverse Saddle Shear Load	1415.56	kgf
Maximum Longitudinal Saddle Shear Load	2831.12	kgf

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	10620.73	kgf
Maximum Transverse Saddle Shear Load	144.66	kgf
Maximum Longitudinal Saddle Shear Load	133.23	kgf

Weights:

Fabricated - Bare w/o removable internals	25918.4	kg
Shop Test - Fabricated + water (full)	31996.1	kg
Shipping - Fab. + rem. intls.+ shipping app.	26346.7	kg
Erected - Fab. + rem. intls.+ insul. (etc)	26346.7	kg
Empty - Fab. + intls. + details + wghts.	26346.7	kg
Operating - Empty + operating liquid (no ca)	29977.2	kg

Nozzle Calculation Summary (kgf/cm² & mm):

Description	MAWP	Ext	MAPNC	Tbl 4.5.2	Weld	Pass/Fail
M1-Man Hole	200	...	...	OK	OK	Passed
A1-Tube Inlet	198	...	...	OK	OK	Passed
B1-Tube Outlet	198	...	...	OK	OK	Passed
A2-Shell inlet	86	...	...	OK	OK	Passed
B2-Shell outlet	86	...	...	OK	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 86 Nozzle : B2-Shell outlet [Shellside]
 Minimum MAWP Nozzles : 198 Nozzle : B1-Tube Outlet [Tubeside]

Note: MAWPs (Internal Case) shown above are at the High Point.

VIII-2 Nozzle to Gross Structural Discontinuity Check:

No Nozzles were too close to any gross structural discontinuities per 4.5.13.1

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Inside Radius mm
10	M1-Man Hole	0.000	0.000	88.525
20	A1-Tube Inlet	700.000	90.000	247.288
20	B1-Tube Outlet	700.000	270.000	247.288
30	A2-Shell inlet	2375.000	90.000	71.863
30	B2-Shell outlet	2375.000	270.000	71.863

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia mm	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
A2-Shell inlet	150 mm	120	None	168.3	14.275	...	...	249	...
B2-Shell outlet	150 mm	120	None	168.3	14.275	...	...	249	...
M1-Man Hole	200 mm	160	WNF	219.1	23.012	...	...	217	1500
A1-Tube Inlet	600 mm	160	None	609.6	59.512	...	...	581	...
B1-Tube Outlet	600 mm	160	None	609.6	59.512	...	...	581	...

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
A2-Shell in	SA-266 2	38.000	10.000	...	...	...
B2-Shell ou	SA-266 2	38.000	10.000	...	...	...
M1-Man Hole	SA-336 F11	44.000	10.000	...	...	...
A1-Tube Inl	SA-336 F11	88.000	10.000	...	...	...
B1-Tube Out	SA-336 F11	88.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum cm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
A2-Shell inlet	237.500	90.0	200.00	0.00	Main Shell
B2-Shell outlet	237.500	270.0	200.00	0.00	Main Shell
M1-Man Hole	...	0.0	150.00	0.00	Ch Dish Head
A1-Tube Inlet	70.000	90.0	300.00	0.00	Ch Shell
B1-Tube Outlet	70.000	270.0	300.00	0.00	Ch Shell

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Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	SPHERICAL HEAD: 52.0mm THK X 1220.0mm ID X 0.0cm	SA-387 11
1	ELLIPTICAL HEAD: 2.0 X 1, 45.0mm THK X 1220.0mm ID X 5.1cm	SA-387 11
1	INSULATION: 609mm X 60mm THK	Hot Insulation
1	INSULATION: 1775mm X 60mm THK	Hot Insulation
10	BAFFLES: 1156.0mm DIA X 8.0mm THK	SA-516 70
8	TIE RODS: 25mm DIA X 2500mm	SA-193 B7M
1	TUBESHEET: 1396.0mm X 300.0mm THK	SA-336 F11
1012	TUBES: 3010.0mm X 25.4mm DIA X 2.8mm THK	SA-213 T11
2	SADDLE: 200.0mm X 120 DEG	SA-387 11
1	INSULATION: 3300mm X 90mm THK	Hot Insulation
1	INSULATION: 355mm X 90mm THK	Hot Insulation
1	CYLINDRICAL SEGMENT 88.0mm THK X 3048.0mm X 177.5cm	SA-387 11
1	CYLINDRICAL SEGMENT 88.0mm THK X 784.7mm X 177.5cm	SA-387 11
1	CYLINDRICAL SEGMENT 38.0mm THK X 3044.9mm X 330.0cm	SA-387 11
1	CYLINDRICAL SEGMENT 38.0mm THK X 787.8mm X 330.0cm	SA-387 11
1	NAMEPLATE	...

**Minimum Design Metal Temperature Results Summary (°C):**

Description	Curve	Basic MDMT	Reduced MDMT	PWHT	Thickness ratio	Gov Thk mm	E*
Notes							
Main Shell [8]	!	-10		No			
Shell Dish En[10]	!	-10		No			
Shell Dish End[7]	!	-10		No			
A2-Shell inlet[1]	B	22	22	No	0.993	38.000	1.00
B2-Shell outle[1]	B	22	22	No	0.994	38.000	1.00
Tubesheet: SS[13]	B	39	39	No	1.000	75.000	1.00
Warmest MDMT:		39	39				
Ch Dish Head [9]	!	-10		Yes			
Ch Shell [8]	!	-10		Yes			
M1-Man Hole [1]	B	7	5	Yes	0.971	44.000	1.00
Nozzle Flg [5]	A	-18	-32	Yes			
A1-Tube Inlet [1]	B	19	18	Yes	0.980	88.000	1.00
B1-Tube Outlet[1]	B	19	18	Yes	0.980	88.000	1.00
Tubesheet: CS[14]	B	19	13	Yes	0.882	88.000	1.00
Warmest MDMT:		19	18				

Exchanger Side	Computed MDMT	Required MDMT	Pass/Fail
Shell	39.0	-10.0	Fail
Channel/Tube	18.0	-10.0	Fail

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per equation 3.1.
- [5] - ASME Flange MDMT Calcs; Thickness ratio per equation 3.3.
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per 3.11.2.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.

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**Units used in this Analysis (Kgfc/m2):**

Name	System Unit	Constant	User Unit
Length	Feet	30.4800	cm
Force	Pounds	0.4536	kgf
Mass	Pounds	0.4536	kg
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	kgf-m.
Stress	lbs./sq.in.	0.0703	kgf/cm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0703	kgf/cm ²
Elast. Modulus	lbs./sq.in.	0.0703	kgf/cm ²
Pipe Density	lbs./cu.in.	0.0277	kg/cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Fluid Density	lbs./cu.ft.	16.0183	kg/m ³
Wind Speed	miles/hr	1.6093	km/hr
Tray Weight	lbs./sq.ft.	0.0005	kgf/cm ²
Inertia	in.4	41.6231	cm ⁴
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8824	kgf/m ²
Elevation	Feet	0.3048	m
Volume	in.3	16.3871	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data**Exchanger Design Pressures and Temperatures**

Shell Side Design Pressure	85.655	kgf/cm ²
Channel Side Design Pressure	194.77	kgf/cm ²
Shell Side Design Temperature	280.0	°C
Channel Side Design Temperature	310.0	°C
Radiography, Shell Side	RT-1	
Radiography, Channel Side	RT-1	
Service Type, Shell Side	Air/Water/Steam	
Service Type, Channel Side	Lethal	
MDMT (CET), Shell Side	-10.0	°C
MDMT (CET), Tube Side	-10.0	°C
User defined MAWP, Shell Side	0	kgf/cm ²
User defined MAWP, Channel Side	0	kgf/cm ²
User defined MAPnc, Shell Side	0	kgf/cm ²
User defined MAPnc, Channel Side	0	kgf/cm ²
User defined Test Pres., Shell Side	0	kgf/cm ²
User defined Test Pres., Channel Side	0	kgf/cm ²

Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	N	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	Hydrostatic
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS



Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-22
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	[V] 180 km/hr
Surface Roughness Category	D: Flat, unobstructed
Type of Surface	Moderately Smooth
Base Elevation	0 cm
Percent Wind for Hydrotest	33.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment	H or Hh 0 cm
Distance Upwind of Crest	Lh 0 cm
Distance from Crest to the Vessel	x 0 cm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 cm

Seismic Design Code	ASCE/SEI 7-22
Importance Factor	1.500
Force Modification Factor R	3.000
Site Class	D
Component Elevation Ratio	z/h 0.000
Amplification Factor	Ap 0.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Component resonance duct. fact.	[CAR] 0.000
Component strength factor	[Rpo] 0.000
Overstrength factor	[Omega0] 0.000
Response Acceleration Factor	[SMS] 0.326
Response Acceleration Factor	[SM1] 0.139
Building Importance Factor	[Ie] 1.000
Site Latitude	0.000 deg
Site Longitude	0.000 deg

Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N

Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Using Metric Material Databases, ASME II D	Yes
Using ASME VIII-1 Appendix 46	No
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes

Consider Longitudinal Stress in MDMT Calculations Yes

Complete Listing of Vessel Elements and Details:

Element#	1/4
Element From Node	10
Element To Node	20
Element Type	Spherical
Description	Ch Dish Head
Distance "FROM" to "TO"	0 cm
Inside Diameter	1220 mm
Element Thickness	44 mm
Internal Corrosion Allowance	2 mm
Nominal Thickness	52 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	194.77 kgf/cm ²
Design Temperature Internal Pressure	310 °C
Design External Pressure	0 kgf/cm ²
Design Temperature External Pressure	310 °C
Effective Diameter Multiplier	1.2
Material Name	SA-387 11 +IT
Allowable Stress, Ambient	1509.1 kgf/cm ²
Allowable Stress, Operating	1509.1 kgf/cm ²
Allowable Stress, Hydrotest	2844.9 kgf/cm ²
Material Density	0.00775 kg/cm ³
P Number Thickness	0 mm
Yield Stress, Operating	2492.1 kgf/cm ²
MDMT Curve	Impact Tested
External Pressure Chart Name	CS-3
UNS Number	K11789
Class / Thickness / Grade	2 ::
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No
Element From Node	10
Detail Type	Liquid
Detail ID	Liquid: 10
Dist. from "FROM" Node / Offset dist	0 cm
Height/Length of Liquid	122 cm
Liquid Density	39.986 kg/m ³
Element From Node	10
Detail Type	Insulation
Detail ID	Hot Insulation
Dist. from "FROM" Node / Offset dist	-61 cm
Height/Length of Insulation	61 cm
Thickness of Insulation	60 mm
Density	0.0001918 kg/cm ³
Element From Node	10
Detail Type	Nozzle
Detail ID	M1-Man Hole
Dist. from "FROM" Node / Offset dist	0 mm
Nozzle Diameter	200 mm
Nozzle Schedule	160
Nozzle Class	1500
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	190.03 kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-336 F11

Element#	2/4
Element From Node	20
Element To Node	30
Element Type	Cylinder



Description	Ch Shell	
Distance "FROM" to "TO"	177.5	cm
Inside Diameter	1220	mm
Element Thickness	88	mm
Internal Corrosion Allowance	2	mm
Nominal Thickness	88	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	194.77	kgf/cm ²
Design Temperature Internal Pressure	310	°C
Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	310	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-387 11	+IT
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	20	
Detail Type	Liquid	
Detail ID	Liquid: 20	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Liquid	122	cm
Liquid Density	39.986	kg/m ³

Element From Node	20	
Detail Type	Insulation	
Detail ID	Hot Insulation	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Insulation	177.5	cm
Thickness of Insulation	60	mm
Density	0.0001918	kg/cm ³

Element From Node	20	
Detail Type	Nozzle	
Detail ID	A1-Tube Inlet	
Dist. from "FROM" Node / Offset dist	70	cm
Nozzle Diameter	600	mm
Nozzle Schedule	160	
Nozzle Class	None	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	859.9	kgf
Grade of Attached Flange	None	
Nozzle Matl	SA-336 F11	

Element From Node	20	
Detail Type	Nozzle	
Detail ID	B1-Tube Outlet	
Dist. from "FROM" Node / Offset dist	70	cm
Nozzle Diameter	600	mm
Nozzle Schedule	160	
Nozzle Class	None	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	1033.1	kgf
Grade of Attached Flange	None	
Nozzle Matl	SA-336 F11	

Element From Node	20	
Detail Type	Weight	
Detail ID	Partition plate	
Dist. from "FROM" Node / Offset dist	85	cm
Miscellaneous Weight	499.99	kgf
Offset from Element Centerline	0	mm



Element Type	Cylinder
Description	Main Shell
Distance "FROM" to "TO"	330 cm
Inside Diameter	1220 mm
Element Thickness	38 mm
Internal Corrosion Allowance	2 mm
Nominal Thickness	38 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	85.658 kgf/cm ²
Design Temperature Internal Pressure	280 °C
Design External Pressure	0 kgf/cm ²
Design Temperature External Pressure	280 °C
Effective Diameter Multiplier	1.2
Material Name	SA-387 11 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	30
Detail Type	Saddle
Detail ID	Left Saddle
Dist. from "FROM" Node / Offset dist	80 cm
Width of Saddle	200 mm
Height of Saddle at Bottom	1180 mm
Saddle Contact Angle	120.0
Height of Composite Ring Stiffener	0 mm
Width of Wear Plate	580 mm
Thickness of Wear Plate	12 mm
Contact Angle, Wear Plate (degrees)	132.0
Friction coefficient	0.0
Moment Factor	3.0
Dimension E at base (optional)	0 mm
Circumferential Eff. over Saddle	1.0
Circumferential Eff. at Midspan	1.0
Tangent to Tangent dist. (optional)	0 cm

Element From Node	30
Detail Type	Saddle
Detail ID	New Saddle
Dist. from "FROM" Node / Offset dist	250 cm
Width of Saddle	200 mm
Height of Saddle at Bottom	1180 mm
Saddle Contact Angle	120.0
Height of Composite Ring Stiffener	0 mm
Width of Wear Plate	580 mm
Thickness of Wear Plate	12 mm
Contact Angle, Wear Plate (degrees)	132.0
Friction coefficient	7.9999998e-002
Moment Factor	3.0
Dimension E at base (optional)	0 mm
Circumferential Eff. over Saddle	1.0
Circumferential Eff. at Midspan	1.0
Tangent to Tangent dist. (optional)	0 cm

Element From Node	30
Detail Type	Liquid
Detail ID	Liquid: 30
Dist. from "FROM" Node / Offset dist	0 cm
Height/Length of Liquid	122 cm
Liquid Density	999.54 kg/m ³

Element From Node	30
Detail Type	Insulation
Detail ID	Hot Insulation
Dist. from "FROM" Node / Offset dist	0 cm
Height/Length of Insulation	330 cm
Thickness of Insulation	90 mm
Density	0.0001918 kg/cm ³

Element From Node	30
-------------------	----



Detail Type	Nozzle	
Detail ID	A2-Shell inlet	
Dist. from "FROM" Node / Offset dist	30	cm
Nozzle Diameter	150	mm
Nozzle Schedule	120	
Nozzle Class	None	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	38.245	kgf
Grade of Attached Flange	None	
Nozzle Matl	SA-266 2	

Element From Node	30	
Detail Type	Nozzle	
Detail ID	B2-Shell outlet	
Dist. from "FROM" Node / Offset dist	30	cm
Nozzle Diameter	150	mm
Nozzle Schedule	120	
Nozzle Class	None	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	38.245	kgf
Grade of Attached Flange	None	
Nozzle Matl	SA-266 2	

Element#	4/4	
Element From Node	40	
Element To Node	50	
Element Type	Elliptical	
Description	Shell Dish End	
Distance "FROM" to "TO"	5.08	cm
Inside Diameter	1220	mm
Element Thickness	37	mm
Internal Corrosion Allowance	2	mm
Nominal Thickness	45	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	85.658	kgf/cm ²
Design Temperature Internal Pressure	280	°C
Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	280	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-387 11 +IT	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	

Element From Node	40	
Detail Type	Liquid	
Detail ID	Liquid: 40	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Liquid	122	cm
Liquid Density	999.54	kg/m ³

Element From Node	40	
Detail Type	Insulation	
Detail ID	Hot Insulation	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Insulation	35.58	cm
Thickness of Insulation	90	mm
Density	0.0001918	kg/cm ³

XY Coordinate Calculations:

From	To	X (Horiz.) cm	Y (Vert.) cm	DX (Horiz.) cm	DY (Vert.) cm
Ch Dish Head		...	...	...	...
Ch Shell		177.5	...	177.5	...
Main Shell		537.5	...	330	...
Shell Dish End		542.58	...	5.08	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd kgf/cm ²	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) kgf/cm ²	Element#
Ch Dish Head		194.78	52	2	1220	1509.1	1
Ch Shell		194.78	88	2	1220	1509.1	2
Main Shell		85.78	38	2	1220	1509.1	3
Shell Dish End		85.78	45	2	1220	1509.1	4

Element Required Thickness and MAWP :

From	To	Design Pressure kgf/cm ²	M.A.W.P. Corroded kgf/cm ²	M.A.P. New & Cold kgf/cm ²	Minimum Thickness mm	Required Thickness mm	Element#
Ch Dish Head		194.77	200.328	210.212	44	42.7968	1
Ch Shell		194.77	198.421	203.366	88	86.3133	2
Main Shell		85.6582	86.1354	91.1974	38	37.7951	3
Shell Dish End		85.6582	94.4074	100.021	37	33.7139	4

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:

For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

Note:

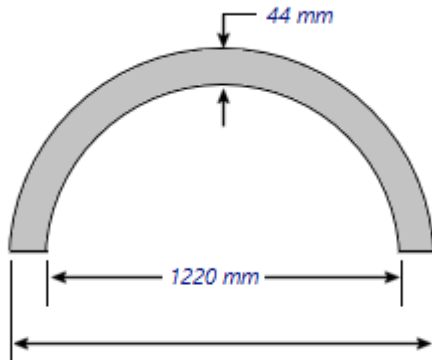
The MAWPs and MAPs for the Exchanger elements all appear to be zero or some could not be calculated. Please fill these values in on the main tubesheet tab.

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Ch Dish Head

Spherical Head Design Information:					
Design Pressure	P	194.775 kgf/cm ²	Int. Design Temperature		310 °C
Material		SA-387 11	Ext. Design Temperature		310 °C
Length		0.000 cm	External Pressure Chart		CS-3
			UNS Number		K11789
Int. Corr. All.	c	2.0000 mm	Allowable Stress (ope)	S	1509.1 kgf/cm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1509.1 kgf/cm ²
Inside Diameter	Di	1220.000 mm	Actual Stress		1372.0 kgf/cm ²
Spec. Min. Thk t _e , or sa		44.0000 mm	Spec. Nominal Thk.		52.000 mm
Surface Area		27535.7 cm ²	SG of Contents		0.040
Element Volume		475389.0 cm ³	Weight of Contents		19.0 kgf
Empty Weight		1024.9 kgf	Operating Weight		1043.9 kgf
			Comp. All. Ope. Stress		932.0 kgf/cm ²
Radiography Information:					
...			...		
Circ. Joint Efficiency		100 %	Long. Joint Eff.	E or v	100 %

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 15.250 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 7.625 mm

Material treatment options: +IT

Spherical Shell Calculation - Section 4.3.5.1

Computed Minimum Required Thickness [t]:
 $= D/2 \cdot (\exp(0.5P/(S \cdot E)) - 1) + c_i + c_o$
 $= 1224.0/2 \cdot (\exp(0.5 \cdot 194.775/(1509.097 \cdot 1.0)) - 1) + 2.0 + 0.0$
 $= 40.7968 + 2.0 + 0.0 = 42.7968 \text{ mm}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= 2 \cdot S \cdot E \cdot \ln((2 \cdot t + D)/D) - P_{liq}$
 $= 2 \cdot 1509.1 \cdot 1.0 \cdot \ln((2 \cdot 42.0 + 1224.0)/1224.0) - 0.005$
 $= 200.333 - 0.005 = 200.328 \text{ kgf/cm}^2$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= 2 \cdot S \cdot E \cdot \ln((2 \cdot t + D)/D)$
 $= 2 \cdot 1509.097 \cdot 1.0 \cdot \ln((2 \cdot 44.0 + 1220.0)/1220.0)$
 $= 210.2124 \text{ kgf/cm}^2$



Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D^2 / (E (D_o^2 - D^2))$$

$$= 194.775 \cdot 1224.0^2 / (1.0 (1308.0^2 - 1224.0^2))$$

$$= 1372.000 \text{ kgf/cm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= 0.5 \cdot P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 0.5 \cdot 194.775 / (1.0 \cdot \ln((2 \cdot 42.0 + 1224.0) / 1224.0))$$

$$= 1467.232 \text{ kgf/cm}^2$$

Percent Fiber Elongation per Table 6.1

For Heads formed from Segments $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 6.132 %

For one Piece doubly Curved Heads $100 \cdot \ln(D_b / (D_f - 2t))$ 42.838 %

Where D_b is the blank diameter for one piece doubly curved surfaces [D_b]:

= square root($4 \cdot \text{Head Surface Area} / 3.14159$)

= square root($4 \cdot 27535.7 / 3.14159$)

= 1872.419 mm

MDMT Calculations in the Spherical Portion:

Note: This Element/Detail was specified as being Impact Tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 15.250 mm

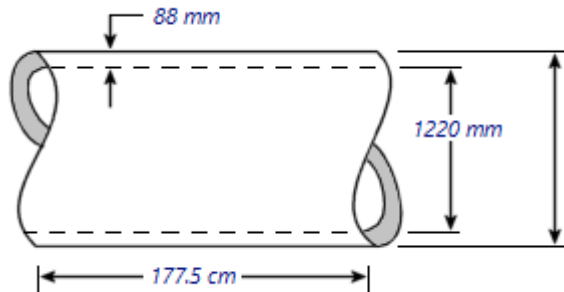
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 7.625 mm

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Ch Shell

Cylindrical Shell Design Information:					
Design Pressure	P	194.775 kgf/cm²	Int. Design Temperature	310 °C	
Material		SA-387 11	Ext. Design Temperature	310 °C	
Length		177.500 cm	External Pressure Chart	CS-3	
			UNS Number	K11789	
Int. Corr. All.	c	2.0000 mm	Allowable Stress (ope)	S	1509.1 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1509.1 kgf/cm²
Inside Diameter	Di	1220.000 mm	Actual Stress	1386.1 kgf/cm²	
Spec. Min. Thk t _e or s _a		88.0000 mm	Spec. Nominal Thk.	88.000 mm	
Surface Area		77845.5 cm²	SG of Contents	0.040	
Element Volume		2074956.4 cm³	Weight of Contents	83.0 kgf	
Empty Weight		4974.6 kgf	Operating Weight	5057.6 kgf	
			Comp. All. Ope. Stress	1019.9 kgf/cm²	

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 1224.0 \cdot (\exp(194.775/(1509.097 \cdot 1.0)) - 1) + 2.0 + 0.0$$

$$= 84.3133 + 2.0 + 0.0 = 86.3133 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1509.097 \cdot 1.0 \cdot \ln((2 \cdot 86.0 + 1224.0) / 1224.0) - 0.005$$

$$= 198.426 - 0.005 = 198.4214 \text{ kgf/cm}^2$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 1509.097 \cdot 1.0 \cdot \ln((2 \cdot 88.0 + 1220.0) / 1220.0)$$

$$= 203.3662 \text{ kgf/cm}^2$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 194.775 \cdot 1224.0 / (1.0 (1396.0 - 1224.0))$$

$$= 1386.0754 \text{ kgf/cm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 194.775 / (1.0 \cdot \ln((2 \cdot 86.0 + 1224.0) / 1224.0))$$
$$= 1481.3295 \text{ kgf/cm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 6.728 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

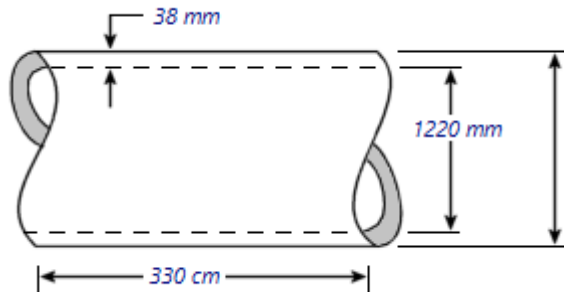
Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Main Shell

Cylindrical Shell Design Information:					
Design Pressure	P	85.780 kgf/cm²	Int. Design Temperature	280 °C	
Material		SA-387 11	Ext. Design Temperature	280 °C	
Length		330.000 cm	External Pressure Chart	CS-3	
			UNS Number	K11789	
Int. Corr. All.	c	2.0000 mm	Allowable Stress (ope)	S	1509.1 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1509.1 kgf/cm²
Inside Diameter	Di	1220.000 mm	Actual Stress	1458.3 kgf/cm²	
Spec. Min. Thk t,e, or sa		38.0000 mm	Spec. Nominal Thk.	38.000 mm	
Surface Area		134359.7 cm²	SG of Contents	1.000	
Element Volume		2008456.8 cm³	Weight of Contents	2007.6 kgf	
Empty Weight		3841.1 kgf	Operating Weight	5848.6 kgf	
			Comp. All. Ope. Stress	959.1 kgf/cm²	

Radiography Information:			
...			...
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 1224.0 \cdot (\exp(85.78/(1509.097 \cdot 1.0)) - 1) + 2.0 + 0.0$$

$$= 35.7951 + 2.0 + 0.0 = 37.7951 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1509.097 \cdot 1.0 \cdot \ln((2 \cdot 36.0 + 1224.0) / 1224.0) - 0.122$$

$$= 86.258 - 0.122 = 86.1354 \text{ kgf/cm}^2$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 1509.097 \cdot 1.0 \cdot \ln((2 \cdot 38.0 + 1220.0) / 1220.0)$$

$$= 91.1974 \text{ kgf/cm}^2$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 85.78 \cdot 1224.0 / (1.0 (1296.0 - 1224.0))$$

$$= 1458.2662 \text{ kgf/cm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 85.78 / (1.0 \cdot \ln((2 \cdot 36.0 + 1224.0) / 1224.0))$$
$$= 1500.7485 \text{ kgf/cm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 3.021 %

Minimum Design Metal Temperature Results:

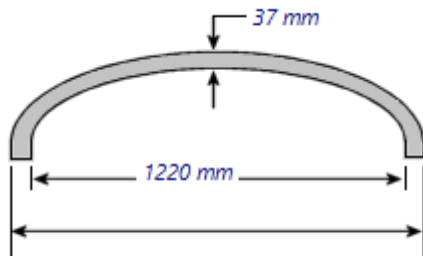
Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +IT

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell Dish End

Elliptical Head Design Information:							
Design Pressure	P	85.780	kgf/cm ²	Int. Design Temperature		280	°C
Material		SA-387	11	Ext. Design Temperature		280	°C
Length		5.080	cm	External Pressure Chart		CS-3	
				UNS Number		K11789	
Int. Corr. All.	c	2.0000	mm	Allowable Stress (ope)	S	1509.1	kgf/cm ²
Ext. Corr. All.	ce	0.0000	mm	Allowable Stress (amb)	Sa	1509.1	kgf/cm ²
Inside Diameter	Di	1220.000	mm	Actual Stress		1369.4	kgf/cm ²
Spec. Min. Thk t _e , or s _a		37.0000	mm	Spec. Nominal Thk.		45.000	mm
Surface Area		20692.9	cm ²	SG of Contents		1.000	
Element Volume		297079.2	cm ³	Weight of Contents		296.9	kgf
Empty Weight		738.4	kgf	Operating Weight		1035.3	kgf
				Comp. All. Ope. Stress		959.1	kgf/cm ²
				Aspect Ratio		2.00	

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 15.250 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 7.625 mm

Material treatment options: +IT

Head Calculation - Section 4.3.6 (and Section 4.3.7 for Ellipsoidal Head)

Computed Minimum Required Thickness [t]:
 $t_r = 31.7139 \text{ mm}$

$$\begin{aligned}
 t &= t_r + c_i + c_o \\
 &= 31.7139 + 2.0 + 0.0 \\
 &= 33.7139 \text{ mm} - \text{see below for the derivation}
 \end{aligned}$$

For Elliptical heads: Compute [h, k, r] per 4.3.7.1

$$\begin{aligned}
 h &= D / (2 \cdot \text{Aspect Ratio}) \\
 &= 1220.0 / (2 \cdot 2.0) \\
 &= 305.000 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 k &= D / (2 \cdot h) \\
 &= 1220.0 / (2 \cdot 305.0) \\
 &= 2.000
 \end{aligned}$$

$$r = D (0.5/k - 0.08) + c_a$$



$$= 1220.0(0.5/2.0 - 0.08) + 2.0$$

$$= 209.400 \text{ mm}$$

$$L = D(0.44 \cdot k + 0.02) + ca$$

$$= 1220.0(0.44 \cdot 2.0 + 0.02) + 2.0$$

$$= 1100.000 \text{ mm}$$

Step 3: Geometric Constants [βTh , ϕTh , Rth]:

Equations 4.3.8 to 4.3.11 as appropriate

t (corroded) and P for the calc.: $t = 31.7139 \text{ mm}$, $P = 85.7803 \text{ kgf/cm}^2$

Angle [βTh]:

$$= \arccos[(0.5 \cdot D - r)/(L - r)]$$

$$= \arccos[(0.5 \cdot 1224.0 - 209.4)/(1100.0 - 209.4)]$$

$$= 1.1017 \text{ radians - Equation 4.3.8}$$

$$\phi Th = \sqrt{(L \cdot t) / r}$$

$$= \sqrt{(1100.0 \cdot 31.714) / 209.4}$$

$$= 0.892 \text{ radians - Equation 4.3.9}$$

$\phi Th < \beta Th$, Thus:

$$Rth = (0.5 \cdot D - r) / \cos(\beta Th - \phi Th) + r$$

$$= (0.5 \cdot 1224.0 - 209.4) / \cos(1.102 - 0.892) + 209.4$$

$$= 621.0234 \text{ mm - Equation 4.3.10}$$

Step 4: Coefficients: Equations 4.3.12 to 4.3.15 [$C1$ and $C2$]:

$r/D > 0.08$, Thus:

$$C1 = 0.692(r/D) + 0.605$$

$$= 0.692(209.4/1224.0) + 0.605$$

$$= 0.7234 \text{ - Equation 4.3.13}$$

$$C2 = 1.46 - 2.6(r/D) = 1.46 - 2.6(209.4/1224.0)$$

$$= 1.0152, \text{ Equation 4.3.15}$$

Step 5: Pressure to produce knuckle elastic buckling [$Peth$]:

$$= C1 \cdot Et \cdot t^2 / (C2 \cdot Rth(0.5 \cdot Rth - r))$$

$$= 0.723 \cdot 1913772 \cdot 31.714^2 / (1.015 \cdot 621.023(0.5 \cdot 621.023 - 209.4))$$

$$= 21842.33 \text{ kgf/cm}^2, \text{ Equation 4.3.16}$$

Step 6: Internal Pressure for max stress in knuckle equals yield [Py]:

$C3 = Sy = 2532.867 \text{ kgf/cm}^2$, Not in the creep range - 4.3.6.1(f)

$$Py = C3 \cdot t / (C2 \cdot Rth(0.5 \cdot Rth / r - 1))$$

$$= 2532.867 \cdot 31.714 / (1.015 \cdot 621.023(0.5 \cdot 621.023 / 209.4 - 1))$$

$$= 263.863 \text{ kgf/cm}^2, \text{ Equation 4.3.17}$$

Step 7: Value of Internal Press. resulting in buckling of knuckle [G and Pck]:

$$G = Peth/Py = 21842.332/263.863 = 82.779$$

$G > 1$, Thus:

Buckling Pressure [Pck]:

$$= ((0.77508 \cdot G - 0.20354 \cdot G^2 + 0.019274 \cdot G^3) /$$

$$(1 + 0.19014 \cdot G - 0.089534 \cdot G^2 + 0.0093965 \cdot G^3)) Py$$

$$= ((0.77508 \cdot 82.779 - 0.20354 \cdot 82.779^2 + 0.019274 \cdot 82.779^3) /$$

$$(1 + 0.19014 \cdot 82.779 - 0.089534 \cdot 82.779^2 + 0.0093965 \cdot 82.779^3)) \cdot 263.863$$

$$= 535.3 \text{ kgf/cm}^2, \text{ Equation 4.3.19}$$

Step 8: Compute Allowable Pressure base on buckling of knuckle [Pak]:

$$= Pck/1.5 = 535.3/1.5 = 356.87 \text{ kgf/cm}^2 \text{ - Equation 4.3.21}$$

Step 9: Compute allowable pressure based on rupture of crown [Pac]:

$$= 2 \cdot S \cdot E / (L/t + 0.5)$$

$$= 2 \cdot 1509.097 \cdot 1.0 / (1100.0/31.714 + 0.5)$$

$$= 85.78 \text{ kgf/cm}^2, \text{ Equation 4.3.22}$$

Step 10: Compute final pressure [Pa]:

$$= \min(Pak, Pac) = \min(356.866, 85.78)$$

$$= 85.78 \text{ kgf/cm}^2, \text{ Equation 4.3.23: (same as original design pressure)}$$



Allowable pressure in Crown at given thickness [Pac]:

$$= 2 \cdot S \cdot E / (L / t + 0.5)$$

$$= 2 \cdot 1509.097 \cdot 1.0 / (1100.0 / 35.0 + 0.5)$$

$$= 94.530 \text{ kgf/cm}^2$$

Computed Maximum Allowable Pressure [MAWP]:

$$= \text{MaxPressure} - \text{Pliq} = 94.5296 - 0.122 = 94.4074 \text{ kgf/cm}^2$$

Computed Maximum Allowable Pressure New and Cold [MAPnc]:

$$= 100.0208 \text{ kgf/cm}^2$$

Computed Membrane Stress Intensity at Design Pressure [Sm]:

$$= 1369.4 \text{ kgf/cm}^2$$

Required Thickness of Straight Flange = 37.795 mm

Percent Fiber Elongation per Table 6.1

For Heads formed from Segments $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$	14.680 %
For one Piece doubly Curved Heads $100 \cdot \ln(D_b / (D_f - 2t))$	28.554 %

Where D_b is the blank diameter for one piece doubly curved surfaces [D_b]:

$$= \text{square root}(4 \cdot \text{Head Surface Area} / 3.14159)$$

$$= \text{square root}(4 \cdot 20692.9 / 3.14159)$$

$$= 1623.178 \text{ mm}$$

MDMT Calculations in the Knuckle Portion:

Note: This Element/Detail was specified as being Impact Tested.

MDMT Calculations in the Head Straight Flange:

Note: This Element/Detail was specified as being Impact Tested.

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 15.250 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 7.625 mm

Material treatment options: +IT

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Hydrostatic Test Pressure Results:

Hydrotest Pressure Calculation [Shell Side]:
 = Test Factor • ShellSide MAWP(St/S)
 = 1.25 • 86.135(1.0)
 = 107.669 kgf/cm²

Hydrotest Pressure Calculation [Channel Side]:
 = Test Factor • Channel Side MAWP(St/S)
 = 1.25 • 198.421(1.0)
 = 248.027 kgf/cm²

Horizontal Test performed per: per VIII-2, Section 8.2

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (kgf/cm²):

From To	Stress	Allowable	Ratio	Pressure
Ch Dish Head	1781.4	2844.9	0.626	248.15
Ch Shell	1841.4	2844.9	0.647	248.15
Main Shell	1783.7	2844.9	0.627	107.79
Shell Dish End	1626.3	2844.9	0.572	107.79

Stress ratios for Nozzle and Pad Materials (kgf/cm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
M1-Man Hole	Nozzle	1509.10	1509.10	1.000
A1-Tube Inlet	Nozzle	1509.10	1509.10	1.000
B1-Tube Outlet	Nozzle	1509.10	1509.10	1.000
A2-Shell inlet	Nozzle	1407.13	1343.95	1.047
B2-Shell outlet	Nozzle	1407.13	1343.95	1.047

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (kgf/cm²):

Description	Ambient	Operating	Ratio
Ch Dish Head	1509.10	1509.10	1.000
Ch Shell	1509.10	1509.10	1.000
Main Shell	1509.10	1509.10	1.000
Shell Dish End	1509.10	1509.10	1.000

Minimum 1.000

Stress ratios for Exchanger Materials (kgf/cm²):

Description	Ambient	Operating	Ratio
Tube Material	1203.20	1107.36	1.087
Tubesheet Material	1509.10	1509.10	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (kgf/cm²):

Description	Ambient	Operating	Ratio
M1-Man Hole	1052.22	1961.83	0.536
A1-Tube Inlet	1142.36	1961.83	0.582
B1-Tube Outlet	1142.36	1961.83	0.582
A2-Shell inlet	579.77	1829.27	0.317
B2-Shell outlet	579.77	1829.27	0.317



Elements Suitable for Test Pressure.

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MDMT Calculations for Stiffening Rings:

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Input Echo, Tubesheet Number 1, Description:

Shell Data:

Main Shell Description: Main Shell

Shell Test Pressure		107.67 kgf/cm ²
Shell Thickness	ts	38.0000 mm
Shell Internal Corrosion Allowance	cas	0.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	1220.000 mm
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-387 11

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Shell Material UNS Number		K11789
Shell Allowable Stress at Temperature	Ss	2110.82 kgf/cm ²
Shell Allowable Stress at Ambient		2110.82 kgf/cm ²

Channel Data:

Channel Description: Ch Shell

Channel Type:		Cylinder
Channel Test Pressure		248.03 kgf/cm ²
Channel Thickness	tc	88.0000 mm
Channel Internal Corrosion Allowance	cac	0.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	1220.000 mm
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-387 11

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Channel Material UNS Number		K11789
Channel Allowable Stress at Temperature	Sc	2110.82 kgf/cm ²
Channel Allowable Stress at Ambient		2110.82 kgf/cm ²

Tube Data:

Number of Tube Holes	Nt	1012
Tube Wall Thickness	et	2.7700 mm
Tube Outside Diameter	D	25.4000 mm
Total Straight Tube Length	Lt	3010.00 mm
Straight Tube Length (bet. inner tubsht faces) L		2710.00 mm
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-213 T11
Tube Material UNS Number		K11597
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		1203.20 kgf/cm ²
Tube Allowable Stress At Ambient		1203.20 kgf/cm ²
Tube Yield Stress At design Temperature	Syt	2090.42 kgf/cm ²
Tube Pitch (Center to Center Spacing)	P	31.7500 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.7700 mm
Groove Weld Leg	ag	2.7700 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		4.21.2
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	570.524 mm
Largest Center-to-Center Tube Distance	Ul	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	6.0000 mm
Area of Tube Layout	Ap	8673.4 cm ²
Perimeter of Tube Layout	Cp	3535.2 mm

Tubesheet Data:

Tubesheet TYPE: U-tube, Integral both sides, Conf. a

Tubesheet Design Metal Temperature T 21.11 °C
 Tubesheet Material SA-336 F11

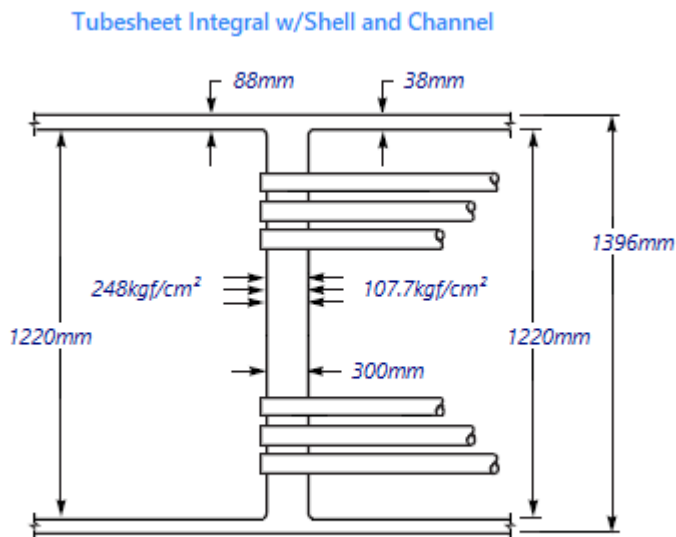
See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Tubesheet Material UNS Number K11572
 Tubesheet Allowable Stress at Temperature S 1509.10 kgf/cm²
 Tubesheet Allowable Stress at Ambient Tt 1509.10 kgf/cm²
 Thickness of Tubesheet h 300.0000 mm
 Tubesheet Corr. Allowance (Shell side) Cats 0.0000 mm
 Tubesheet Corr. Allowance (Channel side) Catc 0.0000 mm
 Tubesheet Outside Diameter A 1396.000 mm

Area of the Untubed Lanes AL 0.0 cm²

Junction Stress Reduction option Perform Plastic Calculation
 Perform Differential Pressure Design No
 Run Multiple Load Cases Yes

Tubesheet Integral with Both
 Tubesheet Extended as Flange No



Configuration a:

ASME Division 2 Tubesheet Results per Part 4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Material Group C

Elastic Mod. at Design Temperature	21.1 °C	0.20811E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20811E+07 kgf/cm ²

Channel - TM-1 Material Group C

Elastic Mod. at Design Temperature	21.1 °C	0.20811E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20811E+07 kgf/cm ²

Tubes - TM-1 Material Group C

Elastic Mod. at Tubsht. Design Temp.	21.1 °C	0.20811E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20811E+07 kgf/cm ²

TubeSheet - TM-1 Material Group C



Elastic Mod. at Design Temperature	21.1 °C	0.20811E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20811E+07 kgf/cm ²

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i$ Equation 4.3.1
 $= 0.5 \cdot 19.86 \cdot (\exp(248.027/(1203.2 \cdot 1.0)) - 1) + 0$
 $= 2.2732 + 0$
 $= 2.2732 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

Determine Parameter [Mx]:

$= L / \sqrt{(R_o \cdot t)}$
 $= 3280.524 / \sqrt{(12.7 \cdot 2.77)}$
 $= 553.097$

Compute the Factor [Ch]:

Since $M_x > 2 \cdot (D_o/t)^{0.94}$, $Ch = 0.55 / (D_o/t)$
 $= 0.55 / (25.4/2.77)$
 $= 0.0600$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$= 1.6 \cdot Ch \cdot E_y (t / D_o)$
 $= 1.6 \cdot 0.06 \cdot 2081087 (2.77/25.4)$
 $= 21780.361 \text{ kgf/cm}^2$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$= Fhe / F_y$
 $= 21780.4/2090.4$
 $= 10.419$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$= \sigma_{ys} / \sigma_{uts}$
 $= 2090.4/4218.4$
 $= 0.496$

Ratio [AE]:

$= Fhe / E$
 $= 21780.4/2081087$
 $= 0.010466$

Factor [m1]:

$= (\ln(r) + (\epsilon_{ps} - \epsilon_{sys})) / (\ln(1 + \epsilon_{ps}) / \ln(1 + \epsilon_{sys}))$
 $= (\ln(0.496) + (0.00002 - 0.002)) / (\ln(1 + 0.00002) / \ln(1 + 0.002))$
 $= 0.153$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$
 $= 2090.4 (1 + 0.002) / \ln(1 + 0.002)^{0.152921}$
 $= 5418.716 \text{ kgf/cm}^2$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$= m2f(m2f2 - R)$
 $= 0.6 (1.0 - 0.496)$
 $= 0.303$

Curve Fitting Constant [A2]:

$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$
 $= (4218.4 \cdot \exp(0.303)) / (0.303^{0.303})$
 $= 8197.734 \text{ kgf/cm}^2$

Material Parameter for Stress-Strain Curve Model [K]:

$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$
 $= 1.5 \cdot 0.4961.5 - 0.5 \cdot 0.4962.5 - 0.4963.5$
 $= 0.351$



Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	2109	-1.950	0.324E-05	0.289E-05	0.883E-05	-0.731E-05	0.123E+06	0.671E-02
2	1054	-4.772	0.697E-07	0.697E-07	0.179E-05	-0.179E-05	0.161E+07	0.981E-02
3	1581	-3.361	0.659E-06	0.655E-06	0.455E-05	-0.451E-05	0.546E+06	0.757E-02
4	1845	-2.655	0.155E-05	0.151E-05	0.649E-05	-0.624E-05	0.264E+06	0.347E-02
5	1977	-2.303	0.227E-05	0.215E-05	0.761E-05	-0.698E-05	0.181E+06	0.462E-03
6	1911	-2.479	0.188E-05	0.181E-05	0.704E-05	-0.664E-05	0.219E+06	0.173E-02
7	1944	-2.391	0.207E-05	0.198E-05	0.732E-05	-0.682E-05	0.199E+06	0.695E-03
8	1960	-2.347	0.216E-05	0.206E-05	0.747E-05	-0.690E-05	0.190E+06	0.133E-03
9	1969	-2.325	0.222E-05	0.211E-05	0.754E-05	-0.694E-05	0.185E+06	0.160E-03
10	1965	-2.336	0.219E-05	0.208E-05	0.750E-05	-0.692E-05	0.188E+06	0.125E-04
11	1962	-2.341	0.218E-05	0.207E-05	0.748E-05	-0.691E-05	0.189E+06	0.606E-04
12	1963	-2.338	0.218E-05	0.208E-05	0.749E-05	-0.692E-05	0.188E+06	0.241E-04
13	1964	-2.337	0.219E-05	0.208E-05	0.750E-05	-0.692E-05	0.188E+06	0.582E-05
14	1964	-2.336	0.219E-05	0.208E-05	0.750E-05	-0.692E-05	0.188E+06	0.333E-05
15	1964	-2.337	0.219E-05	0.208E-05	0.750E-05	-0.692E-05	0.188E+06	0.125E-05
16	1964	-2.336	0.219E-05	0.208E-05	0.750E-05	-0.692E-05	0.188E+06	0.104E-05

Factor of Safety [FS]:
 Since $Fic > 0.55 F_y$ and $Fic < F_y$
 $= 2.407 - 0.741 \cdot Fic / F_y$
 $= 2.407 - 0.741 \cdot 1964.7/2090.42$
 $= 1.711$

Allowable Circumferential Compressive Stress [Fha]:
 $= Fic/FS$
 $= 1964.704/1.711$
 $= 1148.571 \text{ kgf/cm}^2$

Allowable External Pressure at the given Thickness [Pa]:
 $= 2 \cdot Fha(t / Do)$
 $= 2 \cdot 1148.571 (2.77/25.4)$
 $= 250.515 \text{ kgf/cm}^2$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	2.2732 mm
Allowable Internal Pressure at Corroded thickness	296.04 kgf/cm ²
Required Internal Design Pressure	248.03 kgf/cm ²
Allowable External Pressure at Corroded thickness	250.52 kgf/cm ²
Required External Design Pressure	107.67 kgf/cm ²
Required Thickness due to Shell Side pressure	1.5488 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:
 $= 2 \cdot ro + dt$
 $= 2 \cdot 570.524 + 25.4$
 $= 1166.448 \text{ mm}$

Basic ligament efficiency for shear [μ]:
 $= (p - dt) / p$
 $= (31.75 - 25.4) / 31.75$
 $= 0.200$

ASME Section VIII-2 Part 4.18.7.4 Step 2:



Ratio [ps]:

$$= D_s / D_o \text{ (Configurations a, b, c)}$$

$$= 1220.0/1166.448$$

$$= 1.0459$$

Ratio [pc]:

$$= D_c / D_o \text{ (Configurations a, e, f)}$$

$$= 1220.0/1166.448$$

$$= 1.0459$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [p]:

$$= l_{tx} / h$$

$$= 0/300.0$$

$$= 0.000 \text{ (must be } 0 \leq p \leq 1 \text{)}$$

Effective tube hole Diameter [d*]:

$$= \max(dt - 2t(E_t/E) \cdot (S_t/S) \cdot (\rho); dt - 2t)$$

$$= \max(25.4 - 2 \cdot 2.77 (2081087/2081087) \cdot (1203/1509) \cdot (0); 25.4 - 2 \cdot 2.77)$$

$$= 25.4000 \text{ mm}$$

Effective tube pitch [p*]:

$$= p / \sqrt{ (1 - 4 \cdot \min(A; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2)) }$$

$$= 31.75 / \sqrt{ (1 - 4 \cdot \min(0; 4 \cdot 1166.448 \cdot 31.75) / (\pi \cdot 1166.448^2)) }$$

$$= 31.7500 \text{ mm}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (31.75 - 25.4) / 31.75$$

$$= 0.2000$$

$$h/p = 9.448819 ; \mu^* = 0.200000 ; E^* = 323295 \text{ kgf/cm}^2$$

$$E^*/E = 0.155349 ; \nu^* = 0.458572$$

Note: As h/p (9.449) is > 2, data values for h/p = 2 were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient [β_s]:

$$= ((12 (1 - \nu_s^2))^{1/4}) / \sqrt{ (D_s + t_s) t_s }$$

$$= ((12 (1 - 0.3^2))^{1/4}) / \sqrt{ (1220.0 + 38.0) 38.0 }$$

$$= 0.0083 \text{ 1/mm}$$

Shell coefficient [ks]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6 (1 - \nu_s^2))$$

$$= 0.008 \cdot 2081087 \cdot 38.0^3 / (6 (1 - 0.3^2))$$

$$= 173888960.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Shell coefficient [λ_s]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(1220.0 + 38.0) \cdot 0.17389E+09 / 300.0^3 \cdot (1 + 300.0 \cdot 0.008 + (300.0 \cdot 0.008)^2 / 2)$$

$$= 321079.2 \text{ kgf/cm}^2$$

Shell coefficient [δ_s]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot \nu_s / (2(D_s + t_s)))$$

$$= 1220.0^2 / (4 \cdot 2081087 \cdot 38.0) \cdot (1 - 1220.0 \cdot 0.3 / (2(1220.0 + 38.0)))$$

$$= 0.40208E-02 \text{ mm/kgf/cm}^2$$

Parameter [ω_s]:

$$= p_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$$

$$= 1.0459 \cdot 0.17389E+09 \cdot 0.0083 \cdot 0.004021 (1 + 300.0 \cdot 0.0083)$$

$$= 21245.2148 \text{ mm}^2$$

Channel coefficient [β_c]:

$$= ((12 (1 - \nu_c^2))^{1/4}) / \sqrt{ (D_c + t_c) t_c }$$

$$= ((12 (1 - 0.3^2))^{1/4}) / \sqrt{ (1220.0 + 88.0) 88.0 }$$

$$= 0.0054 \text{ 1/mm}$$

Channel coefficient [kc]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - \nu_s^2))$$

$$= 0.005 \cdot 2081087 \cdot 88.0^3 / (6(1 - 0.3^2))$$

$$= 1391733632.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λc]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (Cm\Delta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot Cq\Delta)$$

$$= 6(1220.0 + 88.0 / \cos(0)) \cdot 0.13917E+1 / 300.0^3 \cdot (1.0 + 300.0 \cdot 0.005 \cdot (1.0 + 1.0) / 2 + (300.0 \cdot 0.005)^2 / 2 \cdot 1.0)$$

$$= 1577403.1 \text{ kgf/cm}^2$$

Channel coefficient [δc]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot \nu_c / (2(D_c + t_c)))$$

$$= 1220.0^2 / (4 \cdot 2081087 \cdot 88.0) \cdot (1 - 1220.0 \cdot 0.3 / (2(1220.0 + 88.0)))$$

$$= 0.17476E-02 \text{ mm/kgf/cm}^2$$

Channel coefficient [zetac]:

$$zetac = 0$$

Parameter [ωc]:

$$= \rho_c \cdot k_c [\delta_c (CmTheta \cdot \beta_c + Cq\Delta \cdot h \cdot \beta_c^2) - \zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)]$$

$$= 1.046 \cdot 0.13917E+1 [0.001748(1.0 + 0.005 \cdot 1.0 \cdot 300.0) - 0(1.0/300.0 + 1.0 \cdot 0.005/2)]$$

$$= 35538.9805 \text{ mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o$$

$$= 1396.0 / 1166.448$$

$$= 1.1968$$

Coefficient [F]:

$$= (1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.46) / 323295 \cdot (321079.25 + 1577403 + 2081087 \cdot \ln(1.2))$$

$$= 3.8055$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= D_o^2 / 16 \cdot [(ps - 1) \cdot (ps^2 + 1) \cdot Ps - (pc - 1) \cdot (pc^2 + 1) \cdot Pt]$$

$$= 1166.448^2 / 16 \cdot [(1.046 - 1) \cdot (1.046^2 + 1) \cdot 107.669 - (1.046 - 1) \cdot (1.046^2 + 1) \cdot 248.027]$$

$$= -1147411.3750 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]:

$$= Mts + \omega_c \cdot Pt - \omega_s \cdot Ps$$

$$= -1147411.3 + 35538.98 \cdot 248.027 - 21245.215 \cdot 107.669$$

$$= 5379753.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= (M^*) - D_o^2 / 32 \cdot F(Ps - Pt) / (1 + F)$$

$$= (5379753) - 1166.448^2 / 32 \cdot 3.806(107.67 - 248.03) / (1 + 3.81)$$

$$= 5845457.5000 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$= Mp + D_o^2 / 64 \cdot (3 + \nu^*) \cdot (Ps - Pt)$$

$$= 5845457 + 1166.448^2 / 64 \cdot (3 + 0.459) \cdot (107.67 - 248.03)$$

$$= -4474632.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= \max(|Mp|; |Mo|)$$



$$= \max(|5845457|; |-4474632.0|)$$
$$= 5845457.5000 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu \cdot (h - h_g')^2)$$
$$= 6 \cdot 5845457 / (0.2 \cdot (300.0 - 6.0)^2)$$
$$= 2028.8274 \text{ kgf/cm}^2$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot \beta T \cdot S_y$$
$$= 2 \cdot 0.95 \cdot 3161.0$$
$$= 6005.81 \text{ kgf/cm}^2$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 \cdot M / (\mu \cdot (h - h_g')^2)$$
$$= 6 \cdot 5772747 / (0.2 \cdot (175.8846 - 6.0)^2)$$
$$= 6000.6143 \text{ kgf/cm}^2$$

Required tubesheet thickness for Bending Stress [H_{reqB}]:

$$= h + C_{ats} + C_{atc}$$
$$= 175.8846 + 0 + 0$$
$$= 175.8846 \text{ mm}$$

Required tubesheet thickness for given loadings (includes CA) [H_{req}]:

$$= \max(H_{reqB}; H_{reqS})$$
$$= \max(175.8846; 71.671)$$
$$= 175.8846 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [$|P_s - P_t|$]:

$$= |P_s - P_t|$$
$$= |107.67 - 248.03|$$
$$= 140.358 \text{ kgf/cm}^2$$

Shear Stress check [τ_{limit}]:

$$= 4 \cdot \mu \cdot h / D_o \cdot 0.8 \cdot \beta T \cdot S_y$$
$$= 4 \cdot 0.2 \cdot 300.0 / 1166.45 \cdot 0.8 \cdot 0.95 \cdot 3160.954$$
$$= 494.285 \text{ kgf/cm}^2$$

$|P_s - P_t|$ is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [H_{reqS}]:

$$= 1 / (4 \cdot \mu \cdot (0.8 \cdot \beta T \cdot S_y)) \cdot (4 \cdot A_p / C_p) \cdot |P_s - P_t| + C_{ats} + C_{atc}$$
$$= 1 / (4 \cdot 0.2 \cdot 2402.325) \cdot (4 \cdot 8673.4 / 3535.24) \cdot |107.67 - 248.03| + 0$$
$$= 71.6710 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Shell thickness of: 38.000mm

Min. Shell len. adjacent to the tubesheet is: 387.565 mm

Note: For a given Channel thickness of: 88.000 mm

Min. Channel len. adjacent to the tubesheet is: 589.785mm

Tube Metal Area [A_t]:

$$= \pi (d_o - t) t$$
$$= \pi (25.4 - 2.77) 2.77$$
$$= 1.969 \text{ cm}^2$$

Tube Weld Size Results per VIII-2 4.21.2:

Tube Strength [F_t]:

$$= \pi \cdot t (d_o - t) S$$
$$= \pi \cdot 2.77 (25.4 - 2.77) 1985.902$$
$$= 3910.852 \text{ kgf}$$

Groove Weld Strength [Fg]:
 $= 0.85 \cdot \pi \cdot ag (do + 0.67 \cdot ag) Sw$ (but not > Ft)
 $= 0.85 \cdot \pi \cdot 2.77 (25.4 + 0.67 \cdot 2.77) 1985.9$
 $= 3910.8521 \text{ kgf}$

Weld Strength Factor [fw]:
 $= S / Sw$ where Sw = min(S and St)
 $= 1985.902/1985.902$
 $= 1.000$

Design Strength Ratio [fd]:
 $= 1$ for full strength welds

Fillet Weld Strength [Ff]:
 $= 0.55 \cdot \pi \cdot af (do + 0.67 \cdot af) Sw$ (but not > Ft)
 $= 0.55 \cdot \pi \cdot 2.77 (25.4 + 0.67 \cdot 2.77) 1985.9$
 $= 2590.6577 \text{ kgf}$

Minimum Length of Weld [ar]:
 $= 2 \left(\sqrt{((0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t) fw \cdot fd)} - 0.75 \cdot do \right)$
 $= 2 \left(\sqrt{((0.75 \cdot 25.4)^2 + 1.07 \cdot 2.77 \cdot (25.4 - 2.77) 1.0 \cdot 1.0)} - 0.75 \cdot 25.4 \right)$
 $= 3.372 \text{ mm}$

Minimum Required Fillet Weld Leg [afr]: 1.6859 mm
 Minimum Required Groove Weld Leg [agr]: 1.6859 mm

Tube-Tubesheet Jt allowable, 3910.85 is >= tube strength 3910.85 kgf

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	2028.8 <=	6005.8 kgf/cm ²	Ok
Tubesheet shear stress	573.9 <=	2402.3 kgf/cm ²	Ok

Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	175.885	300.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.686	2.770	Ok
Tube-Tubesheet Groove Weld Leg :	1.686	2.770	Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 3 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend Allwd	Stresses Shear Allwd	Case Type	Pass/ Fail
D1uc	227.731 ...	3389	6006	1014	Ps+Pt D1	Ok
D2uc	152.664 ...	1388	6006	440	Ps+Pt D2	Ok
D3uc	175.885 ...	2029	6006	574	Ps+Pt D3	Ok
Max:	227.7308 ... mm	0.564		0.422	(Str. Ratio)	

Load Case Definitions:

[Ps & Pt]:
 Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[c]:
 With or Without Corrosion Allowance



[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Shell and Channel Stress Summary:

-- Integral Cyl. Req'd. Thk.		Shell	Stress	Channel	Stress	Pass	
Case#	Shell	Channel	Stress	Allwd	Stress	Allwd	Fail
D1uc	...	...	...	4505	...	4505	Ok
D2uc	...	...	...	4505	...	4505	Ok
D3uc	...	...	...	4505	...	4505	Ok
Max	...	...	...	...	...	...	...

Note:

Because there was no net pressure differential across the tubesheet due to the specified pressures psd,min and ptd,min; load case D4 was not processed as the resulting tubesheet bending stress would be zero.

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	227.731	300.000	Ok
Tubesheet thickness flanged extension :	0.000	64.000	Ok
Tube thickness :	2.273	2.770	Ok
Tube-tubesheet fillet weld leg :	1.686	2.770	Ok
Tube-tubesheet groove weld leg :	1.686	2.770	Ok

Given Shell Thickness: 38.0000 mm

Given Channel Thickness: 88.0000 mm

Min Shell length of thk, (38.000) adj. to tubesheet: 387.565 mm

Min Channel length of thk, (88.000) adj. to tubesheet: 589.785 mm

Note: This is a full strength Tube to Tubesheet Joint.

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 2, 2025

External Pressure Calculations:

From	To	Section Length cm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B kgf/cm ²
10	20	No Calc	1308	42	No Calc	No Calc
20	Tsht	197.83	1396	86	No Calc	No Calc
Tsht	30	No Calc	...	86	No Calc	No Calc
30	40	345.25	1296	36	No Calc	No Calc
40	50	No Calc	1294	35	No Calc	No Calc

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure kgf/cm ²	External M.A.W.P. kgf/cm ²
10	20	44	No Calc	...	145.791
20	Tsht	88	No Calc	...	174.539
Tsht	30	88	No Calc	...	No Calc
30	40	38	No Calc	...	48.7384
40	50	37	No Calc	...	60.3856

Minimum

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External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners cm	Allowable Length Bet. Stiffeners cm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	Tsht	197.83	No Calc	No Calc	No Calc
Tsht	30	No Calc	No Calc	No Calc	No Calc
30	40	345.25	No Calc	No Calc	No Calc
40	50	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg	Element ID Volume cm ³	Corroded Metal Wgt. kg	Corroded ID Volume cm ³	Extra due Misc % kg
10	20	1024.85	475389	988.493	480080	163.976
20	30	4974.63	2074956	4869.01	2088585	795.941
30	40	3841.05	2008456	3644.68	2033794	614.568
40	50	738.38	297079	705.563	299815	118.141
Total		10578	4855881.50	10207	4902274.50	1692



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg	X Offset, Dtl. Cent. cm	Y Offset, Dtl. Cent. cm	Z Offset, Dtl. Cent. cm	Description
10	Liqd	19.0089	-20.3333	...	...	Liquid: 10
10	Insl	37.4698	-30.5	...	...	Hot Insulation
10	Nozl	220.438	-61	...	...	M1-Man Hole
20	Liqd	82.9691	88.75	...	...	Liquid: 20
20	Insl	93.5332	88.75	...	...	Hot Insulation
20	Nozl	997.483	70	85.5288	...	A1-Tube Inlet
20	Nozl	1198.44	70	-85.5288	...	B1-Tube Outlet
20	Wght	499.991	85	...	...	Partition plate
30	Sadl	299.305	80	87	...	Left Saddle
30	Sadl	299.305	250	87	...	New Saddle
30	Liqd	2007.55	165	...	...	Liquid: 30
30	Insl	248.299	165	...	...	Hot Insulation
30	Nozl	44.3641	30	67.9863	...	A2-Shell inlet
30	Nozl	44.3641	30	-67.9863	...	B2-Shell outlet
40	Liqd	296.945	15.2467	...	...	Liquid: 40
40	Insl	48.9897	17.79	...	...	Hot Insulation
20	FTsh	2745.16	192.5	...	...	
20	Tube	6733.52	328	...	...	Tubes
30	Bafl	56.3498	51	...	...	Baffles
30	Bafl	56.3498	76	...	...	Baffles
30	Bafl	56.3498	101	...	...	Baffles
30	Bafl	56.3498	126	...	...	Baffles
30	Bafl	56.3498	151	...	...	Baffles
30	Bafl	56.3498	176	...	...	Baffles
30	Bafl	56.3498	201	...	...	Baffles
30	Bafl	56.3498	226	...	...	Baffles
30	Bafl	56.3498	251	...	...	Baffles
30	Bafl	57.3477	276	...	...	Baffles

Total Weight of Each Detail Type:

Saddles	598.6
Liquid	2406.5
Insulation	428.3
Nozzles	2505.1
Weights	500.0
Exchanger Components	10043.2
Liquid in Tubes	1224.0
Sum of the Detail Weights	17705.6 kg

Weight Summation Results: (kg)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
--	------------	-----------	----------	---------	-------	-----------

Main Elements	12271.5	12271.5	12271.5	12271.5	12271.5	12271.5
Saddles	598.6	598.6	598.6	598.6	598.6	598.6
Nozzles	2505.1	2505.1	2505.1	2505.1	2505.1	2505.1
Wld Weights	500.0	500.0	500.0	500.0	500.0	500.0
Exchanger	10043.2	10043.2	10043.2	10043.2	10043.2	10043.2
Insulation	...	...	428.3	428.3	428.3	428.3
Ope. Liquid	...	...	...	...	...	2406.5
Tube Ope Lqd	...	...	...	...	...	1224.0
Test Liquid	...	4853.7	...	...	...	...
Tube Tst Lqd	...	1224.0	...	...	...	...
Totals	25918.4	31996.1	26346.7	26346.7	26346.7	29977.2

Field Installation Options:

Miscellaneous Weight Percent: 16.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	25918.4 kg
Shop Test Wt.	- Fabricated weight + water (full)	31996.1 kg
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	26346.7 kg
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	26346.7 kg
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	26346.7 kg
Operating Wt.	- Empty weight + operating liq. uncorroded	29977.2 kg
Oper. Wt. + CA	- Corr wt. + operating liquid	29546.6 kg

Exchanger Tube Data

Volume of Exchanger tubes :	1224565.0 cm ³
Weight of Ope Liq in tubes :	1224.0 kg
Weight of Water in tubes :	1224.0 kg

The Corroded Weight and thickness are used in the Horizontal Vessel Analysis (Ope Case) and Earthquake Load Calculations.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	27535.7	23379.7
20	30	77845.5	68031.2
30	40	134360	126481
40	50	20692.9	18081.1

Total	260433.8	235972.5 cm ²
Total	280.3	254.0 ft ²

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Nozzle Flange MAWP Results: (kgf/cm² & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
M1-Man Hole	200.95	260.14	310	1500	GR 1.1	...	...	...	...

Shellside Flange Rating

Channelside Flange Rating

Lowest Flange Pressure Rating was (Ope)[TubeSide]: 200.952 kgf/cm²
 Lowest Flange Pressure Rating was (Amb)[TubeSide]: 260.136 kgf/cm²

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Warning:

There are nozzles in this model, but no flange MAWPs were de-rated. Be sure this is what you intended. There is a check box in the nozzle dialog that instructs PV Elite to perform the de-rating for each nozzle flange. See ASME VIII-2, 4.16.12 for more information.

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Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-22
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	[V] 180 km/hr
Surface Roughness Category	D: Flat, unobstructed
Type of Surface	Moderately Smooth
Base Elevation	0 cm
Percent Wind for Hydrotest	33.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 cm
Distance Upwind of Crest Lh	0 cm
Distance from Crest to the Vessel x	0 cm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 cm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot g_Q \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot g_V \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.194 \cdot 0.984) / (1 + 1.7 \cdot 3.4 \cdot 0.194)))$
 $= \min(0.85, 0.917)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	33.000 Hz
Natural Frequency of Vessel (Empty)	33.000 Hz
Natural Frequency of Vessel (Test)	33.000 Hz

User Entered Importance Factor is	1.000
Force Coefficient	[Cf] 0.554
Structure Height to Diameter ratio	4.217

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of [α] and [Zg]:
 Exposure Category: D from Table 26.11-1
 $\alpha = 11.5: Z_g = 58978.801 \text{ cm}$

Effective Height [z]:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 118.0 + 0.0 = 118.0 \text{ cm}$
 $= 3.871 \text{ ft. Imperial Units}$

Velocity Pressure coefficient evaluated at height z [Kz]:
 Because $z (3.871 \text{ ft.}) < 15 \text{ ft.}$
 $= 2.41(15 / Z_g)^{2/\alpha}$
 $= 2.41(15 / 1935.0)^{2/11.5}$
 $= 1.035$

Type of Hill: No Hill

Wind Directionality Factor [Kd]:
 $= 1, \text{ per Table 26.6-1}$

As there is No Hill Present: [Kzt]:
 $K_1 = 0, K_2 = 0, K_3 = 0$

Topographical Factor [Kzt]:

$$= (1 + K1 \cdot K2 \cdot K3)^2$$

$$= (1 + 0.0 \cdot 0.0 \cdot 0.0)^2$$

$$= 1.0$$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$$

$$= \max(16, 0.00256 \cdot 1.035 \cdot 1.0 \cdot 1.0 \cdot 1.0 \cdot 111.85^2)$$

$$= 33.1 \text{ psf } [161.847 \text{ kgf/m}^2]$$

Force on the first element [F]:

$$= qz \cdot G \cdot Cf \cdot \text{WindArea}$$

$$= 33.149 \cdot 0.85 \cdot 0.554 \cdot 8.62$$

$$= 134.5 \text{ lbs. } [61.0 \text{ kgf}]$$

Element	Hgt (z) cm	K1	K2	K3	Kz	Kzt	qz kgf/m ²
Ch Dish Head	118.0	0.000	0.000	0.000	1.035	1.000	161.847
Ch Shell	118.0	0.000	0.000	0.000	1.035	1.000	161.847
Main Shell	118.0	0.000	0.000	0.000	1.035	1.000	161.847
Shell Dish End	118.0	0.000	0.000	0.000	1.035	1.000	161.847

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation cm	Pressure kgf/m ²	Force kgf
Partition pla	0.00	118.00	161.85	0.00

Wind Load Calculation:

From	To	Wind Height cm	Wind Diameter cm	Wind Area cm ²	Wind Pressure kgf/m ²	Element Wind Load kgf
10	20	118	171.36	8008.38	161.847	67.0922
20	30	118	181.92	32290.8	161.847	270.524
30	40	118	177.12	58449.6	161.847	489.676
40	50	118	176.88	5899.71	161.847	49.4263

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code	ASCE/SEI 7-22
Importance Factor	1.500
Force Modification Factor R	3.000
Site Class	D
Component Elevation Ratio	z/h
Amplification Factor	Ap
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Component resonance duct. fact.	[CAR]
Component strength factor	[Rpo]
Overstrength factor	[Omega0]
Response Acceleration Factor	[SMS]
Response Acceleration Factor	[SM1]
Building Importance Factor	[Ie]
Site Latitude	0.000 deg
Site Longitude	0.000 deg

Seismic Analysis Results:

$$SDS = 2/3 * Sms = 2/3 * 0.326 = 0.217$$

$$SD1 = 2/3 * Sm1 = 2/3 * 0.139 = 0.093$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 = $C_t \cdot h_{nx}$ where $C_t = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 = $0.020(5.87270.75)$
 = 0.075 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 = (1/Natural Frequency) = (1/33.0)
 = 0.030

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 = Minimum Value of ($1.7 \cdot 0.075$, 0.03) per 12.8.2
 = 0.030

As the time period is < 0.06 second, use section 15.4.2.

Base Shear per equation 15.4-5, [V]:
 = $0.3 \cdot Sds \cdot W \cdot I$
 = $0.3 \cdot 0.217 \cdot 28948 \cdot 1.5$
 = 2831.116 kgf

Vertical load per 12.4-4, [Ev]:
 = $0.2 \cdot Sds \cdot W$
 = $0.2 \cdot 0.217 \cdot 28948$
 = 1258.3 kgf

Earthquake Load Calculation:

From	To	Earthquake Height cm	Earthquake Weight kgf	Element Ope Load kgf
10	20	61	4824.67	471.853
20	30	61	4824.67	471.853
30	Sad1	61	4824.67	471.853
Sad1	40	61	4824.67	471.853
30	40	61	4824.67	471.853
40	50	61	4824.67	471.853



The Earthquake Weight does not include the weight of the saddles.

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Shop/Field Installation Options :

Insulation is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	372.500 cm
Center of Gravity of the Liquid	381.855 cm
Center of Gravity of the Insulation	296.184 cm
Center of Gravity of the Nozzles	64.405 cm
Center of Gravity of the Added Weights (Operating)	85.000 cm
Center of Gravity of the Added Weights (Empty)	85.000 cm
Center of Gravity of the Tubesheet(s)	192.500 cm
Center of Gravity of the Tubes	328.000 cm
Center of Gravity of the Baffles	371.199 cm
Center of Gravity of Bare Shell New and Cold	212.670 cm
Center of Gravity of Bare Shell Corroded	210.647 cm
Vessel CG in the Operating Condition	247.572 cm
Vessel CG in the Fabricated (Shop/Empty) Condition	231.909 cm
Vessel CG in the Test Condition	234.183 cm

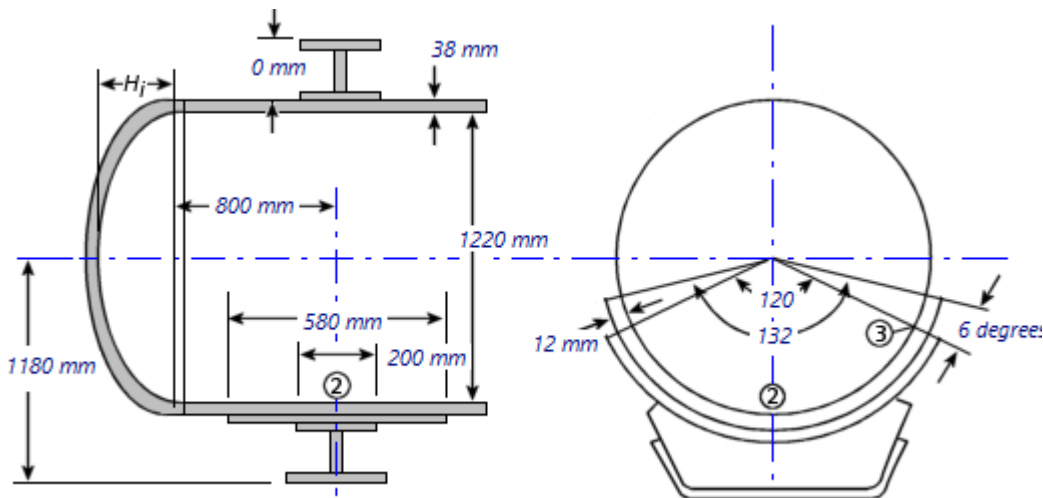
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Input and Calculated Values:

Vessel Mean Radius	Rm	630.00	mm
Shell Thickness used in this Case	t	38.000	mm
Stiffened Vessel Length per 4.15.6	L	335.08	cm
Distance from Saddle to Vessel tangent	a1 or a	800.00	mm
Saddle Width	b1 or b	200.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	580.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	12.0	mm
Wear Plate Allowable Stress	fw or Sr	1509.10	kgf/cm ²
Inside or Mean Depth of Head	Hi or hm	63.30	cm
Shell Allowable Stress used in Calculation		1509.10	kgf/cm ²
Head Allowable Stress used in Calculation		1509.10	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1180.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case	41199.63	kgf
Pressure used in Analysis	P	85.719 kgf/cm ²

Horizontal Vessel Analysis Results:		Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan		762.09	1509.10
Long. Stress at Bottom of Midspan		737.99	1509.10
Long. Stress at Top of Saddles		1020.35	1509.10
Long. Stress at Bottom of Saddles		600.22	1509.10
Tangential Shear in Shell		88.76	1207.28
Circ. Stress at Horn of Saddle		262.56	1886.37
Circ. Compressive Stress in Shell		15.00	1509.10

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:



Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0 (876.7/2 + 0) \cdot 1180.0/1143.1537$
 $= 1357.5 \text{ kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(403.72, 0) \cdot 1180.0/1700.0$
 $= 280.2 \text{ kgf}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(F_{l,eq}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(2831.12, 0) \cdot 1180.0/1700.0$
 $= 1965.1 \text{ kgf}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0 (2831/2 + 0) \cdot 1180.0/1143.1537$
 $= 4383.6 \text{ kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 36816 + \max(280, 1357, 1965, 4384)$
 $= 41199.6 \text{ kgf}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1587.169 (0.0 + \pi/4 (1.416 \cdot 1.2)^2)$
 $= 3959.112 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	41498.94	kgf
Transverse Shear Load	1415.56	kgf
Longitudinal Shear Load	2831.12	kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2 (\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$
 $= \min(2 (1220.0/2 + 38.0 + 12.0) \sin(120.0/2), 2 \cdot 630.0)$
 $= 1143.154 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$
 $= -41200 \cdot 80.0 [1 - (1 - 80.0/335.08 + (63.0^2 - 63.3^2)/(2 \cdot 80.0 \cdot 335.08))/(1 + (4 \cdot 63.3)/(3 \cdot 335.08))]$
 $= -12936.0 \text{ kgf-m.}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$
 $= 41200 \cdot 335/4 ((1 + 2(63^2 - 63^2)/(335^2))/(1 + (4 \cdot 63)/(3 \cdot 335)) - 4 \cdot 80/335)$
 $= -5409.4 \text{ kgf-m.}$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 $= P \cdot R_m/(2t) - M_2/(\pi \cdot R_m^2 t)$
 $= 85.719 \cdot 630.0/(2 \cdot 36.0) - 5409.4/(\pi \cdot 630.0^2 \cdot 36.0)$
 $= 762.09 \text{ kgf/cm}^2$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 85.719 \cdot 630.0 / (2 \cdot 36.0) + -5409.4 / (\pi \cdot 630.0^2 \cdot 36.0)$$

$$= 737.99 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 85.719 \cdot 630.0 / (2 \cdot 36.0) - 12936.0 / (0.1066 \cdot \pi \cdot 630.0^2 \cdot 36.0)$$

$$= 1020.35 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 85.719 \cdot 630.0 / (2 \cdot 36.0) + -12936.0 / (0.1923 \cdot \pi \cdot 630.0^2 \cdot 36.0)$$

$$= 600.22 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 41200(335.08 - 2 \cdot 80.0) / (335.08 + (4 \cdot 63.3 / 3))$$

$$= 17195.7 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 17195.65 / (630.0 \cdot 36.0)$$

$$= 88.76 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(630.0 \cdot 36.0)}$$

$$= 117.467 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(200.0 + 1.56 \cdot \sqrt{(630.0 \cdot 36.0)}, 2 \cdot 80.0)$$

$$= 434.93 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(S_r / S, 1)$$

$$= \min(1509.097 / 1509.097, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.7603 \cdot 41200 \cdot 0.1 / (434.934(36.0 + 1.0 \cdot 12.0))$$

$$= -15.00 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [σ_7, r_{*}]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 12 \cdot K_7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2)$$

$$= -41200 / (4(36.0 + 1.0 \cdot 12.0)434.934) -$$

$$12 \cdot 0.053 \cdot 41200 \cdot 630.0 / (335.08(36.0 + 1.0 \cdot 12.0)^2)$$

$$= -262.56 \text{ kgf/cm}^2$$

Distance between Saddle Supports [Ls]:

$$= 1700.0 \text{ mm}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot L_s(\text{Design Temperature} - \text{Ambient Temperature})$$

$$= 0.000013 \cdot 1700.0(280.0 - 21.1)$$

$$= 5.783 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1278.0000	mm
Baseplate Thickness	Bpthk	20.0000	mm
Baseplate Width	Bpwid	225.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	12.0000	mm
Web Thickness	Webtk	12.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	2492.1	kgf/cm ²
Height of Web at Center	Hw,c	570.0	mm



Friction Coefficient mu 0.000

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(612.0 \cdot 36.0)} + 580.0)^3) / 12$$

$$= 315.532 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 315.532 + 292.159(109.633 - 18.0)^2$$

$$= 24846.781 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	811.6	36.0	18.0	292.2	525886.8	0.248E+05
Wearplate	580.0	12.0	42.0	69.6	292320.0	0.319E+04
Web	12.0	500.0	298.0	60.0	1787999.0	0.338E+05
BasePlate	225.0	20.0	558.0	45.0	2510999.2	0.905E+05
Totals	...	...	...	466.8	5117205.5	0.152E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 5117206 / 466.759$$

$$= 109.633 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 41199.629$$

$$= 8385.0186 \text{ kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 48.0242 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 1495.2588 \text{ kgf/cm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1180.0 - 612.0 \cdot \sin(120.0/2) / 1.0472$$

$$= 673.880 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 5650.5029 \text{ kgf-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 40.6728 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 1661.3987 \text{ kgf/cm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(41200 + 299) \cdot 225.0) / (4 \cdot 1278.0 \cdot 1661.399)}$$

$$= 18.161 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (610.0 + 38.0 + 12.0)$$

$$= 1143.154 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (N_{\text{ribs}} - 1)$$

$$= 1143.1536 / (4 - 1)$$

$$= 381.051 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot B_{pwid} / 2$$

$$= 381.0512 \cdot 225.0 / 2$$

$$= 428.683 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 41498.938 / (1278.0 \cdot 225.0)$$

$$= 14.432 \text{ kgf/cm}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 428.7 \cdot 14.43$$

$$= 6186.705 \text{ kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 22.56 + 22.863$$

$$= 45.423 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 6186.7 / 45.4231$$

$$= 136.202 \text{ kgf/cm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	12.0	200.0	...	24.0	...	800.

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 200.0 - 12.0$$

$$= 188.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 45.423$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 200.0 / 2$$

$$= 100.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(800.0 / 45.423)}$$

$$= 41.967 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1180.0 - \cos(120.0/2) (610.0 + 38.0 + 12.0) - 20.0$$

$$= 830.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 2038900 / 2492.1)}$$

$$= 127.081$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 830.0 / 41.967$$

$$= 19.777$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 2831.1 / (2 \cdot 1278.0) \cdot 381.051 \cdot 830.0 / 2$$

$$= 175.157 \text{ kgf-m.}$$

Compressive Allowable, $KL/r < C_c$ (19.7775 < 127.0809) per AISC E2-1 [Sca]:

$$= (1 - (Klr)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (Klr) / (8 \cdot Cc) - (Klr^3) / (8 \cdot Cc^3))$$

$$= (1 - (19.78)^2 / (2 \cdot 127.08^2)) 2492 /$$

$$(5/3 + 3 \cdot (19.78) / (8 \cdot 127.08) - (19.78^3) / (8 \cdot 127.08^3))$$

$$= 1428 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 136.2 / 1427.57 + (175.16 \cdot 100.0 / 8000001) / 1661.4$$

$$= 0.227$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	12.0	188.0	0.0	22.6	0.0	800.
Web	381.1	12.0	0.0	45.7	0.0	5.49
Totals	...	...	...	68.3	...	805.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 68.286$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 200.0 / 2$$

$$= 100.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{ (Ro + W_{pdthk})^2 - (Pitch/2)^2 } - B_{pthk}$$

$$= 1180.0 - \sqrt{ (660.0 + 12.0)^2 - (381.051/2)^2 } - 20.0$$

$$= 528.098 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ \text{Total Inertia} / \text{Total Area} }$$

$$= \sqrt{ 805.3 / 68.286 }$$

$$= 34.341 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 528.098 / 34.341$$

$$= 15.378$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 2831.116 / (2 \cdot 1278.0)$$

$$= 1.108 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.108 \cdot 381.051 \cdot 528.098$$

$$= 222.892 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 222.892 / 80531.43$$

$$= 276.777 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < C_c$ (15.3779 < 127.0809) per AISC E2-1 [Sca]:

$$= (1 - (Klr)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (Klr) / (8 \cdot Cc) - (Klr^3) / (8 \cdot Cc^3))$$

$$= (1 - (15.38)^2 / (2 \cdot 127.08^2)) 2492 /$$

$$(5/3 + 3 \cdot (15.38) / (8 \cdot 127.08) - (15.38^3) / (8 \cdot 127.08^3))$$

= 1445 kgf/cm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= $Sc/Sca + (M_{base,c} \cdot C1/I) / Sba$
 = $181.33/1445.16 + (222.89 \cdot 100.0/805.314) / 1661.4$
 = 0.292

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1529.00 kgf/cm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	4.0400 cm ²
BasePlate Allowable Stress	S	1509.10 kgf/cm ²
Area Available in a Single Bolt	BlArea	3.0428 cm ²
Saddle Load QO (Weight)	QO	37115.4 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	1965.1 kgf
Maximum Transverse Force	Ft	1415.6 kgf
Maximum Longitudinal Force	Fl	2831.1 kgf
Saddle Bolted to Steel Foundation	No	No

Friction force due to thermal growth, [Ff]:
 = 605.5 kgf

Shear Stress in a Single Bolt, Longitudinal Direction [τ_{b,l}]:
 = $\max(F_f, F_l) / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 = $2831.1 / (3.043 \cdot 2)$
 = 465.2 kgf/cm². Must be less than 1019.0 kgf/cm².

Shear Stress in a Single Bolt, Transverse Direction [τ_{b,t}]:
 = $F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 = $1415.6 / (3.043 \cdot 2)$
 = 232.6 kgf/cm². Must be less than 1019.0 kgf/cm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 = 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 = $F_l / (\text{BoltShearAllowable} \cdot \text{Nbolts})$
 = $2831.12 / (1019.0 \cdot 2.0)$
 = 1.3892 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 = $B \cdot F_t + \text{Sum of X Moments}$
 = $1180.0 \cdot 1415.56 + 0.0$
 = 1670.36 kgf-m.

Eccentricity (e):
 = R_{mom} / QO
 = $1670.36 / 37115.39$
 = 45.00 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 = 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 = $\max[Bltarearl, Bltarears, Bltareart]$
 = $\max[0.0, 1.3892, 0.0]$
 = 1.3892 cm²

Note: Saddle and wear plate material are both specified as impact tested.

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	630.00	mm
Shell Thickness used in this Case	t	38.000	mm
Stiffened Vessel Length per 4.15.6	L	335.08	cm
Distance from Saddle to Vessel tangent	a1 or a	800.00	mm
Saddle Width	b1 or b	200.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	580.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	12.0	mm
Wear Plate Allowable Stress	fw or Sr	1509.10	kgf/cm ²
Inside or Mean Depth of Head	Hi or hm	32.45	cm
Shell Allowable Stress used in Calculation		1509.10	kgf/cm ²
Head Allowable Stress used in Calculation		1509.10	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1180.00	mm
Coefficient of Friction	mu	0.08	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		11653.01	kgf
Pressure used in Analysis	P	85.719	kgf/cm ²

Horizontal Vessel Analysis Results:	Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan	750.55	1509.10
Long. Stress at Bottom of Midspan	749.53	1509.10
Long. Stress at Top of Saddles	804.13	1509.10
Long. Stress at Bottom of Saddles	720.07	1509.10
Tangential Shear in Shell	27.83	1207.28
Circ. Stress at Horn of Saddle	74.26	1886.37
Circ. Compressive Stress in Shell	4.24	1509.10

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0 (876.7/2 + 0) \cdot 1180.0 / 1143.1537$
 $= 1357.5 \text{ kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(403.72, 0) \cdot 1180.0 / 1700.0$
 $= 280.2 \text{ kgf}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(F_{l, \text{eq}}, \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(2831.12, 0) \cdot 1180.0 / 1700.0$
 $= 1965.1 \text{ kgf}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0 (2831/2 + 0) \cdot 1180.0 / 1143.1537$
 $= 4383.6 \text{ kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$



$$= 7269 + \max(280, 1357, 1965, 4384)$$

$$= 11653.0 \text{ kgf}$$

Longitudinal Wind Force [FI]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1587.169(0.0 + \pi/4(1.416 \cdot 1.2)^2)$$

$$= 3959.112 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	11952.32	kgf
Transverse Shear Load	1415.56	kgf
Longitudinal Shear Load	2831.12	kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(1220.0/2 + 38.0 + 12.0)\sin(120.0/2), 2 \cdot 630.0)$$

$$= 1143.154 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$$

$$= -11653 \cdot 80.0 [1 - (1 - 80.0/335.08 + (63.0^2 - 32.45^2)/(2 \cdot 80.0 \cdot 335.08))/(1 + (4 \cdot 32.45)/(3 \cdot 335.08))]$$

$$= -2588.2 \text{ kgf-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 11653 \cdot 335/4((1 + 2(63^2 - 32^2)/(335^2))/(1 + (4 \cdot 32)/(3 \cdot 335)) - 4 \cdot 80/335)$$

$$= -227.9 \text{ kgf-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t)$$

$$= 85.719 \cdot 630.0/(2 \cdot 36.0) - -227.9/(\pi \cdot 630.0^2 \cdot 36.0)$$

$$= 750.55 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$$

$$= 85.719 \cdot 630.0/(2 \cdot 36.0) + -227.9/(\pi \cdot 630.0^2 \cdot 36.0)$$

$$= 749.53 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m/(2t) - M1/(K1 \cdot \pi \cdot R_m^2 t)$$

$$= 85.719 \cdot 630.0/(2 \cdot 36.0) - -2588.2/(0.1066 \cdot \pi \cdot 630.0^2 \cdot 36.0)$$

$$= 804.13 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m/(2t) + M1/(K1^* \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 85.719 \cdot 630.0/(2 \cdot 36.0) + -2588.2/(0.1923 \cdot \pi \cdot 630.0^2 \cdot 36.0)$$

$$= 720.07 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot h_m/3))$$

$$= 11653(335.08 - 2 \cdot 80.0)/(335.08 + (4 \cdot 32.45/3))$$

$$= 5392.4 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 5392.43 / (630.0 \cdot 36.0)$$

$$= 27.83 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x1,x2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(630.0 \cdot 36.0)}$$

$$= 117.467 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(200.0 + 1.56 \cdot \sqrt{(630.0 \cdot 36.0)}, 2 \cdot 80.0)$$

$$= 434.93 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= \min(S_r/S, 1)$$

$$= \min(1509.097/1509.097, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.7603 \cdot 11653 \cdot 0.1 / (434.934(36.0 + 1.0 \cdot 12.0))$$

$$= -4.24 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, L<8Rm (4.15.29) [σ_7, r^*]:

$$= -Q / (4(t + \eta \cdot tr) b1) - 12 \cdot K7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2)$$

$$= -11653 / (4(36.0 + 1.0 \cdot 12.0) 434.934) -$$

$$12 \cdot 0.053 \cdot 11653 \cdot 630.0 / (335.08(36.0 + 1.0 \cdot 12.0)^2)$$

$$= -74.26 \text{ kgf/cm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1278.0000	mm
Baseplate Thickness	Bpthk	20.0000	mm
Baseplate Width	Bpwid	225.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	12.0000	mm
Web Thickness	Webtk	12.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	2492.1	kgf/cm ²
Height of Web at Center	Hw,c	570.0	mm
Friction Coefficient	mu	0.080	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth}) t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(612.0 \cdot 36.0)} + 580.0) 36.0^3) / 12$$

$$= 315.532 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 315.532 + 292.159(109.633 - 18.0)^2$$

$$= 24846.781 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	811.6	36.0	18.0	292.2	525886.8	0.248E+05
Wearplate	580.0	12.0	42.0	69.6	292320.0	0.319E+04
Web	12.0	500.0	298.0	60.0	1787999.0	0.338E+05
BasePlate	225.0	20.0	558.0	45.0	2510999.2	0.905E+05
Totals	...	...	...	466.8	5117205.5	0.152E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 5117206/466.759$$

$$= 109.633 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120.0/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 11653.012$$

$$= 2371.6406 \text{ kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 13.5833 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 1495.2588 \text{ kgf/cm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1180.0 - 612.0 \cdot \sin(120.0/2) / 1.0472$$

$$= 673.880 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 1598.2030 \text{ kgf-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 11.5040 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 1661.3987 \text{ kgf/cm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(11653 + 299) \cdot 225.0) / (4 \cdot 1278.0 \cdot 1661.399)}$$

$$= 9.746 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (610.0 + 38.0 + 12.0)$$

$$= 1143.154 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 1143.1536 / (4 - 1)$$

$$= 381.051 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 381.0512 \cdot 225.0/2$$

$$= 428.683 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 11952.316 / (1278.0 \cdot 225.0)$$

$$= 4.157 \text{ kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 428.7 \cdot 4.16$$

$$= 1781.864 \text{ kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 22.56 + 22.863$$

$$= 45.423 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 1781.9 / 45.4231$$

$$= 39.228 \text{ kgf/cm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	12.0	200.0	...	24.0	...	800.

Rib dimension [D]:

= Saddle Width - Web Thickness
 = 200.0 - 12.0
 = 188.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A
 = 0.0/45.423
 = 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2
 = 200.0/2
 = 100.000 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{800.0/45.423}$
 = 41.967 mm

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 1180.0 - cos(120.0/2)(610.0 + 38.0 + 12.0) - 20.0
 = 830.000 mm

Intermediate Term [Cc]:

= $\sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$
 = $\sqrt{(2 \cdot \pi^2 \cdot 2038900/2492.1)}$
 = 127.081

Slenderness ratio [KL/r]:

= KL/r
 = 1 • 830.0/41.967
 = 19.777

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2
 = 2831.1 / (2 • 1278.0) • 381.051 • 830.0/2
 = 175.157 kgf-m.

Compressive Allowable, KL/r < Cc (19.7775 < 127.0809) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 = $(1 - (19.78)^2 / (2 \cdot 127.08^2)) 2492 /$
 $(5/3 + 3 \cdot (19.78) / (8 \cdot 127.08) - (19.78^3) / (8 \cdot 127.08^3))$
 = 1428 kgf/cm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I)/Sba
 = 39.23/1427.57 + (175.16 • 100.0/8000001)/1661.4
 = 0.159

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	12.0	188.0	0.0	22.6	0.0	800.
Web	381.1	12.0	0.0	45.7	0.0	5.49
Totals	...	...	...	68.3	...	805.

Distance to Centroid from Datum [ytot]:

= AY / A
 = 0.0/68.286
 = 0.000 mm

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 200.0 / 2$$

$$= 100.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(\text{Ro} + \text{Wpdthk})^2 - (\text{Pitch}/2)^2} - \text{Bpthk}$$

$$= 1180.0 - \sqrt{(660.0 + 12.0)^2 - (381.051/2)^2} - 20.0$$

$$= 528.098 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(805.3/68.286)}$$

$$= 34.341 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= \text{KL}/r$$

$$= 1 \cdot 528.098 / 34.341$$

$$= 15.378$$

Unit Force [Force,u]:

$$= \text{FI} / (2 \cdot \text{Baseplate Length})$$

$$= 2831.116 / (2 \cdot 1278.0)$$

$$= 1.108 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.108 \cdot 381.051 \cdot 528.098$$

$$= 222.892 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 222.892 / 80531.43$$

$$= 276.777 \text{ kgf/cm}^2$$

Compressive Allowable, $\text{KL}/r < C_c$ (15.3779 < 127.0809) per AISC E2-1 [Sca]:

$$= (1 - (\text{KL}/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (\text{KL}/r) / (8 \cdot C_c) - (\text{KL}/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (15.38)^2 / (2 \cdot 127.08^2)) 2492 /$$

$$(5/3 + 3 \cdot (15.38) / (8 \cdot 127.08) - (15.38^3) / (8 \cdot 127.08^3))$$

$$= 1445 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= \text{Sc}/\text{Sca} + (\text{Mbase,c} \cdot \text{C1}/I) / \text{Sba}$$

$$= 52.22 / 1445.16 + (222.89 \cdot 100.0 / 805.314) / 1661.4$$

$$= 0.203$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1529.00 kgf/cm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	4.0400 cm ²
BasePlate Allowable Stress	S	1509.10 kgf/cm ²
Area Available in a Single Bolt	BlArea	3.0428 cm ²
Saddle Load Q0 (Weight)	Q0	7568.8 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	1965.1 kgf
Maximum Transverse Force	Ft	1415.6 kgf
Maximum Longitudinal Force	Fl	2831.1 kgf
Saddle Bolted to Steel Foundation	No	

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= \text{FI} / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 2831.1 / (3.043 \cdot 2)$$

$$= 465.2 \text{ kgf/cm}^2. \text{ Must be less than } 1019.0 \text{ kgf/cm}^2.$$



Shear Stress in a Single Bolt, Transverse Direction [Tb,t]:
 $= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 1415.6 / (3.043 \cdot 2)$
 $= 232.6 \text{ kgf/cm}^2$. Must be less than 1019.0 kgf/cm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 $= 0.0$ (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 $= F_l / (\text{BoltShearAllowable} \cdot \text{Nbols})$
 $= 2831.12 / (1019.0 \cdot 2.0)$
 $= 1.3892 \text{ cm}^2$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 $= B \cdot F_t + \text{Sum of X Moments}$
 $= 1180.0 \cdot 1415.56 + 0.0$
 $= 1670.36 \text{ kgf-m}$.

Eccentricity (e):
 $= R_{mom} / QO$
 $= 1670.36 / 7568.76$
 $= 220.69 \text{ mm} > B_{plen} / 6$ --> Uplift in Transverse direction

$f = B_{plen} / 2 - \text{Edgedis}$
 $= 1278.0 / 2 - 50.8$
 $= 588.20 \text{ mm}$

Modular Ratio Of Steel/Concrete (n1):
 $= E_S / E_C$
 $= 2074057 / 219499.77$
 $= 9.45$

$K1 = 3 (e - 0.5 \cdot B_{plen})$
 $= 3 (220.69 - 0.5 \cdot 1278.0)$
 $= -1254.93 \text{ mm}$

$K2 = 6 \cdot n1 \cdot A_t / B_{pwid} \cdot (f + e)$
 $= 6 \cdot 9.45 \cdot 3.04 / 225.0 \cdot (588.2 + 220.69)$
 $= 62017.63 \text{ mm}^2$

$K3 = -K2 \cdot (0.5 \cdot B_{plen} + f)$
 $= -62017.62 \cdot (0.5 \cdot 1278.0 + 588.2)$
 $= -76108027.50 \text{ mm}^3$

Iteratively Solving for the Effective Bearing Length:
 $Y^3 + K1 \cdot Y^2 + K2 \cdot Y + K3 = 0$
 $Y^3 + -1254.93 \cdot Y^2 + 62017.62 \cdot Y + -0.8E+08 = 0$
 $Y = 1253.87 \text{ mm}$

$\text{Num} = (B_{plen} / 2 - Y / 3 - e)$
 $= (1278.0 / 2 - 1253.87 / 3 - 220.69)$
 $= 0.35$

$\text{Denom} = (B_{plen} / 2 - Y / 3 + f)$
 $= (1278.0 / 2 - 1253.87 / 3 + 588.2)$
 $= 809.24$

Total Bolt Tension Force [Tforce]:
 $= - QO \cdot \text{Num} / \text{Denom}$
 $= - 7568.76 \cdot 0.35 / 809.24$
 $= -3.28 \text{ kgf}$

Bolt Area Required due to Transverse Load [Bltareart]:
 $= T_{force} / (Stba \cdot Nbt)$
 $= -3.28 / (1529.0 \cdot 1.0)$
 $= -0.0021 \text{ cm}^2$

Required Area of a Single Bolt [Bltarear]:
= max[Bltarearl, Bltarears, Bltareart]
= max[0.0, 1.3892, -0.0021]
= 1.3892 cm²

Baseplate Thickness Calculation per D. Moss:

Bearing Pressure (fc)
= 2(QO + Tforce) / (Y • Bpwid)
= 2(7568.76 + -3.28)/(1253.87 • 225.0)
= 5.36 kgf/cm²

Distance from Baseplate Edge to the Web [ADIST]:
= (Bplen - WebIngh) / 2
= (1278.0 - 1227.2)/2
= 25.4000 mm

Overturning Moment due To Bolt Tension [Mt]:
= Tforce • Adist
= -3.28 • 2.54
= -0.08 kgf-m.

Equivalent Bearing Pressure (f1):
= fc • (Y - Adist) / Y
= 5.36 • (1253.87 - 25.4)/1253.87
= 5.25 kgf/cm²

Overturning Moment due to Bearing Pressure [Mc]:
= (Adist² • Bpwid / 6) • (f1 + 2 • fc)
= (25.4² • 225.0/6) • (5.25 + 2 • 5.36)
= 3.87 kgf-m.

Baseplate Required Thickness [Treq]:
= $\sqrt{(6 \cdot \max(Mt, Mc) / (Bpwid \cdot Sba))}$
= $\sqrt{(6 \cdot \max(-0.08, 3.87) / (225.0 \cdot 2263.65))}$
= 2.1342 mm

Note: Saddle and wear plate material are both specified as impact tested.

Review notes about nozzle loadings and supports in the Warnings report.

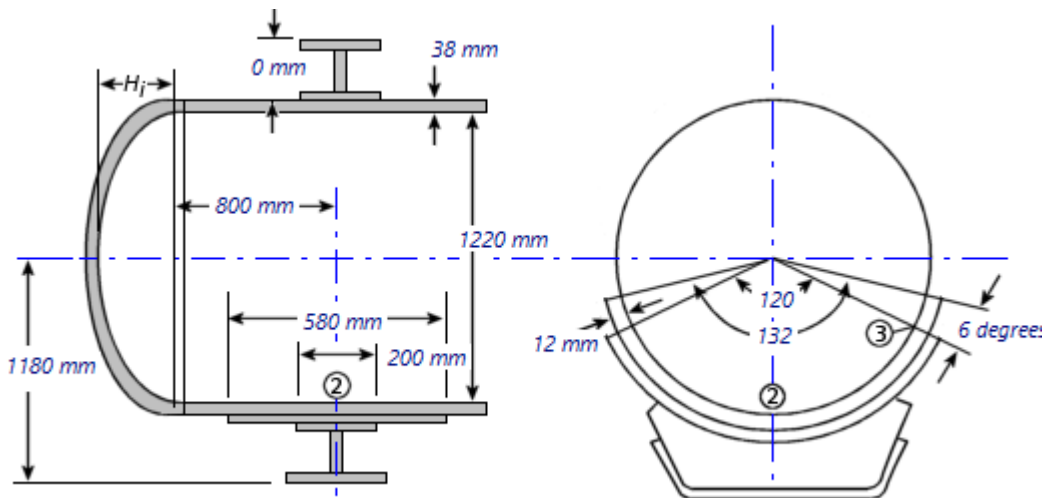
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Input and Calculated Values:

Vessel Mean Radius	Rm	629.00	mm
Shell Thickness used in this Case	t	38.000	mm
Stiffened Vessel Length per 4.15.6	L	335.08	cm
Distance from Saddle to Vessel tangent	a1 or a	800.00	mm
Saddle Width	b1 or b	200.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	580.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	12.0	mm
Wear Plate Allowable Stress	fw or Sr	1509.10	kgf/cm ²
Inside or Mean Depth of Head	Hi or hm	63.20	cm
Shell Allowable Stress used in Calculation		3003.07	kgf/cm ²
Head Allowable Stress used in Calculation		3003.07	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1180.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		10321.42	kgf
Pressure used in Analysis	P	107.732	kgf/cm ²

Horizontal Vessel Analysis Results:		Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan		894.49	3003.07
Long. Stress at Bottom of Midspan		888.76	3003.07
Long. Stress at Top of Saddles		955.95	3003.07
Long. Stress at Bottom of Saddles		855.97	3003.07
Tangential Shear in Shell		21.11	1684.88
Circ. Stress at Horn of Saddle		60.85	4504.61
Circ. Compressive Stress in Shell		3.56	3003.07

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0 (289.3/2 + 0) \cdot 1180.0/1143.1537$
 $= 448.0 \text{ kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(133.23, 0) \cdot 1180.0/1700.0$
 $= 92.5 \text{ kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 9873 + \max(92, 448, 0, 0)$
 $= 10321.4 \text{ kgf}$

Longitudinal Wind Force [F_l]:
 $= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1587.169 (0.0 + \pi/4 (1.416 \cdot 1.2)^2)$
 $= 3959.112 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	10620.73	kgf
Transverse Shear Load	144.66	kgf
Longitudinal Shear Load	133.23	kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$
 $= \min(2(1220.0/2 + 38.0 + 12.0) \sin(120.0/2), 2 \cdot 629.0)$
 $= 1143.154 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$
 $= -10321 \cdot 80.0 [1 - (1 - 80.0/335.08 + (62.9^2 - 63.2^2) / (2 \cdot 80.0 \cdot 335.08)) / (1 + (4 \cdot 63.2) / (3 \cdot 335.08))]$
 $= -3239.1 \text{ kgf-m.}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/(3L)) - 4a/L)$
 $= 10321 \cdot 335/4 ((1 + 2(63^2 - 63^2)/(335^2)) / (1 + (4 \cdot 63)/(3 \cdot 335)) - 4 \cdot 80/335)$
 $= -1353.0 \text{ kgf-m.}$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 $= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 t)$
 $= 107.732 \cdot 629.0 / (2 \cdot 38.0) - -1353.0 / (\pi \cdot 629.0^2 \cdot 38.0)$
 $= 894.49 \text{ kgf/cm}^2$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:
 $= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$
 $= 107.732 \cdot 629.0 / (2 \cdot 38.0) + -1353.0 / (\pi \cdot 629.0^2 \cdot 38.0)$
 $= 888.76 \text{ kgf/cm}^2$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_3]:
 $= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 t)$
 $= 107.732 \cdot 629.0 / (2 \cdot 38.0) - -3239.1 / (0.1066 \cdot \pi \cdot 629.0^2 \cdot 38.0)$
 $= 955.95 \text{ kgf/cm}^2$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 107.732 \cdot 629.0 / (2 \cdot 38.0) + 3239.1 / (0.1923 \cdot \pi \cdot 629.0^2 \cdot 38.0)$$

$$= 855.97 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 10321(335.08 - 2 \cdot 80.0) / (335.08 + (4 \cdot 63.2 / 3))$$

$$= 4309.3 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 4309.26 / (629.0 \cdot 38.0)$$

$$= 21.11 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{R_m \cdot t}$$

$$= 0.78 \cdot \sqrt{629.0 \cdot 38.0}$$

$$= 120.590 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(200.0 + 1.56 \cdot \sqrt{629.0 \cdot 38.0}, 2 \cdot 80.0)$$

$$= 441.18 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.7603 \cdot 10321 \cdot 0.1 / (441.18(38.0 + 1.0 \cdot 12.0))$$

$$= -3.56 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 12 \cdot K7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2)$$

$$= -10321 / (4(38.0 + 1.0 \cdot 12.0)441.18) -$$

$$12 \cdot 0.053 \cdot 10321 \cdot 629.0 / (335.08(38.0 + 1.0 \cdot 12.0)^2)$$

$$= -60.85 \text{ kgf/cm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1278.0000	mm
Baseplate Thickness	Bpthk	20.0000	mm
Baseplate Width	Bpwid	225.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	12.0000	mm
Web Thickness	Webtk	12.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	3161.1	kgf/cm ²
Height of Web at Center	Hw,c	570.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{R \cdot t} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{610.0 \cdot 38.0} + 580.0)38.0^3) / 12$$

$$= 373.820 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 373.82 + 310.654(107.5 - 19.0)^2$$

$$= 24705.016 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	817.5	38.0	19.0	310.7	590241.9	0.247E+05
Wearplate	580.0	12.0	44.0	69.6	306240.0	0.281E+04
Web	12.0	500.0	300.0	60.0	1799999.2	0.347E+05
BasePlate	225.0	20.0	560.0	45.0	2519999.5	0.922E+05



Totals	...	...	...	485.3	5216481.0	0.154E+06
--------	-----	-----	-----	-------	-----------	-----------

Distance to Centroid [C1]:

$$= AY / A$$

$$= 5216481/485.254$$

$$= 107.500 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120.0/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 10321.425$$

$$= 2100.6338 \text{ kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 12.0311 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 1896.6768 \text{ kgf/cm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1180.0 - 610.0 \cdot \sin(120.0/2) / 1.0472$$

$$= 675.534 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 1419.0510 \text{ kgf-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 9.8795 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 2107.4187 \text{ kgf/cm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(10321 + 299) \cdot 225.0) / (4 \cdot 1278.0 \cdot 2107.419)}$$

$$= 8.158 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (610.0 + 38.0 + 12.0)$$

$$= 1143.154 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 1143.1536 / (4 - 1)$$

$$= 381.051 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 381.0512 \cdot 225.0/2$$

$$= 428.683 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 10620.73 / (1278.0 \cdot 225.0)$$

$$= 3.694 \text{ kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 428.7 \cdot 3.69$$

$$= 1583.350 \text{ kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 22.56 + 22.863$$

$$= 45.423 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/A_r$$

$$= 1583.3/45.4231$$

$$= 34.858 \text{ kgf/cm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	12.0	200.0	...	24.0	...	800.

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 200.0 - 12.0$$

$$= 188.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/45.423$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 200.0/2$$

$$= 100.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{800.0/45.423}$$

$$= 41.967 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1180.0 - \cos(120.0/2) (610.0 + 38.0 + 12.0) - 20.0$$

$$= 830.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$$

$$= \sqrt{2 \cdot \pi^2 \cdot 2038900/3161.1}$$

$$= 112.834$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 830.0/41.967$$

$$= 19.777$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{pln}) \cdot e \cdot L / 2$$

$$= 133.2 / (2 \cdot 1278.0) \cdot 381.051 \cdot 830.0/2$$

$$= 8.243 \text{ kgf-m.}$$

Compressive Allowable, $KL/r < C_c$ ($19.7775 < 112.8345$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (19.78)^2 / (2 \cdot 112.83^2)) 3161 /$$

$$(5/3 + 3 \cdot (19.78) / (8 \cdot 112.83) - (19.78^3) / (8 \cdot 112.83^3))$$

$$= 1797 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 34.86/1797.38 + (8.24 \cdot 100.0/8000001) / 2107.42$$

$$= 0.024$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
--	---	---	---	---	----	----

Rib	12.0	188.0	0.0	22.6	0.0	800.
Web	381.1	12.0	0.0	45.7	0.0	5.49
Totals	...	...	...	68.3	...	805.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/68.286$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 200.0/2$$

$$= 100.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(\text{Ro} + \text{Wpdthk})^2 - (\text{Pitch}/2)^2} - \text{Bpthk}$$

$$= 1180.0 - \sqrt{(660.0 + 12.0)^2 - (381.051/2)^2} - 20.0$$

$$= 528.098 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(805.3/68.286)}$$

$$= 34.341 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 528.098/34.341$$

$$= 15.378$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 133.226/(2 \cdot 1278.0)$$

$$= 0.052 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.052 \cdot 381.051 \cdot 528.098$$

$$= 10.489 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 10.489/80531.43$$

$$= 13.025 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < C_c$ (15.3779 < 112.8345) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (15.38)^2 / (2 \cdot 112.83^2)) 3161 /$$

$$(5/3 + 3 \cdot (15.38) / (8 \cdot 112.83) - (15.38)^3 / (8 \cdot 112.83^3))$$

$$= 1823 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 46.41/1823.49 + (10.49 \cdot 100.0/805.314) / 2107.42$$

$$= 0.032$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1753.82 kgf/cm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	4.0400 cm ²
BasePlate Allowable Stress	S	2110.82 kgf/cm ²
Area Available in a Single Bolt	BltArea	3.0428 cm ²
Saddle Load Q0 (Weight)	Q0	10172.8 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	92.5 kgf
Maximum Transverse Force	Ft	144.7 kgf



Maximum Longitudinal Force F1 133.2 kgf
Saddle Bolted to Steel Foundation No

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:

$$= F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 133.2 / (3.043 \cdot 2)$$

$$= 21.9 \text{ kgf/cm}^2. \text{ Must be less than } 1052.3 \text{ kgf/cm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:

$$= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 144.7 / (3.043 \cdot 2)$$

$$= 23.8 \text{ kgf/cm}^2. \text{ Must be less than } 1052.3 \text{ kgf/cm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} QO > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F_l / (\text{BoltShearAllowable} \cdot \text{Nbols})$$

$$= 133.23 / (1052.29 \cdot 2.0)$$

$$= 0.0633 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot F_t + \text{Sum of X Moments}$$

$$= 1180.0 \cdot 144.66 + 0.0$$

$$= 170.70 \text{ kgf-m.}$$

Eccentricity (e):

$$= R_{mom} / QO$$

$$= 170.7 / 10172.77$$

$$= 16.78 \text{ mm} < B_{plen}/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0.0, 0.0633, 0.0]$$

$$= 0.0633 \text{ cm}^2$$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	629.00	mm
Shell Thickness used in this Case	t	38.000	mm
Stiffened Vessel Length per 4.15.6	L	335.08	cm
Distance from Saddle to Vessel tangent	a1 or a	800.00	mm
Saddle Width	b1 or b	200.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	580.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	12.0	mm
Wear Plate Allowable Stress	fw or Sr	1509.10	kgf/cm ²
Inside or Mean Depth of Head	Hi or hm	32.35	cm
Shell Allowable Stress used in Calculation		3003.07	kgf/cm ²
Head Allowable Stress used in Calculation		3003.07	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1180.00	mm
Coefficient of Friction	mu	0.08	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces 10317.94 kgf
 Pressure used in Analysis P 107.732 kgf/cm²

Horizontal Vessel Analysis Results:	Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan	892.05	3003.07
Long. Stress at Bottom of Midspan	891.20	3003.07
Long. Stress at Top of Saddles	937.11	3003.07
Long. Stress at Bottom of Saddles	866.41	3003.07
Tangential Shear in Shell	23.39	1684.88
Circ. Stress at Horn of Saddle	60.83	4504.61
Circ. Compressive Stress in Shell	3.56	3003.07

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= \text{Ftr} (\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$
 $= 3.0 (289.3/2 + 0) \cdot 1180.0/1143.1537$
 $= 448.0 \text{ kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max (\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$
 $= \max (133.23, 0) \cdot 1180.0/1700.0$
 $= 92.5 \text{ kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max (\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$
 $= 9870 + \max (92, 448, 0, 0)$
 $= 10317.9 \text{ kgf}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1587.169 (0.0 + \pi/4 (1.416 \cdot 1.2)^2)$
 $= 3959.112 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 10617.25 kgf
 Transverse Shear Load 144.66 kgf
 Longitudinal Shear Load 133.23 kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:
 $= \min (2 (\text{ShellID}/2 + t + \text{WearPadThickness}) \sin (\theta/2), 2 \cdot \text{Rm})$
 $= \min (2 (1220.0/2 + 38.0 + 12.0) \sin (120.0/2), 2 \cdot 629.0)$
 $= 1143.154 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011
 K5 = 0.7603 K6 = 0.0529 K7 = 0.0529 K8 = 0.3405
 K9 = 0.2711 K10 = 0.0581 K1* = 0.1923 K6p = 0.0434
 K7p = 0.0434

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (\text{Rm}^2 - \text{hm}^2) / (2a \cdot L)) / (1 + (4\text{hm}) / (3L))]$
 $= -10318 \cdot 80.0 [1 - (1 - 80.0/335.08 + (62.9^2 - 32.35^2) / (2 \cdot 80.0 \cdot 335.08)) / (1 + (4 \cdot 32.35) / (3 \cdot 335.08))]$
 $= -2290.4 \text{ kgf-m.}$

Moment per Equation 4.15.2 [M2]:



$$\begin{aligned}
 &= Q \cdot L / 4 \cdot ((1 + 2(Rm^2 - hm^2) / (L^2)) / (1 + (4hm^2) / (3L)) - 4a / L) \\
 &= 10318 \cdot 335 / 4 \cdot ((1 + 2(63^2 - 32^2) / (335^2)) / (1 + (4 \cdot 32) / (3 \cdot 335)) - 4 \cdot 80 / 335) \\
 &= -199.8 \text{ kgf-m.}
 \end{aligned}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 \cdot t) \\
 &= 107.732 \cdot 629.0 / (2 \cdot 38.0) - 199.8 / (\pi \cdot 629.0^2 \cdot 38.0) \\
 &= 892.05 \text{ kgf/cm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t) \\
 &= 107.732 \cdot 629.0 / (2 \cdot 38.0) + 199.8 / (\pi \cdot 629.0^2 \cdot 38.0) \\
 &= 891.20 \text{ kgf/cm}^2
 \end{aligned}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t) \\
 &= 107.732 \cdot 629.0 / (2 \cdot 38.0) - 2290.4 / (0.1066 \cdot \pi \cdot 629.0^2 \cdot 38.0) \\
 &= 937.11 \text{ kgf/cm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t) \\
 &= 107.732 \cdot 629.0 / (2 \cdot 38.0) + 2290.4 / (0.1923 \cdot \pi \cdot 629.0^2 \cdot 38.0) \\
 &= 866.41 \text{ kgf/cm}^2
 \end{aligned}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$\begin{aligned}
 &= Q(L - 2a) / (L + (4 \cdot hm / 3)) \\
 &= 10318(335.08 - 2 \cdot 80.0) / (335.08 + (4 \cdot 32.35 / 3)) \\
 &= 4776.3 \text{ kgf}
 \end{aligned}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$\begin{aligned}
 &= K2 \cdot T / (Rm \cdot t) \\
 &= 1.1707 \cdot 4776.31 / (629.0 \cdot 38.0) \\
 &= 23.39 \text{ kgf/cm}^2
 \end{aligned}$$

Decay Length (4.15.20) [x_1, x_2]:

$$\begin{aligned}
 &= 0.78 \cdot \sqrt{(Rm \cdot t)} \\
 &= 0.78 \cdot \sqrt{(629.0 \cdot 38.0)} \\
 &= 120.590 \text{ mm}
 \end{aligned}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$\begin{aligned}
 &= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a) \\
 &= \min(200.0 + 1.56 \cdot \sqrt{(629.0 \cdot 38.0)}, 2 \cdot 80.0) \\
 &= 441.18 \text{ mm}
 \end{aligned}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$\begin{aligned}
 &= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr)) \\
 &= -0.7603 \cdot 10318 \cdot 0.1 / (441.18(38.0 + 1.0 \cdot 12.0)) \\
 &= -3.56 \text{ kgf/cm}^2
 \end{aligned}$$

Circ. Comp. Stress at Horn of Saddle, $L < 8Rm$ (4.15.29) [σ_7, r^*]:

$$\begin{aligned}
 &= -Q / (4(t + \eta \cdot tr)b_1) - 12 \cdot K7 \cdot Q \cdot Rm / (L(t + \eta \cdot tr)^2) \\
 &= -10318 / (4(38.0 + 1.0 \cdot 12.0)441.18) - \\
 &\quad 12 \cdot 0.053 \cdot 10318 \cdot 629.0 / (335.08(38.0 + 1.0 \cdot 12.0)^2) \\
 &= -60.83 \text{ kgf/cm}^2
 \end{aligned}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1278.0000	mm
Baseplate Thickness	Bpthk	20.0000	mm
Baseplate Width	Bpwid	225.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	12.0000	mm
Web Thickness	Webtk	12.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	3161.1	kgf/cm ²
Height of Web at Center	Hw,c	570.0	mm

Friction Coefficient mu 0.080

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(610.0 \cdot 38.0)} + 580.0)38.0^3)/12$$

$$= 373.820 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 373.82 + 310.654(107.5 - 19.0)^2$$

$$= 24705.016 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	817.5	38.0	19.0	310.7	590241.9	0.247E+05
Wearplate	580.0	12.0	44.0	69.6	306240.0	0.281E+04
Web	12.0	500.0	300.0	60.0	1799999.2	0.347E+05
BasePlate	225.0	20.0	560.0	45.0	2519999.5	0.922E+05
Totals	...	...	...	485.3	5216481.0	0.154E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 5216481/485.254$$

$$= 107.500 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120.0/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0)\cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 10317.939$$

$$= 2099.9246 \text{ kgf}$$

Tension Stress, St = (Fh/As) = 12.0271 kgf/cm²
 Allowed Stress, Sa = 0.6 * Yield Str = 1896.6768 kgf/cm²

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 1180.0 - 610.0 \cdot \sin(120.0/2) / 1.0472$$

$$= 675.534 \text{ mm}$$

Bending Moment, M = Fh * d = 1418.5720 kgf-m.

Bending Stress, Sb = (M * C1 / I) = 9.8762 kgf/cm²
 Allowed Stress, Sa = 2/3 * Yield Str = 2107.4187 kgf/cm²

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt})\text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$$

$$= \sqrt{(3(10318 + 299)225.0 / (4 \cdot 1278.0 \cdot 2107.419))}$$

$$= 8.156 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (610.0 + 38.0 + 12.0)$$

$$= 1143.154 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 1143.1536 / (4 - 1)$$

$$= 381.051 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot B_{pwid} / 2$$

$$= 381.0512 \cdot 225.0 / 2$$

$$= 428.683 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 10617.245 / (1278.0 \cdot 225.0)$$

$$= 3.692 \text{ kgf/cm}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 428.7 \cdot 3.69$$

$$= 1582.830 \text{ kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 22.56 + 22.863$$

$$= 45.423 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 1582.8 / 45.4231$$

$$= 34.846 \text{ kgf/cm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	12.0	200.0	...	24.0	...	800.

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 200.0 - 12.0$$

$$= 188.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 45.423$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 200.0 / 2$$

$$= 100.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(800.0 / 45.423)}$$

$$= 41.967 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1180.0 - \cos(120.0/2) (610.0 + 38.0 + 12.0) - 20.0$$

$$= 830.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 2038900 / 3161.1)}$$

$$= 112.834$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 830.0 / 41.967$$

$$= 19.777$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 133.2 / (2 \cdot 1278.0) \cdot 381.051 \cdot 830.0 / 2$$

$$= 8.243 \text{ kgf-m.}$$

Compressive Allowable, $KL/r < C_c$ (19.7775 < 112.8345) per AISC E2-1 [Sca]:

$$= (1 - (Klr)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (Klr) / (8 \cdot Cc) - (Klr^3) / (8 \cdot Cc^3))$$

$$= (1 - (19.78)^2 / (2 \cdot 112.83^2)) 3161 /$$

$$(5/3 + 3 \cdot (19.78) / (8 \cdot 112.83) - (19.78^3) / (8 \cdot 112.83^3))$$

$$= 1797 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 34.85 / 1797.38 + (8.24 \cdot 100.0 / 8000001) / 2107.42$$

$$= 0.024$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	12.0	188.0	0.0	22.6	0.0	800.
Web	381.1	12.0	0.0	45.7	0.0	5.49
Totals	...	...	...	68.3	...	805.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 68.286$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 200.0 / 2$$

$$= 100.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{ (Ro + W_{pdthk})^2 - (Pitch/2)^2 } - B_{pthk}$$

$$= 1180.0 - \sqrt{ (660.0 + 12.0)^2 - (381.051/2)^2 } - 20.0$$

$$= 528.098 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ \text{Total Inertia} / \text{Total Area} }$$

$$= \sqrt{ 805.3 / 68.286 }$$

$$= 34.341 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 528.098 / 34.341$$

$$= 15.378$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 133.226 / (2 \cdot 1278.0)$$

$$= 0.052 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.052 \cdot 381.051 \cdot 528.098$$

$$= 10.489 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 10.489 / 80531.43$$

$$= 13.025 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < C_c$ (15.3779 < 112.8345) per AISC E2-1 [Sca]:

$$= (1 - (Klr)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (Klr) / (8 \cdot Cc) - (Klr^3) / (8 \cdot Cc^3))$$

$$= (1 - (15.38)^2 / (2 \cdot 112.83^2)) 3161 /$$

$$(5/3 + 3 \cdot (15.38) / (8 \cdot 112.83) - (15.38^3) / (8 \cdot 112.83^3))$$

= 1823 kgf/cm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= $Sc/Sca + (M_{base,c} \cdot C1/I) / Sba$
 = $46.39/1823.49 + (10.49 \cdot 100.0/805.314) / 2107.42$
 = 0.032

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1753.82 kgf/cm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	4.0400 cm ²
BasePlate Allowable Stress	S	2110.82 kgf/cm ²
Area Available in a Single Bolt	BltArea	3.0428 cm ²
Saddle Load QO (Weight)	QO	10169.3 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	92.5 kgf
Maximum Transverse Force	Ft	144.7 kgf
Maximum Longitudinal Force	Fl	133.2 kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

= $Fl / (Bolt\ Area \cdot Number\ of\ Bolts)$
 = $133.2 / (3.043 \cdot 2)$
 = 21.9 kgf/cm². Must be less than 1052.3 kgf/cm².

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

= $Ft / (Bolt\ Area \cdot Number\ of\ Bolts)$
 = $144.7 / (3.043 \cdot 2)$
 = 23.8 kgf/cm². Must be less than 1052.3 kgf/cm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:

= $Fl / (BoltShearAllowable \cdot Nbolts)$
 = $133.23 / (1052.29 \cdot 2.0)$
 = 0.0633 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

= $B \cdot Ft + Sum\ of\ X\ Moments$
 = $1180.0 \cdot 144.66 + 0.0$
 = 170.70 kgf-m.

Eccentricity (e):

= $Rmom / QO$
 = $170.7/10169.28$
 = 16.79 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:

= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:

= $\max[Bltarearl, Bltarears, Bltareart]$
 = $\max[0.0, 0.0633, 0.0]$
 = 0.0633 cm²

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: M1-Man Hole

From: 10

Pressure for Reinforcement Calculations	P	194.773	kgf/cm ²
Temperature for Internal Pressure	Temp	310	°C
Parent Material		SA-387 11	+IT
Parent Allowable Stress at Temperature	Sv	1509.10	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1509.10	kgf/cm ²
Inside Diameter of Hemispherical Head	D	1220.00	mm
Head Finished (Minimum) Thickness	t	44.0000	mm
Head Internal Corrosion Allowance	c	2.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

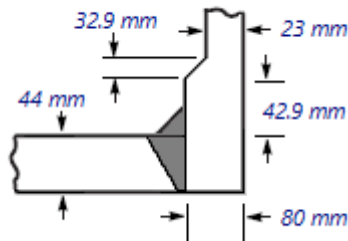
Material		SA-336 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1509.10	kgf/cm ²
Allowable Stress At Ambient	Sna	1509.10	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		200.0000	mm
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	42.9018	mm
Height of Beveled Transition	L'	32.9018	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		80.0000	mm
Corrosion Allowance	can	2.0000	mm
Outside Projection	ho	150.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	44.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		1500	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: M1-Man Hole

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	173.050	mm.
Actual Thickness Used in Calculation	23.012	mm

**Note:**

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 177.05 \cdot (\exp(194.773/(1509.1 \cdot 1.0)) - 1) + 2.0 + 0.0$$

$$= 12.1956 + 2.0 + 0.0 = 14.1956 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1509.1 \cdot 1.0 \cdot \ln((2 \cdot 21.012 + 177.05) / 177.05) - 0.0$$

$$= 321.41 - 0.0 = 321.4095 \text{ kgf/cm}^2$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 9.160 mm

Min. required shell thickness, internal p + c : 42.811 mm

Nozzle neck minimum required thickness : 9.160 mm

Minimum nozzle neck thickness 20.142 mm, is greater than required thickness 9.160 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[1509.1/1509.1, 1]$$

$$= 1.000$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 80.0 - 2.0$$

$$= 78.000 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 23.012 - 2.0$$

$$= 21.012 \text{ mm}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 1220.0/2 + 2.0$$

$$= 612.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{612.0 \cdot 42.0}, 2 \cdot 88.525)$$

$$= 160.325 \text{ mm}$$

Intermediate value [Cn]:

$$= \min(((t + t_e)/t_n)^{0.35}, 1)$$

$$= \min(((42.0 + 0.0)/78.0)^{0.35}, 1)$$

$$= \min(0.805, 1)$$



= 0.805

Intermediate value [Cp]:
 = 9999999.000

Effective Nozzle Wall Length Outside the Vessel [LH]:
 = $\min(t + t_e + F_p \cdot \sqrt{(R_n \cdot t_n)}, L_{pr1} + t)$
 = $\min(42.0 + 0.0 + 0.805 \cdot \sqrt{(88.525 \cdot 78.0)}, 150.0 + 42.0)$
 = 108.909 mm

Effective Vessel Thickness [teff]:
 = $t + ((A5 \cdot frp) / LR)$
 = $42.0 + ((0.0 \cdot 1.0) / 160.325)$
 = 42.000 mm

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:
 = $t \cdot LR$
 = $42.0 \cdot 160.325$
 = 67.336 cm²

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:
 = A2A + A2B
 = 66.223 + 13.734
 = 79.958 cm²

Area Contributed by the Outside Fillet Weld [A41]:
 = $0.5 \cdot Leg^2$
 = $0.5 \cdot 10.0^2$
 = 0.500 cm²

The total area contributed by A1 through A43 [AT]:
 = $A1 + frn(A2 + A3) + A41 + A42 + A43$
 = $67.336 + 1.0(79.958 + 0.0) + 0.5 + 0.0 + 0.0$
 = 147.794 cm²

Nozzle Radius for Force Calculation [Rxn]:
 = $t_n / \ln[1 + t_n/R_n]$
 = $78.0 / \ln[1 + 78.0/88.525]$
 = 123.445 mm

Shell Radius for Force Calculation [Rxs]:
 = $teff / \ln[1 + teff/Reff]$
 = $42.0 / \ln[1 + 42.0/612.0]$
 = 632.767 mm

Allowable Local Primary Membrane Stress [Sallow]:
 = $1.5 \cdot S \cdot E$
 = $1.5 \cdot 1509.097 \cdot 1.0$
 = 2263.6 kgf/cm²

Force acting on the Nozzle [fN]:
 = $P \cdot Rxn \cdot LH$
 = $194.773 \cdot 123.445 \cdot 108.909$
 = 26185.8 kgf

Force acting on the Shell [fS]:
 = $P \cdot Rxs \cdot (LR + t_n) / 2$
 = $194.773 \cdot 632.767 \cdot (160.325 + 78.0) / 2$
 = 146862.5 kgf

Discontinuity Force from Internal Pressure [fY]:
 = $(P \cdot Rxs \cdot R_{nc}) / 2$
 = $(194.773 \cdot 632.767 \cdot 88.525) / 2$
 = 54551.7 kgf

Area Resisting Internal Pressure [Ap]:
 = $(fS + fY + fN) / P$
 = $(146862.5 + 54551.7 + 26185.8) / 194.773$



$$= 1168.5 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= \text{Sallow} / (2 \cdot \text{Ap}/\text{AT} - \text{Rxs}/(2 \cdot \text{teff}))$$

$$= 2263.646 / (2 \cdot 1168.542/147.794 - 632.767/(2 \cdot 42.0))$$

$$= 273.4 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= 2 \cdot \text{S}[t/\text{Rxs}]$$

$$= 2 \cdot 1509.097[42.0/632.767]$$

$$= 200.3 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(\text{Pmax1}, \text{Pmax2})$$

$$= \min(273.382, 200.333)$$

$$= 200.333 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / \text{AT}$$

$$= (26185.771 + 146862.5 + 54551.703) / 147.794$$

$$= 1539.982 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot \text{Rxs} / (2 \cdot \text{teff})$$

$$= 194.773 \cdot 632.767 / (2 \cdot 42.0)$$

$$= 1467.2 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{\text{Avg}} - \sigma_{\text{Circ}}, \sigma_{\text{Circ}})$$

$$= \max(2 \cdot 1539.982 - 1467.213, 1467.213)$$

$$= 1612.8 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2263.65	kgf/cm ²
Local Primary Membrane Stress	PL	1612.75	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	200.33	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (R_{nc} + t_n) / R_{nc}$$

$$= (88.525 + 78.0) / 88.525$$

$$= 1.881 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi/2 \cdot (R_n + t_n)$$

$$= \pi/2 \cdot (88.525 + 78.0)$$

$$= 261.577 \text{ mm}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$t_c, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 78.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$$

$$= \min(54552 \cdot 1.88, 1.5 \cdot 1509.1(83.584 + 0.0), \pi/4 \cdot 194.8 \cdot 88.53^2 \cdot 1.88^2)$$

$$= 42420.625 \text{ kgf}$$

Weld Stress Value [τ]:

$$= \text{fwelds} / (L_t(0.49 \cdot L41T + 0.6 \cdot t_w1 + 0.49 \cdot L43T))$$

$$= 42420.625 / (261.577(0.49 \cdot 7.071 + 0.6 \cdot 42.0 + 0.49 \cdot 0.0))$$

$$= 565.756 < \text{or} = \text{to } 1509.097 \text{ Weld Stress Passed}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

This element is Post Weld Heat Treated PWHT



Computed Minimum Design Metal Temperature : -6 °C
 User entered Minimum Design Temperature : -10 °C
 The Curve for this Material is : A
 From Figure 3.8M- with PWHT, the MDMT is : 13 °C
 Temperature reduction per Graph 3.13 : 18 °C
 The computed stress ratio is : 0.672

The computed thickness ratio [sr]:

$$= (tr \cdot E^*) / (tnom - ci - cext)$$

$$= (12.196 \cdot 1.0) / (20.136 - 2.0 - 0.0)$$

$$= 0.672$$

Warning: Impact Testing Is Mandatory!

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature : 5 °C

User entered Minimum Design Temperature : -10 °C

The Curve for this Material is : B

From Figure 3.8M- with PWHT, the MDMT is : 7 °C

Temperature reduction per Graph 3.13 : 1 °C

The computed stress ratio is : 0.971

The computed thickness ratio [sr]:

$$= (tr \cdot E^*) / (tnom - ci - cext)$$

$$= (40.796 \cdot 1.0) / (44.0 - 2.0 - 0.0)$$

$$= 0.971$$

Warning: Impact Testing Is Mandatory!

Gov. MDMT of the nozzle to shell joint welded assembly : 5 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction -32 °C

The Drop for this Nozzle is : 23.1700 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 217.1700 mm

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Input, Nozzle Desc: A1-Tube Inlet

From: 20

Pressure for Reinforcement Calculations	P	194.770	kgf/cm ²
Temperature for Internal Pressure	Temp	310	°C
Parent Material		SA-387 11	+IT
Parent Allowable Stress at Temperature	Sv	1509.10	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1509.10	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	1220.00	mm
Shell Finished (Minimum) Thickness	t	88.0000	mm
Shell Internal Corrosion Allowance	c	2.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		70.00	cm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

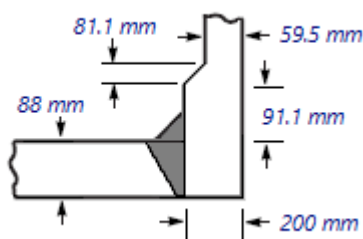
Material		SA-336 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1509.10	kgf/cm ²
Allowable Stress At Ambient	Sna	1509.10	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		600.0000	mm
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Hub Height of Integral Nozzle	h	91.1106	mm
Height of Beveled Transition	L'	81.1106	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		200.0000	mm
Corrosion Allowance	can	2.0000	mm
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	88.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: A1-Tube Inlet

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	490.576	mm.
Actual Thickness Used in Calculation	59.512	mm



Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 494.576 \cdot (\exp(194.77/(1509.1 \cdot 1.0)) - 1) + 2.0 + 0.0$
 $= 34.0671 + 2.0 + 0.0 = 36.0671 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 1509.1 \cdot 1.0 \cdot \ln((2 \cdot 57.512 + 494.576) / 494.576) - 0.0$
 $= 315.557 - 0.0 = 315.5575 \text{ kgf/cm}^2$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 10.340 mm
 Min. required shell thickness, internal p + c : 86.436 mm

Nozzle neck minimum required thickness : 10.340 mm

Minimum nozzle neck thickness 52.070 mm , is greater than required thickness 10.340 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 414.3,	Allowable	: 1509.1 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 3358.5 kgf/cm ²	Passed
Occasional	: 436.7,	Allowable	: 2007.1 kgf/cm ²	Passed
Shear	: 63.1,	Allowable	: 1056.4 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[1509.1/1509.1, 1]$
 $= 1.000$

Thickness of Nozzle at Shell [tn]:
 $= \text{hub thickness} - \text{corrosion allowance}$
 $= 200.0 - 2.0$
 $= 198.000 \text{ mm}$

Thickness of Nozzle at Top [tn2]:
 $= \text{thickness} - \text{corrosion allowance}$
 $= 59.512 - 2.0$
 $= 57.512 \text{ mm}$

Shell Diameter to Thickness ratio [D/t]:
 $= D_i/t$
 $= 1224.0/86.0$
 $= 14.233 \text{ must be less than } 400.$

Effective Pressure Radius [Reff]:
 $= D_i/2 + \text{corrosion allowance}$
 $= 1220.0/2 + 2.0$
 $= 612.000 \text{ mm}$

Effective Length of Vessel Wall [LR]:
 $= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$
 $= \min(\sqrt{612.0 \cdot 86.0}, 2 \cdot 247.288)$



$$= 229.417 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)}$$

$$= \min[1.5 \cdot 86.0, 0.0] + \sqrt{(247.288 \cdot 198.0)}$$

$$= 221.276 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 300.0$$

$$= 300.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(86.0 + 0.0)$$

$$= 688.000 \text{ mm}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles}$$

$$= \min[221.276, 300.0, 688.0] + 86.0$$

$$= 307.276 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 86.0 + ((0.0 \cdot 1.0) / 229.417)$$

$$= 86.000 \text{ mm}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}}))$$

$$= \min(12, (2 \cdot 247.29 + 198.0) / (\sqrt{(1224.0 + 86.0) \cdot 86.0}))$$

$$= 2.063$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 86.0 \cdot 229.417 \cdot \max((2.063/5)^{0.85}, 1)$$

$$= 197.298 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= A2A + A2C$$

$$= 350.679 + 53.498$$

$$= 404.177 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + frn(A2 + A3) + A41 + A42 + A43$$

$$= 197.298 + 1.0(404.177 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 601.975 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 198.0 / \ln[1 + 198.0/247.288]$$

$$= 336.639 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 86.0 / \ln[1 + 86.0/612.0]$$

$$= 654.058 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 1509.097 \cdot 1.0$$

$$= 2263.6 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:



$$\begin{aligned}
 &= P \cdot R_{xn} \cdot L_H \\
 &= 194.77 \cdot 336.639 \cdot 307.276 \\
 &= 201471.9 \text{ kgf}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Force acting on the Shell [fS]:} \\
 &= P \cdot R_{xs} (L_R + t_n) \\
 &= 194.77 \cdot 654.058 (229.417 + 198.0) \\
 &= 544490.2 \text{ kgf}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Discontinuity Force from Internal Pressure [fY]:} \\
 &= P \cdot R_{xs} \cdot R_{nc} \\
 &= 194.77 \cdot 654.058 \cdot 247.288 \\
 &= 315022.3 \text{ kgf}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Area Resisting Internal Pressure [Ap]:} \\
 &= (f_S + f_Y + f_N) / P \\
 &= (544490 + 315022.3 + 201471.9) / 194.77 \\
 &= 5447.4 \text{ cm}^2
 \end{aligned}$$

$$\begin{aligned}
 &\text{Maximum Allowable Working Pressure Candidate [Pmax1]:} \\
 &= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff}) \\
 &= 2263.646 / (2 \cdot 5447.368 / 601.975 - 654.058 / 86.0) \\
 &= 215.7 \text{ kgf/cm}^2
 \end{aligned}$$

$$\begin{aligned}
 &\text{Maximum Allowable Working Pressure Candidate [Pmax2]:} \\
 &= S / R_{xs} \\
 &= 1509.097 [86.0 / 654.058] \\
 &= 198.4 \text{ kgf/cm}^2
 \end{aligned}$$

$$\begin{aligned}
 &\text{Maximum Allowable Working Pressure [Pmax]:} \\
 &= \min(P_{max1}, P_{max2}) \\
 &= \min(215.729, 198.426) \\
 &= 198.426 \text{ kgf/cm}^2
 \end{aligned}$$

$$\begin{aligned}
 &\text{Average Primary Membrane Stress [\sigma_{Avg}]:} \\
 &= (f_N + f_S + f_Y) / A_T \\
 &= (201471.891 + 544490 + 315022.281) / 601.975 \\
 &= 1762.508 \text{ kgf/cm}^2
 \end{aligned}$$

$$\begin{aligned}
 &\text{General Primary Membrane Stress [\sigma_{Circ}]:} \\
 &= P \cdot R_{xs} / t_{eff} \\
 &= 194.77 \cdot 654.058 / 86.0 \\
 &= 1481.3 \text{ kgf/cm}^2
 \end{aligned}$$

$$\begin{aligned}
 &\text{Maximum Local Primary Membrane Stress [PL]:} \\
 &= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\
 &= \max(2 \cdot 1762.508 - 1481.292, 1481.292) \\
 &= 2043.7 \text{ kgf/cm}^2
 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2263.65	kgf/cm ²
Local Primary Membrane Stress	PL	2043.72	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	198.43	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

$$\begin{aligned}
 &\text{Discontinuity Force Factor [ky]:} \\
 &= (R_{nc} + t_n) / R_{nc} \\
 &= (247.288 + 198.0) / 247.288 \\
 &= 1.801 \text{ For set-in Nozzles}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Weld Length of Nozzle to Shell Weld [Lr]:} \\
 &= \pi / 2 \cdot (R_n + t_n) \\
 &= \pi / 2 \cdot (247.288 + 198.0) \\
 &= 699.456 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Weld Throat Dimensions, (0.7071 \cdot Leg Dimensions) [L41T, L42T, L43T]:} \\
 &= 7.071, 0.000, 0.000, \text{ mm}
 \end{aligned}$$

Required Weld Throat Dimensions :

tc, L41T = 7.071, must be $\geq \min(0.7 * 198.000, 6.000)$ Passed

Weld Load Value [fwelds]:
 $= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$
 $= \min(315022 \cdot 1.8, 1.5 \cdot 1509.1(404.177 + 0.0), \pi/4 \cdot 194.8 \cdot 247.29^2 \cdot 1.8^2)$
 $= 303314.875 \text{ kgf}$

Weld Stress Value [τ]:
 $= \text{fwelds} / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$
 $= 303314.875 / (699.456(0.49 \cdot 7.071 + 0.6 \cdot 86.0 + 0.49 \cdot 0.0))$
 $= 787.516 < \text{or} = \text{to } 1509.097 \text{ Weld Stress Passed}$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : 18 °C
 User entered Minimum Design Temperature : -10 °C
 The Curve for this Material is : B
 From Figure 3.8M- with PWHT, the MDMT is : 19 °C
 Temperature reduction per Graph 3.13 : 1 °C
 The computed stress ratio is : 0.980

The computed thickness ratio [sr]:
 $= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$
 $= (84.311 \cdot 1.0) / (88.0 - 2.0 - 0.0)$
 $= 0.980$

Warning: Impact Testing Is Mandatory!

Gov. MDMT of the nozzle to shell joint welded assembly : 18 °C

The Drop for this Nozzle is : 193.0841 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 581.0842 mm

Input Echo, WRC297 Item 1, Description: A1-Tube Inlet :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	2.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1220.000	mm
Shell Thickness	88.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-387 11	
Vessel Cold S.I. Allowable	Smc 1509.10	kgf/cm ²
Vessel Hot S.I. Allowable	Smh 1509.10	kgf/cm ²
Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	2.0000	mm
Nozzle Diameter	490.576	mm
Nozzle Thickness	200.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-336 F11	
Nozzle Cold S.I. Allowable	SNmc 1509.10	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh 1509.10	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:
 External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	194.77	kgf/cm ²
Radial Load	P	-19579.53	kgf
Circumferential Shear	Vc	19579.53	kgf
Longitudinal Shear	Vl	14684.63	kgf
Circumferential Moment	Mc	30.53	kgf-m.
Longitudinal Moment	Ml	30.53	kgf-m.



Torsional Moment Mt 35.23 kgf-m.

Include Axial Pressure Thrust No
 Include Pressure Stress Indices per Div. 2 No
 Local Loads applied at end of Nozzle/Attachment No

Warning - The ratio, D_v/T_v (15.864) is < 20
 check the Limitations of WRC bulletin 297.
 Warning - The ratio, D_n/D_v (.638) is > 0.5
 check the Limitations of WRC bulletin 297.
 Warning - The ratio, D_n/T_n (4.453) is < 10
 check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - c_a))}$$

$$= 890.576 + 2 \cdot 1.65 \cdot \sqrt{(655.0(88.0 - 2.0))}$$

$$= 1673.796 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 1310.000 mm
 Nozzle Outside Diameter (d) = 890.576 mm
 Vessel Thickness used (T) = 86.000 mm
 Nozzle Thickness used (t) = 198.000 mm
 $T/t = 0.434$
 $d/t = 10.000$

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.653$$

Nr/P = 0.062
 Mr/P = 0.115
 M0/P = 0.036
 N0/P = 0.084
 MrD/Mc = 0.271
 NrDL/Mc = 0.099
 M0d/Mc = 0.082
 N0DL/Mc = 0.082
 MrD/Ml = 0.098
 NrDL/Ml = 0.056
 M0D/Ml = 0.028
 N0DL/Ml = 0.099

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	22	22	22	22
Outplane Bending (P)	57	-57	57	-57
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	0	0	0	0
Outplane Bending (Ml)	0	0	0	0
Normal Pressure Stress	1294	1489	1294	1489
Outplane Stress Summary	1374	1454	1374	1454

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	16	16	16	16
Inplane Bending (P)	182	-182	182	-182
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (Ml)	0	0	0	0
Inplane Bending (Ml)	0	0	0	0



Inplane Pressure Stress	647	647	647	647
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Inplane Stress Summary	846	481	846	481
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Shear stress normal to longitudinal plane)	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	16	16	-16	-16
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0

Shear Stress Summary	16	16	-16	-16
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stress Intensities)	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	1374	1455	1374	1454
------------------------	------	------	------	------

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses Normal to circumferential plane)	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Outplane Membrane (P)	22	22	22	22
Outplane Bending (P)	57	-57	57	-57
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	647	647	647	647

Outplane Stress Summary	726	612	726	612
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses parallel to circumferential plane)	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Inplane Membrane (P)	16	16	16	16
Inplane Bending (P)	182	-182	182	-182
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1294	1489	1294	1489

Inplane Stress Summary	1493	1324	1494	1323
------------------------	------	------	------	------

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Shear stress normal to circumferential plane)	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-12	-12	12	12
Torsional Shear (Mt)	0	0	0	0

Shear Stress Summary	-12	-12	12	12
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stress Intensities)	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	1493	1324	1494	1323
------------------------	------	------	------	------

**Nozzle Stresses (kgf/cm²):**

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	A1 Top Inside	Bu Bottom Outside	B1 Bottom Inside
Hoop Membrane (P)	22	22	22	22
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	208	442	208	442
Hoop Stress Summary	230	464	230	464

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	A1 Top Inside	Bu Bottom Outside	B1 Bottom Inside
Axial Membrane (P)	4	4	4	4
Axial Bending (P)	25	-25	25	-25
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	104	104	104	104
Axial Stress Summary	133	83	133	83

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	A1 Inside	Bu Outside	B1 Inside
Shear due to (Vc)	7	7	-7	-7
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	7	7	-7	-7

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	A1 Inside	Bu Outside	B1 Inside
Two * Max Shear Stress	230	464	230	464

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	C1 Left Inside	Du Right Outside	D1 Right Inside
Hoop Membrane (P)	22	22	22	22
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	208	442	208	442
Hoop Stress Summary	230	464	230	464

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	C1 Left Inside	Du Right Outside	D1 Right Inside
Axial Membrane (P)	4	4	4	4
Axial Bending (P)	25	-25	25	-25
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0



Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	104	104	104	104
Axial Stress Summary	133	83	133	83

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-5	-5	5	5
Shear due to Torsion	0	0	0	0
Shear Stress Summary	-5	-5	5	5

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	230	464	230	464

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1294	1489	1294	1489	1294	1489	1294	1489
Circ. Pl (SUS)	22	22	22	22	16	16	16	16
Circ. Q (SUS)	57	-57	57	-57	182	-182	182	-182
Long. Pm (SUS)	647	647	647	647	647	647	647	647
Long. Pl (SUS)	16	16	16	16	22	22	22	22
Long. Q (SUS)	182	-182	182	-182	57	-57	57	-57
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	16	16	-16	-16	-12	-12	12	12
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	1294	1489	1294	1489	1294	1489	1294	1489
Pm+Pl (SUS)	1316	1511	1316	1511	1310	1505	1310	1505
Pm+Pl+Q (Total)	1373	1454	1373	1454	1492	1323	1492	1323

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1489.00	1509.10	Passed
Pm+Pl (SUS)	1511.30	2263.65	Passed
Pm+Pl+Q (TOTAL)	1492.19	4527.29	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	208	442	208	442	208	442	208	442
Circ. Pl (SUS)	22	22	22	22	22	22	22	22

Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	104	104	104	104	104	104	104	104
Long. Pl (SUS)	4	4	4	4	4	4	4	4
Long. Q (SUS)	25	-25	25	-25	25	-25	25	-25
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	7	7	-7	-7	-5	-5	5	5
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	208.0	442.0	208.0	442.0	208.0	442.0	208.0	442.0
Pm+Pl (SUS)	230.4	464.1	230.4	464.1	230.2	464.1	230.2	464.1
Pm+Pl+Q (Total)	230.5	464.1	230.5	464.1	230.3	464.1	230.3	464.1

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	442.00	1509.10	Passed
Pm+Pl (SUS)	464.14	2263.65	Passed
Pm+Pl+Q (TOTAL)	464.13	4527.30	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: B1-Tube Outlet

From: 20

Pressure for Reinforcement Calculations	P	194.775	kgf/cm ²
Temperature for Internal Pressure	Temp	310	°C
Parent Material		SA-387 11	+IT
Parent Allowable Stress at Temperature	Sv	1509.10	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1509.10	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	1220.00	mm
Shell Finished (Minimum) Thickness	t	88.0000	mm
Shell Internal Corrosion Allowance	c	2.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		70.00	cm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

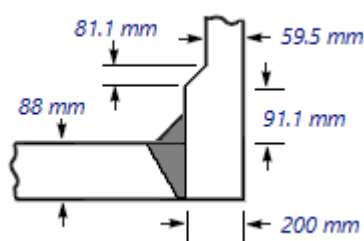
Material		SA-336 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1509.10	kgf/cm ²
Allowable Stress At Ambient	Sna	1509.10	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		600.0000	mm
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Hub Height of Integral Nozzle	h	91.1106	mm
Height of Beveled Transition	L'	81.1106	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		200.0000	mm
Corrosion Allowance	can	2.0000	mm
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	88.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: B1-Tube Outlet

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	490.576	mm.
Actual Thickness Used in Calculation	59.512	mm



Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 494.576 \cdot (\exp(194.775/(1509.1 \cdot 1.0)) - 1) + 2.0 + 0.0$
 $= 34.068 + 2.0 + 0.0 = 36.068 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 1509.1 \cdot 1.0 \cdot \ln((2 \cdot 57.512 + 494.576) / 494.576) - 0.0$
 $= 315.557 - 0.0 = 315.5575 \text{ kgf/cm}^2$

This nozzle or its diameter limit intersects weld seam # [2] on this element. The distance between the nozzle and seam is, 174.01 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 674.70 mm

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 10.340 mm
 Min. required shell thickness, internal p + c : 86.438 mm

Nozzle neck minimum required thickness : 10.340 mm

Minimum nozzle neck thickness 52.070 mm , is greater than required thickness 10.340 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 414.3,	Allowable	: 1509.1 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 3358.5 kgf/cm ²	Passed
Occasional	: 436.7,	Allowable	: 2007.1 kgf/cm ²	Passed
Shear	: 63.1,	Allowable	: 1056.4 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[1509.1/1509.1, 1]$
 $= 1.000$

Thickness of Nozzle at Shell [tn]:
 $= \text{hub thickness} - \text{corrosion allowance}$
 $= 200.0 - 2.0$
 $= 198.000 \text{ mm}$

Thickness of Nozzle at Top [tn2]:
 $= \text{thickness} - \text{corrosion allowance}$
 $= 59.512 - 2.0$
 $= 57.512 \text{ mm}$

Shell Diameter to Thickness ratio [D/t]:
 $= D_i/t$
 $= 1224.0/86.0$
 $= 14.233 \text{ must be less than } 400.$

Effective Pressure Radius [Reff]:
 $= D_i/2 + \text{corrosion allowance}$
 $= 1220.0/2 + 2.0$
 $= 612.000 \text{ mm}$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{(\text{Reff} \cdot t)}, 2 \cdot R_n)$$

$$= \min(\sqrt{(612.0 \cdot 86.0)}, 2 \cdot 247.288)$$

$$= 229.417 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$

$$= \min[1.5 \cdot 86.0, 0.0] + \sqrt{247.288 \cdot 198.0}$$

$$= 221.276 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 300.0$$

$$= 300.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(86.0 + 0.0)$$

$$= 688.000 \text{ mm}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles}$$

$$= \min[221.276, 300.0, 688.0] + 86.0$$

$$= 307.276 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot \text{frp}) / LR)$$

$$= 86.0 + ((0.0 \cdot 1.0) / 229.417)$$

$$= 86.000 \text{ mm}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + \text{teff}) \cdot \text{teff}}))$$

$$= \min(12, (2 \cdot 247.29 + 198.0) / (\sqrt{((1224.0 + 86.0) \cdot 86.0)}))$$

$$= 2.063$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 86.0 \cdot 229.417 \cdot \max((2.063/5)^{0.85}, 1)$$

$$= 197.298 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= A_{2A} + A_{2C}$$

$$= 350.679 + 53.498$$

$$= 404.177 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$

$$= 197.298 + 1.0(404.177 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 601.975 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 198.0 / \ln[1 + 198.0/247.288]$$

$$= 336.639 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= \text{teff} / \ln[1 + \text{teff}/\text{Reff}]$$

$$= 86.0 / \ln[1 + 86.0/612.0]$$

$$= 654.058 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$



$$= 1.5 \cdot 1509.097 \cdot 1.0$$

$$= 2263.6 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 194.775 \cdot 336.639 \cdot 307.276$$

$$= 201476.9 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (L_R + t_n)$$

$$= 194.775 \cdot 654.058 (229.417 + 198.0)$$

$$= 544503.9 \text{ kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 194.775 \cdot 654.058 \cdot 247.288$$

$$= 315030.2 \text{ kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (544504 + 315030.2 + 201476.9) / 194.775$$

$$= 5447.4 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$$

$$= 2263.646 / (2 \cdot 5447.369 / 601.975 - 654.058 / 86.0)$$

$$= 215.7 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S / t / R_{xs}$$

$$= 1509.097 [86.0 / 654.058]$$

$$= 198.4 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(215.729, 198.426)$$

$$= 198.426 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / A_T$$

$$= (201476.938 + 544504 + 315030.219) / 601.975$$

$$= 1762.552 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 194.775 \cdot 654.058 / 86.0$$

$$= 1481.3 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 1762.552 - 1481.33, 1481.33)$$

$$= 2043.8 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2263.65	kgf/cm ²
Local Primary Membrane Stress	PL	2043.77	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	198.43	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (R_{nc} + t_n) / R_{nc}$$

$$= (247.288 + 198.0) / 247.288$$

$$= 1.801 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi / 2 \cdot (R_n + t_n)$$

$$= \pi / 2 \cdot (247.288 + 198.0)$$

$$= 699.456 \text{ mm}$$



Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:
= 7.071, 0.000, 0.000, mm

Required Weld Throat Dimensions :

tc, L41T = 7.071, must be $\geq \min(0.7 * 198.000, 6.000)$ Passed

Weld Load Value [fwelds]:

= $\min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$
= $\min(315030 \cdot 1.8, 1.5 \cdot 1509.1(404.177 + 0.0), \pi/4 \cdot 194.8 \cdot 247.29^2 \cdot 1.8^2)$
= 303322.500 kgf

Weld Stress Value [τ]:

= $fwelds / (L_t(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$
= $303322.5 / (699.456(0.49 \cdot 7.071 + 0.6 \cdot 86.0 + 0.49 \cdot 0.0))$
= 787.536 < or = to 1509.097 Weld Stress Passed

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	:	18 °C
User entered Minimum Design Temperature	:	-10 °C
The Curve for this Material is	:	B
From Figure 3.8M- with PWHT, the MDMT is	:	19 °C
Temperature reduction per Graph 3.13	:	1 °C
The computed stress ratio is	:	0.980

The computed thickness ratio [sr]:

= $(t_r \cdot E_s) / (t_{nom} - c_i - c_{ext})$
= $(84.313 \cdot 1.0) / (88.0 - 2.0 - 0.0)$
= 0.980

Warning: Impact Testing Is Mandatory!

Gov. MDMT of the nozzle to shell joint welded assembly : 18 °C

The Drop for this Nozzle is : 193.0841 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 581.0842 mm

Input Echo, WRC297 Item 1, Description: B1-Tube Outlet:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	2.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1220.000	mm
Shell Thickness	88.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-387 11	
Vessel Cold S.I. Allowable	Smc 1509.10	kgf/cm ²
Vessel Hot S.I. Allowable	Smh 1509.10	kgf/cm ²
Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	2.0000	mm
Nozzle Diameter	490.576	mm
Nozzle Thickness	200.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-336 F11	
Nozzle Cold S.I. Allowable	SNmc 1509.10	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh 1509.10	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSTAINED loads.

Design Internal Pressure	Dp	194.78	kgf/cm ²
Radial Load	P	-19579.53	kgf



Circumferential Shear	Vc	19579.53	kgf
Longitudinal Shear	Vl	14684.63	kgf
Circumferential Moment	Mc	30.53	kgf-m.
Longitudinal Moment	Ml	30.53	kgf-m.
Torsional Moment	Mt	35.23	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, D_v/T_v (15.864) is < 20
check the limitations of WRC bulletin 297.

Warning - The ratio, D_n/D_v (.638) is > 0.5
check the limitations of WRC bulletin 297.

Warning - The ratio, D_n/T_n (4.453) is < 10
check the limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - c_a))}$$

$$= 890.576 + 2 \cdot 1.65 \cdot \sqrt{(655.0(88.0 - 2.0))}$$

$$= 1673.796 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1310.000 mm
Nozzle Outside Diameter	(d) =	890.576 mm
Vessel Thickness used	(T) =	86.000 mm
Nozzle Thickness used	(t) =	198.000 mm
T / t	=	0.434
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.653$$

Nr/P	=	0.062
Mr/P	=	0.115
M0/P	=	0.036
N0/P	=	0.084
MrD/Mc	=	0.271
NrDL/Mc	=	0.099
M0d/Mc	=	0.082
N0DL/Mc	=	0.082
MrD/Ml	=	0.098
NrDL/Ml	=	0.056
M0D/Ml	=	0.028
N0DL/Ml	=	0.099

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	22	22	22	22
Outplane Bending (P)	57	-57	57	-57
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	0	0	0	0
Outplane Bending (Ml)	0	0	0	0
Normal Pressure Stress	1295	1489	1295	1489
Outplane Stress Summary	1374	1454	1374	1454

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	16	16	16	16
Inplane Bending (P)	182	-182	182	-182



Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	647	647	647	647

Inplane Stress Summary	846	481	846	481
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	16	16	-16	-16
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0

Shear Stress Summary	16	16	-16	-16
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	1374	1455	1374	1454
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	22	22	22	22
Outplane Bending (P)	57	-57	57	-57
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	647	647	647	647

Outplane Stress Summary	726	612	726	612
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	16	16	16	16
Inplane Bending (P)	182	-182	182	-182
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1295	1489	1295	1489

Inplane Stress Summary	1493	1324	1494	1323
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-12	-12	12	12
Torsional Shear (Mt)	0	0	0	0

Shear Stress Summary	-12	-12	12	12
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stress Intensities)	Left	Left	Right	Right



	Outside	Inside	Outside	Inside
Two * Max Shear Stress	1493	1324	1494	1323

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	22	22	22	22
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	208	442	208	442
Hoop Stress Summary	230	464	230	464

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	4	4	4	4
Axial Bending (P)	25	-25	25	-25
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	104	104	104	104
Axial Stress Summary	133	83	133	83

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	7	7	-7	-7
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	7	7	-7	-7

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	230	464	230	464

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	22	22	22	22
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	208	442	208	442
Hoop Stress Summary	230	464	230	464

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Axial Membrane (P)	4	4	4	4
Axial Bending (P)	25	-25	25	-25
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	104	104	104	104

Axial Stress Summary	133	83	133	83
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-5	-5	5	5
Shear due to Torsion	0	0	0	0
Shear Stress Summary	-5	-5	5	5

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	230	464	230	464

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1295	1489	1295	1489	1295	1489	1295	1489
Circ. Pl (SUS)	22	22	22	22	16	16	16	16
Circ. Q (SUS)	57	-57	57	-57	182	-182	182	-182
Long. Pm (SUS)	647	647	647	647	647	647	647	647
Long. Pl (SUS)	16	16	16	16	22	22	22	22
Long. Q (SUS)	182	-182	182	-182	57	-57	57	-57
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	16	16	-16	-16	-12	-12	12	12
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	1295	1489	1295	1489	1295	1489	1295	1489
Pm+Pl (SUS)	1317	1511	1317	1511	1311	1505	1311	1505
Pm+Pl+Q (Total)	1374	1454	1374	1454	1493	1323	1493	1323

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1489.00	1509.10	Passed
Pm+Pl (SUS)	1511.30	2263.65	Passed
Pm+Pl+Q (TOTAL)	1493.19	4527.29	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):**

Type of Stress Int.	Stress Values at
------------------------	------------------



Location	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	208	442	208	442	208	442	208	442
Circ. Pl (SUS)	22	22	22	22	22	22	22	22
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	104	104	104	104	104	104	104	104
Long. Pl (SUS)	4	4	4	4	4	4	4	4
Long. Q (SUS)	25	-25	25	-25	25	-25	25	-25
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	7	7	-7	-7	-5	-5	5	5
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	208.0	442.0	208.0	442.0	208.0	442.0	208.0	442.0
Pm+Pl (SUS)	230.4	464.1	230.4	464.1	230.2	464.1	230.2	464.1
Pm+Pl+Q (Total)	230.5	464.1	230.5	464.1	230.3	464.1	230.3	464.1

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	442.00	1509.10	Passed
Pm+Pl (SUS)	464.14	2263.65	Passed
Pm+Pl+Q (TOTAL)	464.13	4527.30	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: A2-Shell inlet

From: 30

Pressure for Reinforcement Calculations	P	85.658	kgf/cm ²
Temperature for Internal Pressure	Temp	280	°C
Parent Material		SA-387 11	+IT
Parent Allowable Stress at Temperature	Sv	1509.10	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1509.10	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	1220.00	mm
Shell Finished (Minimum) Thickness	t	38.0000	mm
Shell Internal Corrosion Allowance	c	2.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		237.50	cm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

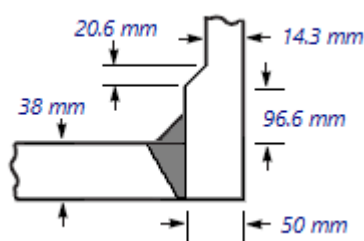
Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1343.95	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.13	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		150.0000	mm
Size and Thickness Basis		Nominal	
Nominal Thickness		120	
Hub Height of Integral Nozzle	h	96.6259	mm
Height of Beveled Transition	L'	20.6259	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm
Corrosion Allowance	can	2.0000	mm
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	38.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: A2-Shell inlet

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	139.725	mm.
Actual Thickness Used in Calculation	14.275	mm



Nozzle input data check completed without errors.



Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 143.725 \cdot (\exp(85.658/(1343.949 \cdot 1.0)) - 1) + 2.0 + 0.0$$

$$= 4.7294 + 2.0 + 0.0 = 6.7294 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1343.949 \cdot 1.0 \cdot \ln((2 \cdot 12.275 + 143.725) / 143.725) - 0.0$$

$$= 211.934 - 0.0 = 211.9342 \text{ kgf/cm}^2$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 8.220 mm

Min. required shell thickness, internal p + c : 37.753 mm

Nozzle neck minimum required thickness : 8.220 mm

Minimum nozzle neck thickness 12.497 mm, is greater than required thickness 8.220 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 202.3,	Allowable	: 1343.9 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 3236.6 kgf/cm ²	Passed
Occasional	: 273.4,	Allowable	: 1787.5 kgf/cm ²	Passed
Shear	: 194.2,	Allowable	: 940.8 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[1343.9/1509.1, 1]$$

$$= 0.891$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 50.0 - 2.0$$

$$= 48.000 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 14.275 - 2.0$$

$$= 12.275 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 1224.0/36.0$$

$$= 34.000 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 1220.0/2 + 2.0$$

$$= 612.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{(Reff \cdot t)}, 2 \cdot R_n)$$

$$= \min(\sqrt{(612.0 \cdot 36.0)}, 2 \cdot 71.863)$$

$$= 143.725 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)}$$

$$= \min[1.5 \cdot 36.0, 0.0] + \sqrt{(71.863 \cdot 48.0)}$$

$$= 58.732 \text{ mm}$$



Thickness Limit Candidate [LH2]:

$$\begin{aligned} &= L_{pr1} \\ &= 200.0 \\ &= 200.000 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned} &= 8(t + t_e) \\ &= 8(36.0 + 0.0) \\ &= 288.000 \text{ mm} \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles} \\ &= \min[58.732, 200.0, 288.0] + 36.0 \\ &= 94.732 \text{ mm} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A5 \cdot frp) / LR) \\ &= 36.0 + ((0.0 \cdot 1.0) / 143.725) \\ &= 36.000 \text{ mm} \end{aligned}$$

Parameter [λ]:

$$\begin{aligned} &= \min\left(12, \left(2 \cdot R_n + t_n\right) / \left(\sqrt{\left(D_i + t_{eff}\right) \cdot t_{eff}}\right)\right) \\ &= \min\left(12, \left(2 \cdot 71.86 + 48.0\right) / \left(\sqrt{\left(1224.0 + 36.0\right) \cdot 36.0}\right)\right) \\ &= 0.900 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \cdot \max\left(\left(\lambda/5\right)^{0.85}, 1\right) \\ &= 36.0 \cdot 143.725 \cdot \max\left(\left(0.9/5\right)^{0.85}, 1\right) \\ &= 51.741 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= t_n \cdot LH \\ &= 48.0 \cdot 94.732 \\ &= 45.471 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg}^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 0.500 \text{ cm}^2 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned} &= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 \\ &= 51.741 + 0.891(45.471 + 0.0) + 0.5 + 0.0 + 0.0 \\ &= 92.736 \text{ cm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 48.0 / \ln[1 + 48.0/71.863] \\ &= 93.825 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 36.0 / \ln[1 + 36.0/612.0] \\ &= 629.828 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 1509.097 \cdot 1.0 \\ &= 2263.6 \text{ kgf/cm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn} \cdot LH \\ &= 85.658 \cdot 93.825 \cdot 94.732 \\ &= 7613.5 \text{ kgf} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot R_{xs}(LR + t_n) \\ &= 85.658 \cdot 629.828(143.725 + 48.0) \end{aligned}$$



= 103435.5 kgf

Discontinuity Force from Internal Pressure [fY]:

= $P \cdot Rxs \cdot Rnc$
= $85.658 \cdot 629.828 \cdot 71.863$
= 38769.8 kgf

Area Resisting Internal Pressure [Ap]:

= $(fS + fY + fN) / P$
= $(103435.5 + 38769.8 + 7613.5) / 85.658$
= 1749.0 cm²

Maximum Allowable Working Pressure Candidate [Pmax1]:

= $Sallow / (2 \cdot Ap / AT - Rxs / teff)$
= $2263.646 / (2 \cdot 1749.034 / 92.736 - 629.828 / 36.0)$
= 111.9 kgf/cm²

Maximum Allowable Working Pressure Candidate [Pmax2]:

= $S[t / Rxs]$
= $1509.097 [36.0 / 629.828]$
= 86.3 kgf/cm²

Maximum Allowable Working Pressure [Pmax]:

= $\min(Pmax1, Pmax2)$
= $\min(111.921, 86.258)$
= 86.258 kgf/cm²

Average Primary Membrane Stress [σ_{Avg}]:

= $(fN + fS + fY) / AT$
= $(7613.475 + 103435.531 + 38769.805) / 92.736$
= 1615.540 kgf/cm²

General Primary Membrane Stress [σ_{Circ}]:

= $P \cdot Rxs / teff$
= $85.658 \cdot 629.828 / 36.0$
= 1498.6 kgf/cm²

Maximum Local Primary Membrane Stress [PL]:

= $\max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$
= $\max(2 \cdot 1615.54 - 1498.609, 1498.609)$
= 1732.5 kgf/cm²

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2263.65	kgf/cm ²
Local Primary Membrane Stress	PL	1732.47	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	86.26	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

= $(Rnc + tn) / Rnc$
= $(71.863 + 48.0) / 71.863$
= 1.668 For set-in Nozzles

Weld Length of Nozzle to Shell Weld [Lt]:

= $\pi / 2 \cdot (Rn + tn)$
= $\pi / 2 \cdot (71.863 + 48.0)$
= 188.280 mm

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

= 7.071, 0.000, 0.000, mm

Required Weld Throat Dimensions :

tc, L41T = 7.071, must be $\geq \min(0.7 \cdot 48.000, 6.000)$ Passed

Weld Load Value [fwelds]:

= $\min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi / 4 \cdot P \cdot Rn^2 \cdot ky^2)$
= $\min(38770 \cdot 1.67, 1.5 \cdot 1343.9(45.471 + 0.0), \pi / 4 \cdot 85.7 \cdot 71.86^2 \cdot 1.67^2)$
= 9665.541 kgf

Weld Stress Value [τ]:
 $= \text{fwelds} / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$
 $= 9665.541 / (188.28(0.49 \cdot 7.071 + 0.6 \cdot 36.0 + 0.49 \cdot 0.0))$
 $= 204.814 < \text{or} = \text{to } 1509.097 \text{ Weld Stress Passed}$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Not Post Weld Heat Treated
 Computed Minimum Design Metal Temperature : 22 °C
 User entered Minimum Design Temperature : -10 °C
 The Curve for this Material is : B
 From Figure 3.7M- Non PWHT, the MDMT is : 22 °C
 Temperature reduction per Graph 3.12 : 1 °C
 The computed stress ratio is : 0.993

The computed thickness ratio [sr]:
 $= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$
 $= (35.743 \cdot 1.0) / (38.0 - 2.0 - 0.0)$
 $= 0.993$

Warning: Impact Testing Is Mandatory!

Gov. MDMT of the nozzle to shell joint welded assembly : 22 °C

The Drop for this Nozzle is : 11.8922 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 249.8922 mm

Input Echo, WRC297 Item 1, Description: A2-Shell inlet:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	2.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1220.000	mm
Shell Thickness	38.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-387 11	
Vessel Cold S.I. Allowable	Smc 1509.10	kgf/cm ²
Vessel Hot S.I. Allowable	Smh 1509.10	kgf/cm ²
Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	2.0000	mm
Nozzle Diameter	139.725	mm
Nozzle Thickness	50.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	
Nozzle Cold S.I. Allowable	SNmc 1407.13	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh 1343.95	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSTAINED loads.

Design Internal Pressure	Dp	85.66	kgf/cm ²
Radial Load	P	-3671.16	kgf
Circumferential Shear	Vc	3671.16	kgf
Longitudinal Shear	Vl	2753.37	kgf
Circumferential Moment	Mc	1.43	kgf-m.
Longitudinal Moment	Ml	1.43	kgf-m.
Torsional Moment	Mt	1.65	kgf-m.
Include Axial Pressure Thrust		No	
Include Pressure Stress Indices per Div. 2		No	
Local Loads applied at end of Nozzle/Attachment		No	

Warning - The ratio, D_n/T_n (4.795) is < 10



check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - c_a))}$$

$$= 239.725 + 2 \cdot 1.65 \cdot \sqrt{(630.0(38.0 - 2.0))}$$

$$= 736.701 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1260.000 mm
Nozzle Outside Diameter	(d) =	239.725 mm
Vessel Thickness used	(T) =	36.000 mm
Nozzle Thickness used	(t) =	48.000 mm
T / t	=	0.750
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.126$$

Nr/P	=	0.109
Mr/P	=	0.191
M0/P	=	0.060
N0/P	=	0.122
MrD/Mc	=	0.291
NrDL/Mc	=	0.119
M0d/Mc	=	0.100
N0DL/Mc	=	0.042
MrD/M1	=	0.182
NrDL/M1	=	0.096
M0D/M1	=	0.063
N0DL/M1	=	0.125

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	34	34	34	34
Outplane Bending (P)	102	-102	102	-102
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	1414	1500	1414	1500
Outplane Stress Summary	1551	1432	1551	1432

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	30	30	30	30
Inplane Bending (P)	324	-324	324	-324
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	707	707	707	707
Inplane Stress Summary	1062	413	1063	412

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	27	27	-27	-27
Outplane Shear (Vl)	0	0	0	0



Outplane Shear (Mt)	0	0	0	0
---------------------	---	---	---	---

Shear Stress Summary	27	27	-27	-27
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Two * Max Shear Stress	1552	1433	1552	1432
------------------------	------	------	------	------

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Membrane (P)	34	34	34	34
Outplane Bending (P)	102	-102	102	-102
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	707	707	707	707

Outplane Stress Summary	843	639	844	639
-------------------------	-----	-----	-----	-----

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Inplane Membrane (P)	30	30	30	30
Inplane Bending (P)	324	-324	324	-324
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1414	1500	1414	1500

Inplane Stress Summary	1769	1207	1770	1205
------------------------	------	------	------	------

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-20	-20	20	20
Torsional Shear (Mt)	0	0	0	0

Shear Stress Summary	-20	-20	20	20
----------------------	-----	-----	----	----

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Two * Max Shear Stress	1769	1207	1771	1206
------------------------	------	------	------	------

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
---	----------------------	---------------------	-------------------------	------------------------

Hoop Membrane (P)	34	34	34	34
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0



Hoop Pressure Stress	115	218	115	218
----------------------	-----	-----	-----	-----

Hoop Stress Summary	149	252	149	252
---------------------	-----	-----	-----	-----

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Axial Membrane (P)	12	12	12	12
Axial Bending (P)	130	-130	130	-130
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	57	57	57	57

Axial Stress Summary	200	-60	201	-60
----------------------	-----	-----	-----	-----

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
--------------------------------------	---------------	--------------	---------------	--------------

Shear due to (Vc)	20	20	-20	-20
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary	20	20	-20	-20
----------------------	----	----	-----	-----

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
---	---------------	--------------	---------------	--------------

Two * Max Shear Stress	207	315	208	315
------------------------	-----	-----	-----	-----

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Hoop Membrane (P)	34	34	34	34
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	115	218	115	218

Hoop Stress Summary	149	252	149	252
---------------------	-----	-----	-----	-----

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Axial Membrane (P)	12	12	12	12
Axial Bending (P)	130	-130	130	-130
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	57	57	57	57

Axial Stress Summary	200	-59	201	-60
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
---	---------------	--------------	---------------	--------------

Shear due to (Vc)	0	0	0	0
-------------------	---	---	---	---



Shear due to (V1)	-15	-15	15	15
Shear due to Torsion	0	0	0	0

Shear Stress Summary	-15	-15	15	15
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	204	313	205	314

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1414	1500	1414	1500	1414	1500	1414	1500
Circ. Pl (SUS)	34	34	34	34	30	30	30	30
Circ. Q (SUS)	102	-102	102	-102	324	-324	324	-324
Long. Pm (SUS)	707	707	707	707	707	707	707	707
Long. Pl (SUS)	30	30	30	30	34	34	34	34
Long. Q (SUS)	324	-324	324	-324	102	-102	102	-102
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	27	27	-27	-27	-20	-20	20	20
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	1414	1500	1414	1500	1414	1500	1414	1500
Pm+Pl (SUS)	1449	1534	1449	1534	1444	1530	1444	1530
Pm+Pl+Q (Total)	1551	1432	1551	1432	1768	1206	1768	1206

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1500.00	1509.10	Passed
Pm+Pl (SUS)	1534.91	2263.65	Passed
Pm+Pl+Q (TOTAL)	1768.43	4527.29	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	115	218	115	218	115	218	115	218
Circ. Pl (SUS)	34	34	34	34	34	34	34	34
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	57	57	57	57	57	57	57	57
Long. Pl (SUS)	12	12	12	12	12	12	12	12
Long. Q (SUS)	130	-130	130	-130	130	-130	130	-130
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	20	20	-20	-20	-15	-15	15	15
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	115.0	218.0	115.0	218.0	115.0	218.0	115.0	218.0



Pm+Pl (SUS)	153.7	254.2	153.7	254.2	151.7	253.2	151.7	253.2
Pm+Pl+Q (Total)	206.0	315.5	206.0	315.5	203.2	314.4	203.2	314.4

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	218.00	1343.95	Passed
Pm+Pl (SUS)	254.16	2015.92	Passed
Pm+Pl+Q (TOTAL)	315.55	4031.85	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: B2-Shell outlet

From: 30

Pressure for Reinforcement Calculations	P	85.780	kgf/cm ²
Temperature for Internal Pressure	Temp	280	°C
Parent Material		SA-387 11	+IT
Parent Allowable Stress at Temperature	Sv	1509.10	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1509.10	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	1220.00	mm
Shell Finished (Minimum) Thickness	t	38.0000	mm
Shell Internal Corrosion Allowance	c	2.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		237.50	cm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

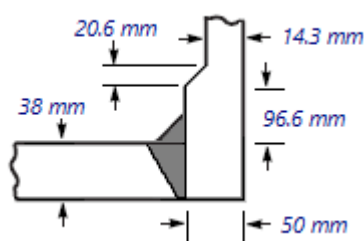
Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1343.95	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.13	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		150.0000	mm
Size and Thickness Basis		Nominal	
Nominal Thickness		120	
Hub Height of Integral Nozzle	h	96.6259	mm
Height of Beveled Transition	L'	20.6259	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm
Corrosion Allowance	can	2.0000	mm
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	38.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: B2-Shell outlet

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	139.725	mm.
Actual Thickness Used in Calculation	14.275	mm



Nozzle input data check completed without errors.



Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 143.725 \cdot (\exp(85.78/(1343.949 \cdot 1.0)) - 1) + 2.0 + 0.0$$

$$= 4.7363 + 2.0 + 0.0 = 6.7363 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1343.949 \cdot 1.0 \cdot \ln((2 \cdot 12.275 + 143.725) / 143.725) - 0.0$$

$$= 211.934 - 0.0 = 211.9342 \text{ kgf/cm}^2$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 8.220 mm

Min. required shell thickness, internal p + c : 37.805 mm

Nozzle neck minimum required thickness : 8.220 mm

Minimum nozzle neck thickness 12.497 mm, is greater than required thickness 8.220 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 202.7,	Allowable	: 1343.9 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 3236.2 kgf/cm ²	Passed
Occasional	: 273.8,	Allowable	: 1787.5 kgf/cm ²	Passed
Shear	: 194.2,	Allowable	: 940.8 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[1343.9/1509.1, 1]$$

$$= 0.891$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 50.0 - 2.0$$

$$= 48.000 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 14.275 - 2.0$$

$$= 12.275 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 1224.0/36.0$$

$$= 34.000 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 1220.0/2 + 2.0$$

$$= 612.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{(Reff \cdot t)}, 2 \cdot R_n)$$

$$= \min(\sqrt{(612.0 \cdot 36.0)}, 2 \cdot 71.863)$$

$$= 143.725 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)}$$

$$= \min[1.5 \cdot 36.0, 0.0] + \sqrt{(71.863 \cdot 48.0)}$$

$$= 58.732 \text{ mm}$$



Thickness Limit Candidate [LH2]:

$$\begin{aligned} &= L_{pr1} \\ &= 200.0 \\ &= 200.000 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned} &= 8(t + t_e) \\ &= 8(36.0 + 0.0) \\ &= 288.000 \text{ mm} \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles} \\ &= \min[58.732, 200.0, 288.0] + 36.0 \\ &= 94.732 \text{ mm} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A5 \cdot frp) / LR) \\ &= 36.0 + ((0.0 \cdot 1.0) / 143.725) \\ &= 36.000 \text{ mm} \end{aligned}$$

Parameter [λ]:

$$\begin{aligned} &= \min\left(12, \left(2 \cdot R_n + t_n\right) / \left(\sqrt{\left(D_i + t_{eff}\right) \cdot t_{eff}}\right)\right) \\ &= \min\left(12, \left(2 \cdot 71.86 + 48.0\right) / \left(\sqrt{\left(1224.0 + 36.0\right) \cdot 36.0}\right)\right) \\ &= 0.900 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \cdot \max\left(\left(\lambda/5\right)^{0.85}, 1\right) \\ &= 36.0 \cdot 143.725 \cdot \max\left(\left(0.9/5\right)^{0.85}, 1\right) \\ &= 51.741 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= t_n \cdot LH \\ &= 48.0 \cdot 94.732 \\ &= 45.471 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg}^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 0.500 \text{ cm}^2 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned} &= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 \\ &= 51.741 + 0.891(45.471 + 0.0) + 0.5 + 0.0 + 0.0 \\ &= 92.736 \text{ cm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 48.0 / \ln[1 + 48.0/71.863] \\ &= 93.825 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 36.0 / \ln[1 + 36.0/612.0] \\ &= 629.828 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 1509.097 \cdot 1.0 \\ &= 2263.6 \text{ kgf/cm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn} \cdot LH \\ &= 85.78 \cdot 93.825 \cdot 94.732 \\ &= 7624.3 \text{ kgf} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot R_{xs}(LR + t_n) \\ &= 85.78 \cdot 629.828(143.725 + 48.0) \end{aligned}$$



= 103582.8 kgf

Discontinuity Force from Internal Pressure [fY]:

= $P \cdot Rxs \cdot Rnc$
= $85.78 \cdot 629.828 \cdot 71.863$
= 38825.0 kgf

Area Resisting Internal Pressure [Ap]:

= $(fS + fY + fN) / P$
= $(103582.8 + 38825.0 + 7624.3) / 85.78$
= 1749.0 cm²

Maximum Allowable Working Pressure Candidate [Pmax1]:

= $Sallow / (2 \cdot Ap / AT - Rxs / teff)$
= $2263.646 / (2 \cdot 1749.034 / 92.736 - 629.828 / 36.0)$
= 111.9 kgf/cm²

Maximum Allowable Working Pressure Candidate [Pmax2]:

= $S[t/Rxs]$
= $1509.097[36.0/629.828]$
= 86.3 kgf/cm²

Maximum Allowable Working Pressure [Pmax]:

= $\min(Pmax1, Pmax2)$
= $\min(111.921, 86.258)$
= 86.258 kgf/cm²

Average Primary Membrane Stress [σAvg]:

= $(fN + fS + fY) / AT$
= $(7624.313 + 103582.789 + 38825.0) / 92.736$
= 1617.840 kgf/cm²

General Primary Membrane Stress [σCirc]:

= $P \cdot Rxs / teff$
= $85.78 \cdot 629.828 / 36.0$
= 1500.7 kgf/cm²

Maximum Local Primary Membrane Stress [PL]:

= $\max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ)$
= $\max(2 \cdot 1617.84 - 1500.742, 1500.742)$
= 1734.9 kgf/cm²

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2263.65	kgf/cm ²
Local Primary Membrane Stress	PL	1734.94	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	86.26	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

= $(Rnc + tn) / Rnc$
= $(71.863 + 48.0) / 71.863$
= 1.668 For set-in Nozzles

Weld Length of Nozzle to Shell Weld [Lt]:

= $\pi/2 \cdot (Rn + tn)$
= $\pi/2 \cdot (71.863 + 48.0)$
= 188.280 mm

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

= 7.071, 0.000, 0.000, mm

Required Weld Throat Dimensions :

tc, L41T = 7.071, must be $\geq \min(0.7 \cdot 48.000, 6.000)$ Passed

Weld Load Value [fwelds]:

= $\min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2)$
= $\min(38825 \cdot 1.67, 1.5 \cdot 1343.9(45.471 + 0.0), \pi/4 \cdot 85.8 \cdot 71.86^2 \cdot 1.67^2)$
= 9679.301 kgf

Weld Stress Value [τ]:
 $= \text{fwelds} / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$
 $= 9679.301 / (188.28(0.49 \cdot 7.071 + 0.6 \cdot 36.0 + 0.49 \cdot 0.0))$
 $= 205.105 < \text{or} = \text{to } 1509.097 \text{ Weld Stress Passed}$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Not Post Weld Heat Treated
 Computed Minimum Design Metal Temperature : 22 °C
 User entered Minimum Design Temperature : -10 °C
 The Curve for this Material is : B
 From Figure 3.7M- Non PWHT, the MDMT is : 22 °C
 Temperature reduction per Graph 3.12 : 1 °C
 The computed stress ratio is : 0.994

The computed thickness ratio [sr]:
 $= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$
 $= (35.795 \cdot 1.0) / (38.0 - 2.0 - 0.0)$
 $= 0.994$

Warning: Impact Testing Is Mandatory!

Gov. MDMT of the nozzle to shell joint welded assembly : 22 °C

The Drop for this Nozzle is : 11.8922 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 249.8922 mm

Input Echo, WRC297 Item 1, Description: B2-Shell outle:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	2.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1220.000	mm
Shell Thickness	38.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-387 11	
Vessel Cold S.I. Allowable	Smc 1509.10	kgf/cm ²
Vessel Hot S.I. Allowable	Smh 1509.10	kgf/cm ²
Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	2.0000	mm
Nozzle Diameter	139.725	mm
Nozzle Thickness	50.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	
Nozzle Cold S.I. Allowable	SNmc 1407.13	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh 1343.95	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSTAINED loads.

Design Internal Pressure	Dp	85.78	kgf/cm ²
Radial Load	P	-3671.16	kgf
Circumferential Shear	Vc	3671.16	kgf
Longitudinal Shear	Vl	2753.37	kgf
Circumferential Moment	Mc	1.43	kgf-m.
Longitudinal Moment	Ml	1.43	kgf-m.
Torsional Moment	Mt	1.65	kgf-m.
Include Axial Pressure Thrust		No	
Include Pressure Stress Indices per Div. 2		No	
Local Loads applied at end of Nozzle/Attachment		No	

Warning - The ratio, D_n/T_n (4.795) is < 10



check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - c_a))}$$

$$= 239.725 + 2 \cdot 1.65 \cdot \sqrt{(630.0(38.0 - 2.0))}$$

$$= 736.701 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1260.000 mm
Nozzle Outside Diameter	(d) =	239.725 mm
Vessel Thickness used	(T) =	36.000 mm
Nozzle Thickness used	(t) =	48.000 mm
T / t	=	0.750
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.126$$

Nr/P	=	0.109
Mr/P	=	0.191
M0/P	=	0.060
N0/P	=	0.122
MrD/Mc	=	0.291
NrDL/Mc	=	0.119
M0d/Mc	=	0.100
N0DL/Mc	=	0.042
MrD/M1	=	0.182
NrDL/M1	=	0.096
M0D/M1	=	0.063
N0DL/M1	=	0.125

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Outplane Membrane (P)	34	34	34	34
Outplane Bending (P)	102	-102	102	-102
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	1416	1502	1416	1502
Outplane Stress Summary	1553	1434	1553	1434

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Inplane Membrane (P)	30	30	30	30
Inplane Bending (P)	324	-324	324	-324
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	708	708	708	708
Inplane Stress Summary	1063	414	1064	413

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Outplane Shear (Vc)	27	27	-27	-27
Outplane Shear (Vl)	0	0	0	0



Outplane Shear (Mt)	0	0	0	0
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Shear Stress Summary	27	27	-27	-27
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Two * Max Shear Stress	1554	1435	1554	1435
------------------------	------	------	------	------

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Membrane (P)	34	34	34	34
Outplane Bending (P)	102	-102	102	-102
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	708	708	708	708

Outplane Stress Summary	844	640	845	640
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Inplane Membrane (P)	30	30	30	30
Inplane Bending (P)	324	-324	324	-324
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1416	1502	1416	1502

Inplane Stress Summary	1771	1209	1772	1207
------------------------	------	------	------	------

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-20	-20	20	20
Torsional Shear (Mt)	0	0	0	0

Shear Stress Summary	-20	-20	20	20
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Two * Max Shear Stress	1771	1209	1773	1208
------------------------	------	------	------	------

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
---	----------------------	---------------------	-------------------------	------------------------

Hoop Membrane (P)	34	34	34	34
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0



Hoop Pressure Stress	115	218	115	218
----------------------	-----	-----	-----	-----

Hoop Stress Summary	149	252	149	252
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Axial Membrane (P)	12	12	12	12
Axial Bending (P)	130	-130	130	-130
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	57	57	57	57

Axial Stress Summary	200	-60	201	-60
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
--------------------------------------	---------------	--------------	---------------	--------------

Shear due to (Vc)	20	20	-20	-20
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary	20	20	-20	-20
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
---	---------------	--------------	---------------	--------------

Two * Max Shear Stress	207	315	208	315
------------------------	-----	-----	-----	-----

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Hoop Membrane (P)	34	34	34	34
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	115	218	115	218

Hoop Stress Summary	149	252	149	252
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Axial Membrane (P)	12	12	12	12
Axial Bending (P)	130	-130	130	-130
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	57	57	57	57

Axial Stress Summary	200	-59	201	-60
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
---	---------------	--------------	---------------	--------------

Shear due to (Vc)	0	0	0	0
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Shear due to (V1)	-15	-15	15	15
Shear due to Torsion	0	0	0	0

Shear Stress Summary	-15	-15	15	15
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	204	314	205	314

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1416	1502	1416	1502	1416	1502	1416	1502
Circ. Pl (SUS)	34	34	34	34	30	30	30	30
Circ. Q (SUS)	102	-102	102	-102	324	-324	324	-324
Long. Pm (SUS)	708	708	708	708	708	708	708	708
Long. Pl (SUS)	30	30	30	30	34	34	34	34
Long. Q (SUS)	324	-324	324	-324	102	-102	102	-102
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	27	27	-27	-27	-20	-20	20	20
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	1416	1502	1416	1502	1416	1502	1416	1502
Pm+Pl (SUS)	1451	1536	1451	1536	1446	1532	1446	1532
Pm+Pl+Q (Total)	1553	1434	1553	1434	1770	1208	1770	1208

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1502.00	1509.10	Passed
Pm+Pl (SUS)	1536.91	2263.65	Passed
Pm+Pl+Q (TOTAL)	1770.43	4527.29	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	115	218	115	218	115	218	115	218
Circ. Pl (SUS)	34	34	34	34	34	34	34	34
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	57	57	57	57	57	57	57	57
Long. Pl (SUS)	12	12	12	12	12	12	12	12
Long. Q (SUS)	130	-130	130	-130	130	-130	130	-130
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	20	20	-20	-20	-15	-15	15	15
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	115.0	218.0	115.0	218.0	115.0	218.0	115.0	218.0



Pm+Pl (SUS)	153.7	254.2	153.7	254.2	151.7	253.2	151.7	253.2
Pm+Pl+Q (Total)	206.0	315.5	206.0	315.5	203.2	314.4	203.2	314.4

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	218.00	1343.95	Passed
Pm+Pl (SUS)	254.16	2015.92	Passed
Pm+Pl+Q (TOTAL)	315.55	4031.85	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input Echo, Tubesheet Number 1, Description:

Shell Data:

Main Shell Description: Main Shell

Shell Maximum Design Pressure	Psd,max	85.66	kgf/cm ²
Shell Maximum Operating Pressure	Psox,max	85.66	kgf/cm ²
Shell Minimum Operating Pressure	Psox,min	0.00	kgf/cm ²
Shell Thickness	ts	38.0000	mm
Shell Internal Corrosion Allowance	cas	2.0000	mm
Shell External Corrosion Allowance	casext	0.0000	mm
Inside Diameter of Shell	Ds	1220.000	mm
Shell Temperature for Internal Pressure	Ts	280.00	°C
Shell Material		SA-387 11	

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Shell Material UNS Number		K11789	
Shell Allowable Stress at Temperature	Ss	1688.67	kgf/cm ²
Shell Allowable Stress at Ambient		2110.82	kgf/cm ²

Channel Data:

Channel Description: Ch Shell

Channel Type:		Cylinder	
Channel Maximum Design Pressure	Ptd,max	194.77	kgf/cm ²
Channel Maximum Operating Pressure	Ptox,max	194.77	kgf/cm ²
Channel Minimum Operating Pressure	Ptox,min	0.00	kgf/cm ²
Channel Thickness	tc	88.0000	mm
Channel Internal Corrosion Allowance	cac	2.0000	mm
Channel External Corrosion Allowance	cacext	0.0000	mm
Inside Diameter of Channel	Dc	1220.000	mm
Channel Design Temperature	TEMPC	310.00	°C
Channel Flange Matl		SA-387 11	

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Channel Material UNS Number		K11789	
Channel Allowable Stress at Temperature	Sc	1660.13	kgf/cm ²
Channel Allowable Stress at Ambient		2110.82	kgf/cm ²

Tube Data:

Number of Tube Holes	Nt	1012	
Tube Wall Thickness	et	2.7700	mm
Tube Outside Diameter	D	25.4000	mm
Total Straight Tube Length	Lt	3010.00	mm
Straight Tube Length (bet. inner tubsht faces) L		2710.00	mm
Design Temperature of the Tubes		310.00	°C
Tube Material		SA-213 T11	
Tube Material UNS Number		K11597	
Is this a Welded Tube		No	
Tube Material Specification used		Smls. tube	
Tube Allowable Stress at Temperature		1107.36	kgf/cm ²
Tube Allowable Stress At Ambient		1203.20	kgf/cm ²
Tube Yield Stress At design Temperature	Syt	1660.04	kgf/cm ²
Tube Pitch (Center to Center Spacing)	P	31.7500	mm
Tube Layout Pattern		Triangular	
Fillet Weld Leg	af	2.7700	mm
Groove Weld Leg	ag	2.7700	mm
Tube-Tubesheet Joint Weld Type		Full Strength	
Method for Tube-Tubesheet Jt. Allowable		4.21.2	
Is the Tube Inset		No	
Radius to Outermost Tube Hole Center	ro	570.524	mm
Largest Center-to-Center Tube Distance	Ul	0.0000	mm
Length of Expanded Portion of Tube	ltx	0.0000	mm
Tube-side pass partition groove depth	hg	6.0000	mm
Area of Tube Layout	Ap	8673.4	cm ²
Perimeter of Tube Layout	Cp	3535.2	mm



Tubesheet Data:

Tubesheet TYPE: U-tube, Integral both sides, Conf. a

Tubesheet Design Metal Temperature T 310.00 °C
 Tubesheet Material SA-336 F11

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

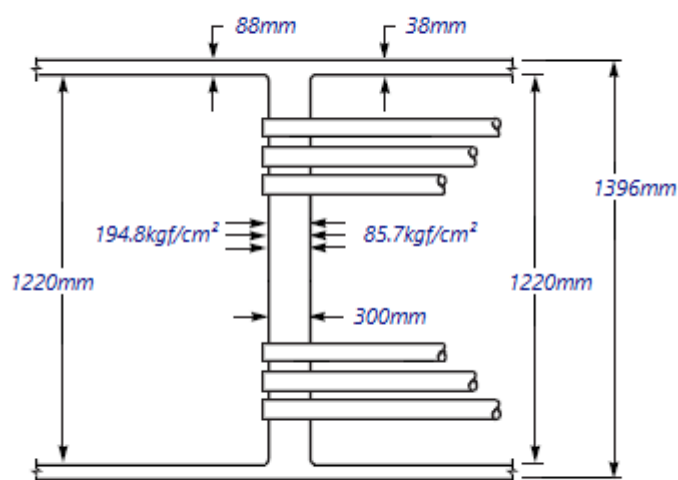
Tubesheet Material UNS Number K11572
 Tubesheet Allowable Stress at Temperature S 1509.10 kgf/cm²
 Tubesheet Allowable Stress at Ambient Tt 1509.10 kgf/cm²
 Thickness of Tubesheet h 300.0000 mm
 Tubesheet Corr. Allowance (Shell side) Cats 3.0000 mm
 Tubesheet Corr. Allowance (Channel side) Catc 3.0000 mm
 Tubesheet Outside Diameter A 1396.000 mm

Area of the Untubed Lanes AL 0.0 cm²

Junction Stress Reduction option Perform Plastic Calculation
 Perform Differential Pressure Design No
 Run Multiple Load Cases Yes

Tubesheet Integral with Both
 Tubesheet Extended as Flange No

Tubesheet Integral w/Shell and Channel



Configuration a:

ASME Division 2 Tubesheet Results per Part 4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Material Group C

Elastic Mod. at Design Temperature	280.0 °C	0.19138E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20811E+07 kgf/cm ²

Channel - TM-1 Material Group C

Elastic Mod. at Design Temperature	310.0 °C	0.18948E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20811E+07 kgf/cm ²

Tubes - TM-1 Material Group C



Elastic Mod. at Tubsht. Design Temp.	310.0 °C	0.18948E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20811E+07 kgf/cm ²

TubeSheet - TM-1 Material Group C

Elastic Mod. at Design Temperature	310.0 °C	0.18948E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20811E+07 kgf/cm ²

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + ci$ Equation 4.3.1
 $= 0.5 \cdot 19.86 \cdot (\exp(194.77/(1107.4 \cdot 1.0)) - 1) + 0$
 $= 1.9096 + 0$
 $= 1.9096 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

Determine Parameter [Mx]:

$= L / \sqrt{(Ro \cdot t)}$
 $= 3280.524 / \sqrt{(12.7 \cdot 2.77)}$
 $= 553.097$

Compute the Factor [Ch]:

Since $Mx > 2 \cdot (Do/t)^{0.94}$, $Ch = 0.55 / (Do/t)$
 $= 0.55 / (25.4/2.77)$
 $= 0.0600$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$= 1.6 \cdot Ch \cdot Ey(t / Do)$
 $= 1.6 \cdot 0.06 \cdot 1894790 (2.77/25.4)$
 $= 19830.605 \text{ kgf/cm}^2$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$= Fhe / Fy$
 $= 19830.6/1660.0$
 $= 11.946$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$= \sigma_{ys} / \sigma_{uts}$
 $= 1660.0/4218.4$
 $= 0.394$

Ratio [AE]:

$= Fhe / E$
 $= 19830.6/1894790$
 $= 0.010466$

Factor [m1]:

$= (\ln(r) + (\epsilon_{ps} - \epsilon_{sys})) / (\ln(\ln(1 + \epsilon_{ps})) / \ln(1 + \epsilon_{sys}))$
 $= (\ln(0.394) + (0.00002 - 0.002)) / (\ln(\ln(1 + 0.00002)) / \ln(1 + 0.002))$
 $= 0.203$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$= \sigma_{ys}(1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys}) m1$
 $= 1660.0 (1 + 0.002) / \ln(1 + 0.002) \cdot 0.202989$
 $= 5874.026 \text{ kgf/cm}^2$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$= m2f(m2f2 - R)$
 $= 0.6 (1.0 - 0.394)$
 $= 0.364$

Curve Fitting Constant [A2]:

$= (\sigma_{uts} \cdot \exp(m2)) / (m2 m2)$
 $= (4218.4 \cdot \exp(0.364)) / (0.364 \cdot 0.364)$
 $= 8768.844 \text{ kgf/cm}^2$



Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.3941.5 - 0.5 \cdot 0.3942.5 - 0.3943.5$$

$$= 0.283$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	2109	-0.761	0.752E-05	-0.393E-06	0.130E-04	0.782E-05	0.351E+05	0.495E-01
2	1054	-3.670	0.494E-06	0.493E-06	0.386E-05	-0.385E-05	0.653E+06	0.885E-02
3	1581	-2.215	0.243E-05	0.227E-05	0.785E-05	-0.709E-05	0.167E+06	0.985E-03
4	1845	-1.488	0.445E-05	0.317E-05	0.103E-04	-0.577E-05	0.790E+05	0.129E-01
5	1713	-1.852	0.333E-05	0.287E-05	0.903E-05	-0.714E-05	0.116E+06	0.429E-02
6	1647	-2.034	0.285E-05	0.258E-05	0.843E-05	-0.722E-05	0.140E+06	0.135E-02
7	1614	-2.125	0.263E-05	0.243E-05	0.814E-05	-0.718E-05	0.153E+06	0.113E-03
8	1598	-2.170	0.253E-05	0.235E-05	0.799E-05	-0.714E-05	0.160E+06	0.452E-03
9	1606	-2.147	0.258E-05	0.239E-05	0.807E-05	-0.716E-05	0.156E+06	0.173E-03
10	1610	-2.136	0.261E-05	0.241E-05	0.810E-05	-0.717E-05	0.154E+06	0.312E-04
11	1612	-2.130	0.262E-05	0.242E-05	0.812E-05	-0.717E-05	0.154E+06	0.407E-04
12	1611	-2.133	0.261E-05	0.241E-05	0.811E-05	-0.717E-05	0.154E+06	0.470E-05
13	1611	-2.135	0.261E-05	0.241E-05	0.811E-05	-0.717E-05	0.154E+06	0.132E-04
14	1611	-2.134	0.261E-05	0.241E-05	0.811E-05	-0.717E-05	0.154E+06	0.427E-05
15	1611	-2.133	0.261E-05	0.241E-05	0.811E-05	-0.717E-05	0.154E+06	0.216E-06
16	1611	-2.134	0.261E-05	0.241E-05	0.811E-05	-0.717E-05	0.154E+06	0.203E-05

Factor of Safety [FS]:

$$\text{Since } Fic > 0.55 Fy \text{ and } Fic < Fy$$

$$= 2.407 - 0.741 \cdot Fic / Fy$$

$$= 2.407 - 0.741 \cdot 1611.58/1660.04$$

$$= 1.688$$

Allowable Circumferential Compressive Stress [Fha]:

$$= Fic/FS$$

$$= 1611.581/1.688$$

$$= 954.936 \text{ kgf/cm}^2$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot Fha(t / Do)$$

$$= 2 \cdot 954.936 (2.77/25.4)$$

$$= 208.281 \text{ kgf/cm}^2$$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	1.9096 mm
Allowable Internal Pressure at Corroded thickness	272.46 kgf/cm ²
Required Internal Design Pressure	194.77 kgf/cm ²
Allowable External Pressure at Corroded thickness	208.28 kgf/cm ²
Required External Design Pressure	85.66 kgf/cm ²
Required Thickness due to Shell Side pressure	1.5740 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$$= 2 \cdot ro + dt$$

$$= 2 \cdot 570.524 + 25.4$$

$$= 1166.448 \text{ mm}$$

Basic ligament efficiency for shear [μ]:

$$= (p - dt) / p$$

$$= (31.75 - 25.4) / 31.75$$



= 0.200

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:

$$= D_s / D_o \text{ (Configurations a, b, c)}$$

$$= 1220.0/1166.448$$

$$= 1.0459$$

Ratio [pc]:

$$= D_c / D_o \text{ (Configurations a, e, f)}$$

$$= 1220.0/1166.448$$

$$= 1.0459$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [p]:

$$= l_{tx} / h$$

$$= 0/300.0$$

$$= 0.000 \text{ (must be } 0 \leq p \leq 1 \text{)}$$

Effective tube hole Diameter [d*]:

$$= \max(dt - 2t_t(E_t T/E) \cdot (St T/S) \cdot (p); dt - 2t_t)$$

$$= \max(25.4 - 2 \cdot 2.77 (1894790/1894790) \cdot$$

$$(1107/1509) \cdot (0); 25.4 - 2 \cdot 2.77)$$

$$= 25.4000 \text{ mm}$$

Effective tube pitch [p*]:

$$= p / \sqrt{ (1 - 4 \cdot \min(A_l; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2)) }$$

$$= 31.75 / \sqrt{ (1 - 4 \cdot \min(0 ; 4 \cdot 1166.448 \cdot 31.75) / (\pi \cdot 1166.448^2)) }$$

$$= 31.7500 \text{ mm}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (31.75 - 25.4) / 31.75$$

$$= 0.2000$$

$h/p = 9.448819$; $\mu^* = 0.200000$; $E^* = 294354 \text{ kgf/cm}^2$

$E^*/E = 0.155349$; $\nu^* = 0.458572$

Note: As h/p (9.449) is > 2, data values for $h/p = 2$ were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient [β_s]:

$$= ((12(1 - \nu_s^2))^{1/4}) / \sqrt{ ((D_s + t_s) t_s) }$$

$$= ((12(1 - 0.3^2))^{1/4}) / \sqrt{ ((1220.0 + 38.0) 38.0) }$$

$$= 0.0083 \text{ 1/mm}$$

Shell coefficient [k_s]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - \nu_s^2))$$

$$= 0.008 \cdot 1913771 \cdot 38.0^3 / (6(1 - 0.3^2))$$

$$= 159908592.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Shell coefficient [λ_s]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(1220.0 + 38.0) \cdot 0.15991E+09 / 300.0^3 \cdot (1 + 300.0 \cdot 0.008 + (300.0 \cdot 0.008)^2 / 2)$$

$$= 295265.0 \text{ kgf/cm}^2$$

Shell coefficient [δ_s]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot \nu_s / (2(D_s + t_s)))$$

$$= 1220.0^2 / (4 \cdot 1913771 \cdot 38.0) \cdot (1 - 1220.0 \cdot 0.3 / (2(1220.0 + 38.0)))$$

$$= 0.43723E-02 \text{ mm/kgf/cm}^2$$

Parameter [ω_s]:

$$= p_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$$

$$= 1.0459 \cdot 0.15991E+09 \cdot 0.0083 \cdot 0.004372 (1 + 300.0 \cdot 0.0083)$$

$$= 21245.2109 \text{ mm}^2$$

Channel coefficient [β_c]:



$$= ((12(1 - v_c^2))^{1/4}) / \sqrt{((D_c + t_c)t_c)}$$

$$= ((12(1 - 0.3^2))^{1/4}) / \sqrt{((1220.0 + 88.0)88.0)}$$

$$= 0.0054 \text{ 1/mm}$$

Channel coefficient [kc]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - v_s^2))$$

$$= 0.005 \cdot 1894790 \cdot 88.0^3 / (6(1 - 0.3^2))$$

$$= 1267147008.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λc]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot kc / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (Cm\Delta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot Cq\Delta)$$

$$= 6(1220.0 + 88.0 / \cos(0)) \cdot 0.12671E+1 / 300.0^3 \cdot (1.0 + 300.0 \cdot 0.005 \cdot (1.0 + 1.0) / 2 + (300.0 \cdot 0.005)^2 / 2 \cdot 1.0)$$

$$= 1436195.8 \text{ kgf/cm}^2$$

Channel coefficient [δc]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot v_c / (2(D_c + t_c)))$$

$$= 1220.0^2 / (4 \cdot 1894790 \cdot 88.0) \cdot (1 - 1220.0 \cdot 0.3 / (2(1220.0 + 88.0)))$$

$$= 0.19194E-02 \text{ mm/kgf/cm}^2$$

Channel coefficient [zetac]:

$$zetac = 0$$

Parameter [wc]:

$$= \rho_c \cdot kc [\delta C (CmTheta \cdot \beta_c + Cq\Delta \cdot h \cdot \beta_c^2) - \zeta c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)]$$

$$= 1.046 \cdot 0.12671E+1 [0.001919 (1.0 + 0.005 \cdot 1.0 \cdot 300.0) - 0 (1.0 / 300.0 + 1.0 \cdot 0.005 / 2)]$$

$$= 35538.9844 \text{ mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o$$

$$= 1396.0 / 1166.448$$

$$= 1.1968$$

Coefficient [F]:

$$= (1 - v^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.46) / 294354 \cdot (295265.03 + 1436195 + 1894790 \cdot \ln(1.2))$$

$$= 3.8109$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= D_o^2 / 16 \cdot [(ps - 1) \cdot (ps^2 + 1) \cdot Ps - (pc - 1) \cdot (pc^2 + 1) \cdot Pt]$$

$$= 1166.448^2 / 16 \cdot [(1.046 - 1) \cdot (1.046^2 + 1) \cdot 85.658 - (1.046 - 1) \cdot (1.046^2 + 1) \cdot 194.77]$$

$$= -891983.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]:

$$= Mts + wc \cdot Pt - \omega_s \cdot Ps$$

$$= -891983.0 + 35538.984 \cdot 194.77 - 21245.211 \cdot 85.658$$

$$= 4210132.5000 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= (M^*) - D_o^2 / 32 \cdot F (Ps - Pt) / (1 + F)$$

$$= (4210132) - 1166.448^2 / 32 \cdot 3.811 (85.66 - 194.77) / (1 + 3.81)$$

$$= 4550107.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$= Mp + D_o^2 / 64 \cdot (3 + v^*) \cdot (Ps - Pt)$$

$$= 4550107 + 1166.448^2 / 64 \cdot (3 + 0.459) \cdot (85.66 - 194.77)$$



$$= -3472599.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$\begin{aligned} &= \max(|M_p|; |M_o|) \\ &= \max(|4550107|; |-3472599.0|) \\ &= 4550107.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$\begin{aligned} &= 6 \cdot M / (\mu \cdot (h - h_g')^2) \\ &= 6 \cdot 4550107 / (0.2 \cdot (300.0 - 6.0)^2) \\ &= 1579.2401 \text{ kgf/cm}^2 \end{aligned}$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$\begin{aligned} &= 2 \cdot S \\ &= 2 \cdot 1509.1 \\ &= 3018.20 \text{ kgf/cm}^2 \end{aligned}$$

Tubesheet bending stress at final thickness [σ]:

$$\begin{aligned} &= 6 \cdot M / (\mu \cdot (h - h_g')^2) \\ &= 6 \cdot 4476304 / (0.2 \cdot (217.0058 - 6.0)^2) \\ &= 3016.1440 \text{ kgf/cm}^2 \end{aligned}$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$\begin{aligned} &= h + Cats + Catc \\ &= 217.0058 + 0 + 0 \\ &= 217.0058 \text{ mm} \end{aligned}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$\begin{aligned} &= \max(HreqB; HreqS) \\ &= \max(217.0058; 110.8675) \\ &= 217.0058 \text{ mm} \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [|Ps - Pt|]:

$$\begin{aligned} &= |Ps - Pt| \\ &= |85.66 - 194.77| \\ &= 109.112 \text{ kgf/cm}^2 \end{aligned}$$

Shear Stress check [τ_{limit}]:

$$\begin{aligned} &= 4 \cdot \mu \cdot h / Do \cdot \min(0.8 \cdot S; 0.533 \cdot Sy) \\ &= 4 \cdot 0.2 \cdot 300.0 / 1166.45 \cdot \min(0.8 \cdot 1509.1; 0.533 \cdot 2492.098) \\ &= 248.401 \text{ kgf/cm}^2 \end{aligned}$$

|Ps - Pt| is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$\begin{aligned} &= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S; 0.533 \cdot Sy)) \cdot (4 \cdot Ap / Cp) \cdot |Ps - Pt| + Cats + Catc \\ &= 1 / (4 \cdot 0.2 \cdot 1207.28) \cdot (4 \cdot 8673.4 / 3535.24) \cdot |85.66 - 194.77| + 0 \\ &= 110.8675 \text{ mm} \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Shell thickness of: 38.000mm

Min. Shell len. adjacent to the tubesheet is: 387.565 mm

Axial Shell Membrane Stress [σ_s, m]:

$$\begin{aligned} &= Ds^2 / (4 \cdot ts(Ds + ts)) \cdot Ps \\ &= 1220.0^2 / (4 \cdot 38.0(1220.0 + 38.0)) \cdot 85.658 \\ &= 666.7519 \text{ kgf/cm}^2 \end{aligned}$$

Axial Shell Bending Stress [σ_s, b]:

$$\begin{aligned} &= 6 \cdot ks / ts^2 \cdot [\beta S \cdot P's + 6(1 - \nu) / E \cdot \\ &\quad Do / h^3 \cdot (1 + h \cdot \beta S / 2) \cdot (Mp + Do^2 / 32 \cdot (Ps - Pt))] \\ &= 6 \cdot 3525378 / 38.0^2 \cdot [0.008 \cdot 0.3725 + 6(1 - 0.459) / 294354 \cdot \\ &\quad 1166.448 / 300.0^3 \cdot (1 + 300.0 \cdot 0.008 / 2) \cdot (4550107 + 1166.448^2 / 32 \cdot \\ &\quad (85.66 - 194.77))] \end{aligned}$$



$$= 2005.500 \text{ kgf/cm}^2$$

Shell Membrane + Bending Stress [σ_s]:

$$= |\sigma_{s,m}| + |\sigma_{s,b}| \text{ (Should be } \leq 1.5 \cdot S_s)$$

$$= 2672.252 \text{ kgf/cm}^2 \text{ (Should be } \leq 2533.01)$$

Note: For a given Channel thickness of: 88.000 mm
 Min. Channel len. adjacent to the tubesheet is: 589.785mm

Axial Channel Membrane Stress [$\sigma_{c,m}$]:

$$= D_c^2 / (4 \cdot t_c (D_c + t_c) \cdot \cos(\theta)) \cdot P_t$$

$$= 1220.0^2 / (4 \cdot 88.0 (1220.0 + 88.0) \cdot \cos(0)) \cdot 194.77$$

$$= 629.640 \text{ kgf/cm}^2$$

Axial Channel Bending Stress [$\sigma_{c,b}$]:

$$= 6 \cdot k_c / t_c^2 \cdot [(\delta_c \cdot C_m \Delta + \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t -$$

$$6((1 - \nu^*) / E^*) \cdot D_o / h^3 \cdot (C_m \theta +$$

$$h \cdot \beta_c / 2 \cdot C_m \Delta) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 27935792 / 88.0^2 \cdot [(0.001919 \cdot 1.0 \cdot 0.005 - 0 \cdot 1.0) \cdot 194.77 -$$

$$6((1 - 0.459) / 294354) \cdot 1166.448 / 300.0^3 \cdot (1.0 +$$

$$0.005 \cdot 300.0 / 2 \cdot 1.0) \cdot (4550107 + 1166.448^2 / 32 \cdot (85.658 - 194.77))]$$

$$= 2041.880 \text{ kgf/cm}^2$$

Channel Membrane + Bending Stress [σ_c]:

$$= |\sigma_{c,m}| + |\sigma_{c,b}| \text{ (Should be } \leq 1.5 \cdot S_c)$$

$$= 2671.521 \text{ kgf/cm}^2 \text{ (Should be } \leq 2490.20)$$

Step 11, The Cylinder-to-Tubesheet Junction is Overstressed.

Performing a simplified Elastic-Plastic calculation

(option 3) to reduce the overstress condition.

Modify E_s and/or E_c and recompute from Step 4 onwards.

Modified Elasticity [E_s]:

$$= E_s \cdot \sqrt{(1.5 \cdot S_s / \sigma_s)}$$

$$= 1913771 \cdot \sqrt{(1.5 \cdot 1688.67 / 2672)}$$

$$= 1863244.4 \text{ kgf/cm}^2$$

Modified Elasticity [E_c]:

$$= E_c \cdot \sqrt{(1.5 \cdot S_c / \sigma_c)}$$

$$= 1894790 \cdot \sqrt{(1.5 \cdot 1660.13 / 2671.52)}$$

$$= 1829359.6 \text{ kgf/cm}^2$$

Tubesheet Bending Stress at Original Thk., after the Elastic-Plastic iteration:

$$= 6 \cdot M / (\mu \cdot (h - h'g)^2)$$

$$= 6 \cdot 4548105 / (0.2 (300.0 - 6.0)^2)$$

$$= 1578.5 \text{ kgf/cm}^2 \text{ (Should be } \leq 3018.2)$$



Note: The tubesheet is Not overstressed after the Elastic-Plastic iteration.

The design is acceptable. Recomputing tubesheet required thickness.

Required Tubesheet Thk., for Bending Stress after the Elastic-Plastic iteration [H_{reqB}]:

$$= h + C_{ats} + C_{atc} = 217.0058 + 0 + 0 = 217.0058 \text{ mm}$$

Required Tubesheet Thk. after the Elastic-Plastic iteration (includes CA) [H_{req}]:

$$= \max(H_{reqB}; H_{reqS}) = \max(217.0058; 110.8675) = 217.0058 \text{ mm}$$

Tube Metal Area [A_t]:

$$= \pi (d_o - t) t$$

$$= \pi (25.4 - 2.77) 2.77$$

$$= 1.969 \text{ cm}^2$$

Tube Weld Size Results per VIII-2 4.21.2:

Tube Strength [F_t]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 2.77 (25.4 - 2.77) 1107.364$$



= 2180.740 kgf

Groove Weld Strength [Fg]:

= $0.85 \cdot \pi \cdot ag (do + 0.67 \cdot ag) Sw$ (but not > Ft)

= $0.85 \cdot \pi \cdot 2.77 (25.4 + 0.67 \cdot 2.77) 1107.36$

= 2180.7405 kgf

Weld Strength Factor [fw]:

= S / Sw where Sw = min(S and St)

= 1107.364/1107.364

= 1.000

Design Strength Ratio [fd]:

= 1 for full strength welds

Fillet Weld Strength [Ff]:

= $0.55 \cdot \pi \cdot af (do + 0.67 \cdot af) Sw$ (but not > Ft)

= $0.55 \cdot \pi \cdot 2.77 (25.4 + 0.67 \cdot 2.77) 1107.36$

= 1444.5836 kgf

Minimum Length of Weld [ar]:

= $2 \cdot (\sqrt{(0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t) fw \cdot fd}) - 0.75 \cdot do$

= $2 \cdot (\sqrt{(0.75 \cdot 25.4)^2 + 1.07 \cdot 2.77 \cdot (25.4 - 2.77) 1.0 \cdot 1.0}) - 0.75 \cdot 25.4$

= 3.372 mm

Minimum Required Fillet Weld Leg [afr]: 1.6859 mm

Minimum Required Groove Weld Leg [agr]: 1.6859 mm

Tube-Tubesheet Jt allowable, 2180.74 is >= tube strength 2180.74 kgf

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	1578.5 <=	3018.2 kgf/cm ²	Ok
Tubesheet shear stress	446.2 <=	1207.3 kgf/cm ²	Ok
Stress in Shell at Tubesheet	2672.3 <=	5066.0 kgf/cm ²	Ok
Stress in Channel at Tubesheet	2671.5 <=	4984.4 kgf/cm ²	Ok

Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	217.006	300.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.686	2.770	Ok
Tube-Tubesheet Groove Weld Leg :	1.686	2.770	Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 6 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Stresses	Case Type	Pass/Fail
		Bend	Allwd Shear Allwd		
D1uc	282.039	...	2648 3018	796 1207	Ps+Pt D1 Ok
D2uc	188.087	...	1106 3018	350 1207	Ps+Pt D2 Ok
D3uc	217.006	...	1579 3018	446 1207	Ps+Pt D3 Ok
D1c	283.987	...	2676 3018	813 1207	Ps+Pt-c D1 Ok
D2c	191.028	...	1141 3018	357 1207	Ps+Pt-c D2 Ok
D3c	219.505	...	1600 3018	455 1207	Ps+Pt-c D3 Ok
Max:	283.9869	... mm	0.887	0.673 (Str. Ratio)	

Load Case Definitions:

[Ps & Pt]:



Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Shell and Channel Stress Summary:

--	Integral Cyl.	Reqd. Thk.	Shell	Stress	Channel	Stress	Pass
Case#	Shell	Channel	Stress	Allwd	Stress	Allwd	Fail
D1uc	...	...	437	2533	3114	4984	Ok
D2uc	...	...	3109	5066	443	2490	Ok
D3uc	...	...	2672	5066	2672	4984	Ok
D1c	...	...	499	2533	3266	4984	Ok
D2c	...	...	3313	5066	492	2490	Ok
D3c	...	...	2814	5066	2775	4984	Ok
Max	...	...	0.654		0.655		

Note:
 Because there was no net pressure differential across the tubesheet due to
 the specified pressures psd,min and ptd,min; load case D4 was not processed as
 the resulting tubesheet bending stress would be zero.

Summary of Thickness Comparisons for 6 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	283.987	300.000	Ok
Tubesheet thickness flanged extension :	0.000	64.000	Ok
Tube thickness :	1.910	2.770	Ok
Tube-tubesheet fillet weld leg :	1.686	2.770	Ok
Tube-tubesheet groove weld leg :	1.686	2.770	Ok

Given Shell Thickness: 38.0000 mm
 Given Channel Thickness: 88.0000 mm

Min Shell length of thk, (38.000) adj. to tubesheet: 387.565 mm

Min Channel length of thk, (88.000) adj. to tubesheet: 589.785 mm

Note: This is a full strength Tube to Tubesheet Joint.

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	218.187	1.000	227.109	1.000
Tubesheet Shear Stress	289.344	1.000	289.344	1.000
Shell Stress (Axial, Junction)	289.344	1.326	130.416	0.996
Tubesheet-Channel Junction	289.344	1.326	295.249	0.680
Tube Pressure Stress	272.457	1.000	208.280	1.000
Minimum MAWP	218.187		130.416	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
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Tubesheet Bending Stress	220.906	1.000	234.164	1.000
Tubesheet Shear Stress	295.249	1.000	295.249	1.000
Shell Stress (Axial, Junction)	295.249	1.337	174.634	1.000
Tubesheet-Channel Junction	295.249	1.337	234.164	0.383
Tube Pressure Stress	296.036	1.000	-0.001	1.000
Minimum MAPnc	220.906		-0.001	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Tubesheet MDMT Calculations:

Shell Side MDMT calculation:

Governing thickness on the shell side per figure 3.10:

$$\begin{aligned}
 &= \max(\text{tubesheet thk}/4, \min(\text{tubesheet thk}, \text{shell thickness})) \\
 &= \max(300.0/4, \min(300.0, 38.0)) \\
 &= 75.000 \text{ mm}
 \end{aligned}$$

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature : 39 °C
 User entered Minimum Design Temperature : -10 °C
 The Curve for this Material is : B
 From Figure 3.7M- Non PWHT, the MDMT is : 39 °C
 Temperature reduction per Graph 3.12 : 0 °C
 The computed stress ratio is : 1.000

The computed thickness ratio [sr]:

$$\begin{aligned}
 &= (tr \cdot E^*) / (tnom - ci - cext) \\
 &= (2175717 \cdot 1.0) / (75.0 - 6.0 - 0) \\
 &= 1.000
 \end{aligned}$$

Warning: Impact Testing Is Mandatory!

Channel Side MDMT calculation:

Governing thickness on the shell side per figure 3.10:

$$\begin{aligned}
 &= \max(\text{tubesheet thk}/4, \min(\text{tubesheet thk}, \text{channel thickness})) \\
 &= \max(300.0/4, \min(300.0, 88.0)) \\
 &= 88.000 \text{ mm}
 \end{aligned}$$

MDMT reduction ratio:

$$\begin{aligned}
 &= \min(pt/\text{Tubeside MAPnc}, 1) \\
 &= \min(194.77/220.91, 1) \\
 &= 0.882
 \end{aligned}$$

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature : 13 °C
 User entered Minimum Design Temperature : -10 °C
 The Curve for this Material is : B
 From Figure 3.8M- with PWHT, the MDMT is : 19 °C
 Temperature reduction per Graph 3.13 : 6 °C
 The computed stress ratio is : 0.882

Warning: Impact Testing Is Mandatory!

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Noted issues for the current model:

Listed below are the known problem areas for the current design. If one or more of the design flags are turned on, please re-run the analysis. Some of these issues may be resolved when using updated input values.

**** Warning: The net force on one of the saddles putting it in tension!**

Please review all reports carefully!

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